

GEI Consultants



Geotechnical Investigation Report

GRCA 318719: Rehabilitate North Rim Utility Infrastructure

Grand Canyon North Rim, Arizona

Submitted to:
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August 11, 2025
Project No. 2305987



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August 11, 2025
Project No. 2305987

VIA EMAIL: matt.gough@hdrinc.com

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Re: Geotechnical Investigation Report
GRCA 318719: Rehabilitate North Rim Utility Infrastructure
Grand Canyon North Rim, Arizona

Mr. Gough:

Submitted herewith is the Geotechnical Engineering Investigation for the North Rim Portion of the Grand Canyon National Park Rehabilitate North Rim Utility Infrastructure project. This study was conducted in accordance with our contract between GEI Consultants, Inc. and HDR Engineering, Inc. This report contains the results of our findings, an engineering interpretation with respect to the available project characteristics, and recommendations to aid design and construction of earth-related phases of this project.

If you have any questions regarding the contents of this report, please contact the undersigned.

Sincerely,

GEI Consultants, Inc.

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1. Introduction and Background

The National Park Service (NPS) is planning several upgrades to their existing water treatment infrastructure as part of the GRCA 318719: Rehabilitate North Rim Infrastructure Project. The purpose of the project is to rehabilitate existing drinking water supply infrastructure and includes improvements and upgrades to existing treatment facilities and replacement of the North Rim Waterline via directional drilling and installation of a cased bore from the North Rim to Roaring Springs. The NPS retained HDR Engineering, Inc. (HDR) as the lead engineer for this project. HDR retained GEI Consultants, Inc. (GEI) to perform geotechnical investigation and design services for the various infrastructure improvements.

The focus of this report is to present the findings of our geotechnical investigation at the North Rim and to provide geotechnical recommendations for design and construction of the various improvements proposed on the North Rim. The report includes the initial Phase 1 and Phase 2 investigations completed in October through November 2022, the Phase 3 investigation completed in November 2023, and the Phase 4 investigation completed in November 2024. Geotechnical reports covering the proposed inner canyon improvements and the proposed bore are provided under separate cover.

The project site is located along Arizona State Route 67 in the North Rim of Grand Canyon National Park as shown on Figure 1 and includes an existing Wastewater Treatment Plant (WWTP), proposed new Water Treatment Plant (WTP), and proposed pipeline alignment. Proposed upgrades/additions to the existing and proposed structures include:

1. Additions to the WWTP including a secondary treatment/UV disinfection building, headworks/EQ building, retaining walls, and ancillary structures as shown in Figure 2. Renovations to the existing control building are not anticipated to require geotechnical recommendations and are therefore not discussed in this report.
2. Addition of a new diesel fuel tank adjacent to the existing generator building east of the WWTP as shown in Figure 3.
3. Construction of a new WTP building and two above-grade raw water tanks as shown in Figure 4.
4. Installation of several new utilities from the proposed WTP to the proposed bore location including a new 1.5-inch PVC waterline, 3.5-inch steel raw waterline, and 2-inch fiber optic line, all of which will be installed in a single open-cut trench as shown in Figure 4 through Figure 6.

2. Geologic Setting

2.1. Regional Geology

Soils at the project site are described as residual materials developed from limestone and other carbonate rocks by Soller et al. (2009). This description is supported by the US Department of Agriculture's Web Soil Survey (USDA, 2022), who describes the surficial deposits as the following:

- Consisting of cobbly fine sandy loam, gravely sandy loam, or clay to sandy clay loam depending on the member of the Kanabownits-Kippers-Kaiparowits complex encountered.
- Soils derived from cherty limestone or calcareous sandstone parent material.

Bedrock at the project site consists of the Kaibab Formation, as mapped by Huntoon et al. (1996). Descriptions of the Kaibab formation have changed over time, as the formation is comprised of several different interbedded rock types and several new members and distinctions within the formation have been added. The uppermost member of the Kaibab Formation is the Harrisburg Member, described by Bassler and Reeside (1921) as 160 ft thick and consisting primarily of cream to gray limestone separated by shale, sandstone, and gypsum. The Harrisburg Member is further defined by Sorauf and Billingsley (1991) as containing interbedded limestone, gypsum, dolomite, chert, and sandstone. Both Sorauf and Billingsley (1991) and McKee (1938) note that in the eastern portion of the Grand Canyon and further to the east the Kaibab Formation transitions to being mostly sandstone, calcareous sandstone, or sandy limestone with red shale beds and gypsum also being noted. The lower, Fossil Mountain Member, meanwhile, contains fossiliferous limestone, white chert, and a yellow sandstone bed (Sorauf and Billingsley, 1991). Altogether, the Kaibab formation is approximately 350 ft thick at the North Rim (Sorauf and Billingsley, 1991; Huntoon et al., 1996) and is underlain by gypsum, shale, and sandstone of the Toroweap Formation.

2.2. Karst Geology

Carbonate rocks (limestone and dolomite) and evaporites (halite and gypsum) are known lithologies that commonly experience dissolution when in contact with water resulting in the formation of karst features. Common karst features include sinkholes, depressions, pinnacles/towers, and dissolution voids (e.g., caves). There are numerous lithologies within the Kaibab formation that are susceptible to karstification, including limestone and gypsum. The Arizona Geologic Survey has documented numerous surficial karst features in the Kaibab Formation on the north rim of the Grand Canyon (Harris, 2002). These include enlarged joints and shallow sinkholes formed through the dissolution of gypsum and, to a lesser extent, limestone. Furthermore, numerous enlarged joints and voids within the shallow subsurface are noted in the Kaibab formation, particularly within the Holbrook Basin located approximately 140 miles southeast of the Grand Canyon. These enlarged joints are commonly encountered when drilling and often result in the loss of circulation down the drill hole, indicating that drilling fluids are entering and following the joints rather than returning to the surface.

GEI performed a limited desktop review to screen for the presence of sinkholes associated with large karst features at the subject sites. GEI reviewed historical aerial imagery available on Google Earth and

HistoricAerials.com dating back to 1954 but was unable to distinguish any closed topographic depressions due to the tree cover at the sites. GEI also reviewed LiDAR-based mapping of karst sinkhole features on the Kaibab presented in Jones et al. (2017). In general, the Jones et al. (2017) sinkhole map was at too coarse a resolution to completely rule out existing sinkhole features at the subject sites, as several sinkhole features were mapped in the area.

3. Geotechnical Investigation

GEI conducted a total of four phases of geotechnical investigation at the project site. GEI conducted the first phase of investigation between October 18, 2022, and October 20, 2022, the second phase between November 1, 2022, and November 7, 2022, the third phase between November 7, 2023, and November 13, 2023, and the fourth phase on November 4, 2024. The geotechnical investigation included site observations, geotechnical drilling, percolation testing, a test pit investigation, and a geotechnical laboratory testing program. GEI completed a total of 5 percolation tests (Phase 1), 1 test pit (Phase 1), and 32 geotechnical borings (17 borings in Phase 2 and 15 borings in Phase 3) to investigate the general subsurface conditions at the WWTP, WTP and pipeline alignment. The percolation test locations, test pit location, and boring locations are shown in Figure 2 through Figure 6 at the end of this report.

3.1. Subsurface Investigation

3.1.1. Geotechnical Borings

Geotechnical borings were advanced to depths between 10.4 feet and 41.5 feet below ground surface (bgs). GEI subcontracted Geomechanics Southwest Inc. (GSI) to drill the borings using a CME-55 truck-mounted drill rig for Phase 2, and a CME-85 truck-mounted drill rig for Phase 3. Drilling and sampling procedures were conducted in general accordance with ASTM D1586 – Standard Test Method for Standard Penetration Test (SPT) and Split-Barrel Sampling of Soils, ASTM D 3550 – Standard Practice for Thick Wall, Ring-Lined, Split Barrel, and Drive Sampling of Soils, and ASTM D2113 Standard Practice for Rock Core Drilling and Sampling of Rock for Site Investigation. Continuous-flight, hollow-stem augers were used to advance the borings through soil terminating in bedrock. During advance of the hollow-stem augers, alternating Modified California samples (2.0-inch inner diameter) and Split Spoon samples (1.4-inch inner diameter) were obtained at intervals of 2.5 to 5 feet. The number of blows by a 140-pound hammer falling 30-inches required for 12 inches of sampler penetration (recorded in 6-inch increments) are presented on the boring logs. Blow counts with less than 6 inches of penetration are presented showing the number of blows for the resulting depth of penetration (50/2” = 50 blows to drive the sampler two inches). After auger refusal was encountered in bedrock, HQ wireline rock coring methods were used to obtain rock core samples until the required depth was achieved. Boring logs are included in Appendix A. Table 3-1 below summarizes the geotechnical borings completed and associated upgrade/improvement.

Table 3-1. Geotechnical Boring Summary

Boring Name	Depth (ft)	Proposed Upgrade/Improvement
GB 1	41.5	WWTP Secondary Treatment Building/UV Disinfection Building
GB 2	40.1	Headworks/EQ Building
GB 3	41.5	WTP Foundation - Alternate Location
GB 4	41.0	WTP Foundation and Waterline
GB 5	10.4	WTP Foundation and Waterline
GB 6	15.3	Waterline
GB 7	10.8	Waterline
GB 8	15.2	Waterline
GB 9	11.5	Waterline
GB 10	16.0	Waterline
GB 11	10.4	Waterline
GB 12	11.3	Waterline
GB 13	16.4	Waterline
GB 14	23.0	Waterline and Bore Feasibility Assessment
GB 15	40.0	WWTP Secondary Treatment Building/UV Disinfection Building
GB 16	40.0	WWTP Secondary Treatment Building/UV Disinfection Building
GB 17	35.0	WWTP Secondary Treatment Building/UV Disinfection Building
GB 18	40.0	WWTP Secondary Treatment Building/UV Disinfection Building
GB 19	40.0	WTP Raw Water Tanks
GB 20	40.0	WTP Raw Water Tanks
GB 21	40.0	WTP Foundation and Waterline
GB 22	41.5	WTP Foundation and Waterline
GB 23	40.0	WTP Foundation and Waterline
GB 24	40.0	WTP Foundation and Waterline
GB 25	15.3	Waterline
GB 26	16.0	Waterline
GB 27	15.8	Waterline
GB 28	16.0	Waterline
GB 29	13.1	Waterline
Ballfield	25.0	Waterline and Potential Bore Drill Pad Foundation
Tank Farm	16.5	Waterline and Potential Bore Drill Pad Foundation
North Rim Lodge	15.3	Waterline and Potential Bore Drill Pad Foundation

3.1.2. Percolation Tests

GEI performed five percolation tests to determine the percolation rate at several potential leach field locations around the WWTP. Percolation tests were conducted using the EPA's falling head percolation test guidelines (EPA, 1980). The location of these percolation tests is shown in Figure 2. Results of the percolation testing are included in Appendix F.

3.1.3. Test Pit Investigation

A test pit investigation was conducted at one location (TP-1) utilizing a mid-size Cat backhoe loader with an approximately 2-foot bucket and extended to a depth of approximately 66 inches below existing grade. GEI provided full time oversight of the test pit investigation and documented encountered subsurface conditions as excavation progressed. The location of the test pit is shown in Figure 2. The resulting test pit log is presented in Appendix C.

3.1.4. Geotechnical Laboratory Testing

A geotechnical laboratory testing program was developed by GEI and conducted by Martinez Associates and Advanced Terra Testing on representative samples collected during the subsurface investigation. A laboratory summary table and graphical testing results are provided in Appendix D. Laboratory tests conducted in general accordance with associated ASTM standards are presented in Table 3-2 below.

Table 3-2. Geotechnical Laboratory Testing

Test	Standard
Moisture Content	ASTM D2216
Dry Density	ASTM D2937
Grain Size Distribution	ASTM D422/ASTM C136
#200 Wash	ASTM D1140
Atterberg Limits	ASTM D4318
Swell/Consolidation	ASTM D4546
UCS (Soil)	ASTM D2166
UCS (Rock)	ASTM D7012 – Method C

3.1.5. Corrosion Potential Testing

In addition to the geotechnical laboratory testing, selected soil samples were tested for corrosion potential. Corrosion laboratory testing was performed by Project X Corrosion Engineering. Laboratory corrosion results are presented in Appendix E. Corrosion testing conducted in general accordance with associated ASTM standards are presented in Table 3-3 below.

Table 3-3. Corrosion Laboratory Testing

Test	Standard
Soil Resistivity	ASTM G187
pH	ASTM G51
Sulfate	ASTM D4327
Redox Potential	ASTM G200
Sulfide	SM 4500-S2
Ammonia	ASTM D4327
Nitrate	ASTM D4327
Fluoride	SM 2320B
Phosphate	ASTM D4327
Lithium	ASTM D4327
Sodium	ASTM D4327
Potassium	ASTM D4327
Magnesium	ASTM D4327
Calcium	ASTM D4327
Bicarbonate	SM 2320B

3.2. Geophysical Investigation

GEI subcontracted Collier Geophysics to perform a geophysical investigation using various geophysical methods to further characterize the subsurface conditions at the project site. GEI summarized the results and interpretation of the geophysical investigation in the North Rim Geophysical Study letter provided to HDR. Collier's report and other supporting information is included with the summary letter.

4. Subsurface Conditions

Subsurface conditions were assessed based on the findings of the geotechnical investigation described in Section 3 above. Soil and rock descriptions noted on the boring logs and below are in general accordance with ASTM D 2487 – Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System) and D 2488 – Standard Practice for Description and Identification of Soils (Visual-Manual Procedure).

4.1. Subsurface Materials

Primary materials encountered during the subsurface investigation include fill, residuum and sandy limestone of the Kaibab Formation. Boring logs and a supplementary boring log key explaining boring log details are provided in Appendix A.

4.1.1. Existing Fill

Existing fill was encountered in proximity to the existing WWTP in borings GB 1, GB 15, GB 16, GB 17, and GB 18 at ground surface and extended to depths of 2 feet to 16.5 feet. Encountered existing fill was described as the following in accordance with the Unified Soil Classification System (USCS):

- Lean Clay with Sand (CL)
- Sandy Lean Clay with Gravel (CL)
- Clayey Sand (SC)
- Clayey Sand with Gravel (SC)

Existing fill was primarily comprised of clay and sand with varying gravel content with uncorrected SPT blow counts ranging from 1 to 17 blows per foot (bpf) indicating a relative consistency of very soft to very stiff for fine soils and a relative density of very loose to medium dense for coarse soils.

4.1.2. Residuum

Residuum generated from in situ weathering of limestone bedrock was encountered in all borings at depths between ground surface or below the existing fill (where present) and extended to depths of between 5 feet bgs and the depth of exploration. For the purposes of this investigation, GEI generally classified the weathered bedrock as soil material where auger drilling was feasible. GEI generally classified the subsurface material as bedrock below the depth where auger refusal occurred. Residuum classifications in accordance with USCS are presented on the boring logs in Appendix A.

Residuum consisted of a wide variety of soil types with varying clay, silt, sand, and gravel content. Uncorrected SPT blow counts in the residuum ranged from 0 bpf to 50 blows for 1 inch of sampler penetration (50/1”) indicating a relative consistency of very soft to hard for fine soils and a relative density of very loose to very dense for coarse soils. Two unconfined compressive strength tests were performed on the residuum with results ranging from 10 to 76 psi as shown in Appendix D.

Swell/collapse testing on five samples of residuum resulted in swell/collapse percent between -0.5% and 0.5% when inundated at a confining pressure of 500 psf as shown in Appendix D.

4.1.3. Kaibab Formation Bedrock

Limestone bedrock of the Kaibab Formation was encountered in 14 of the 32 borings below the residuum at depths ranging from 5 to 40 feet bgs and extended to the depth of exploration. Bedrock was generally described as soft to medium hard, fresh to highly weathered, light brownish gray to light gray, very fine to fine grained sandy limestone with thin to thick bedding, very close to wide joint spacing, with hard chert nodules throughout. Unconfined compressive strength (UCS) tests were performed on 12 samples of bedrock resulting in UCS values ranging from 1,460 to 9,273 psi as shown in Appendix D. Low UCS values may be attributed to fractures within the rock mass and associated planes of weakness in the tested sample.

4.1.4. Karst Features

Though no shallow sinkholes or other surficial karst features were observed at the project site, Phase 2 and Phase 3 geotechnical investigation results indicated the potential presence of karstic voids associated with the Kaibab Formation. During Phase 2 of the geotechnical investigation, boring GB 4 encountered a 2-foot-thick void followed by soils with relatively low blow counts. Relatively softer materials within the residuum and bedrock were encountered in several other borings during the Phase 3 investigation at the WTP and WWTP sites as indicated by low blow counts, zones of no recovery during rock coring activities, and loss of drilling fluid circulation during rock coring.

To better understand the presence, extent, and size of potential karst features in the vicinity of the proposed WTP improvements, a geophysical ERT survey was performed as summarized in the North Rim Geophysical Study Letter. ERT data from the geophysical investigation conducted at the WTP does not indicate the presence of large, laterally extensive void structures as indicated by relatively low electrical resistivity measurements throughout the site (approximately 10 to 8,000 Ohm-m). Were large void structures present at the site, associated high electrical resistivity readings would be expected to be about 100,000 Ohm-m which is significantly higher than shown in the ERT data. Overall, the geophysical investigation did not produce evidence for the presence of significant karst voids within approximately 100 feet of the ground surface.

GEI interprets the structure encountered in Boring GB-4 at a depth of 20 to 22 feet bgs to represent an enlarged bedrock joint where minor dissolution has occurred over time. The size of this structure is smaller and generally beyond the limits of detection with the ERT methods used. It is likely that other similar size bedrock joints exist within the subsurface at the subject sites.

GEI interprets the significantly softer residuum soils (with blow counts less than approximately 10 blows per foot) to be a result of the in-situ weathering of the bedrock through networks of fractures resulting in large, clay-filled fracture zones within the bedrock. This interpretation is consistent with observations of nearby exposures of the Kaibab Formation, where zones of more heavily jointed and thin-bedded rock are typical and presumably more prone to weathering.

Further explanation of the methods, results, and conclusions of the geophysical investigation are presented in the North Rim Geophysical Study Letter and Collier's report.

4.2. Groundwater

Groundwater was not encountered in any of the borings during the geotechnical investigation.

5. Geotechnical Design Recommendations

The following sections are included primarily for the engineer performing design. If additional geotechnical design recommendations are necessary, GEI should be contacted to provide the required information.

5.1. Foundation Recommendations

Based on data presented in this report, our experience, and the proposed foundation elevations, GEI recommends the foundation schemes shown in Table 5-1 below for the various structures proposed:

Table 5-1. Foundation Recommendations

Proposed Structure	Representative Borings	Assumed Foundation Depth[1]	Recommendation[2]
Secondary Treatment/UV Disinfection Building	GB 1, GB 15, GB 16, GB 17, GB 18	13 feet	Existing fill is present below proposed structure and should be removed. Support mat or spread footing foundation on a minimum 5-foot zone of Select Fill extending to residuum.
Headworks/EQ Building	GB 2	At Grade to 15 feet	Residuum anticipated at structure bottom. Support mat or spread footing foundation on a minimum 1-foot zone of Select Fill extending to residuum. Use CLSM in place of Select Fill if bedrock is encountered during foundation excavation. Structurally tie-in headworks footing to EQ tank.
Retaining Walls	Various	At Grade	Based on results from WWTP borings and visual inspection, residuum anticipated below retaining walls. Support retaining wall foundations on a minimum 1-foot zone of Select Fill extending to residuum.
Ancillary Structures	Various	10 feet to 15 feet	Based on results from WWTP borings and visual inspection, residuum anticipated below foundations. Support mat foundations on a minimum 1-foot zone of Select Fill extending to residuum.
Diesel Fuel Tanks	None	At Grade	Based on results from WWTP borings and visual inspection during Phase 4 of the investigation, residuum anticipated at pad bottom. Support mat foundation on a minimum 1-foot zone of Select Fill extending to residuum. Remove all organic material from base of excavation prior to fill placement.
North Rim WTP Building	GB 4, GB 5, GB 21, GB 22, GB 23, GB 24	At Grade	Residuum anticipated at structure bottom. Support mat or spread footing foundation on a minimum 1-foot zone of Select Fill extending to residuum.
Raw Water Tanks	GB 19, GB 20	At Grade	Residuum anticipated at structure bottom. Support mat or ringwall footing foundation on a minimum 1-foot zone of Select Fill extending to residuum.

Notes:
1. Assumed foundation depth is as measured from existing ground surface.
2. Select Fill (see Section 6.2.2) zone dimensions are the thickness below the bottom of the foundation.

Based on our desktop review, subsurface investigation, and the results of the geophysical investigation, we do not regard karst-related subsidence as a geologic hazard for the proposed structures. Therefore, we recommend the structures be supported on shallow foundations.

Existing fill was encountered in five borings during the subsurface investigation at the WWTP. Foundations constructed on existing fill may experience unacceptable differential settlement and therefore such foundation footprints should be overexcavated to remove the existing fill to firm residual soil and be backfilled with Select Fill.

Additionally, the residuum encountered during the subsurface investigation was generally stiff with high sand and gravel content which is generally favorable for support of shallow foundations. However, due to the potential for boulder-size bedrock particles or bedrock pinnacle features within the residuum, there is a risk of point loading effects on the proposed structure foundations and unacceptable differential settlement. Therefore, we recommend a minimum 1-foot overexcavation and backfill with Select Fill to remove potential boulder-size bedrock particles and bedrock pinnacles from below structure foundation depths.

Shallow foundations are recommended for the proposed structures. Shallow foundations may consist of conventional spread footing or mat foundations with overexcavation depths as described in Table 5-1 above. Overexcavations should extend laterally beyond the structure footprints in all directions a distance equal to the overexcavation depth. The proposed structures may be then founded on conventional spread footing or mat foundations bearing on Select Fill material meeting the requirements outlined in Section 6.2.2. GEI recommends the following criteria be utilized in design of the spread footings or mat foundation elements:

- The foundations should be designed using a maximum allowable bearing pressure of 3,000 psf This value includes appropriate allowable-stress-design (ASD) safety factors.
- The foundations should be designed to withstand a total settlement of 1 inch and differential settlements of 0.5 inches.
- It is acceptable to increase the net allowable bearing capacity by one-third if the load combination utilized considers infrequent loads such as wind or earthquake. Snow loading is not considered an infrequent load condition.
- Sliding friction between anticipated subgrade and foundation materials should be evaluated using a coefficient of 0.35.
- The coefficient of subgrade reaction (k) is typically used for flexible foundation analysis and describes the load intensity per unit of displacement. For compacted Select Fill, a k value of 125 psi/in is recommended.

5.2. Lateral Earth Pressures

Lateral earth pressures presented in Table 5-2 and Table 5-3 below should be considered during the structural design process for foundation elements extending below grade. Lateral earth pressure values are a function of the properties and the geometry of the retained fill, soil, and/or bedrock and anticipated magnitude of lateral deflection. Groundwater was not encountered within the geotechnical

investigation, therefore, additional loads due to groundwater were not considered in the lateral earth pressures.

Additional lateral earth pressure due to seismic loading was assessed using a site modified peak ground acceleration of 0.2g as determined using the ASCE 7 hazard tool with inputs consisting of the ASCE 7-22 Standard, Site Class C, and a Structure Risk Category of III.

Table 5-2. Static Lateral Earth Pressures

Backfill Material Type	Unit Weight (pcf)	Friction Angle (degrees)	Static Earth Pressure Coefficients			Static Equivalent Fluid Pressure (pcf)		
			Active	At-Rest	Passive	Active	At-Rest	Passive[1]
Imported ADOT Class 4 Aggregate	130	34	0.28	0.44	3.54	37	57	230
Select Fill	130	31	0.32	0.48	3.12	42	63	200

Notes:
1. Values shown are half of theoretical calculated value due to large strains required to mobilize passive earth pressure.

Table 5-3. Additional Lateral Earth Pressures due to Seismic Loading

Backfill Material Type	Point of Application of Additional Loading[2]	Seismic Earth Pressure Coefficients			Static Equivalent Fluid Pressure (pcf)		
		Active	At-Rest	Passive	Active	At-Rest	Passive[1]
Imported ADOT Class 4 Aggregate	0.6H	0.06	0.10	-0.19	8	13	-12
Select Fill		0.07	0.10	-0.18	9	14	-12

Notes:
1. Values shown are half of theoretical calculated value due to large strains required to mobilize passive earth pressure.
2. As measured from the base of the wall where H = total wall height.

For the lateral earth pressures presented above to remain applicable, backfill material placement adjacent to below-grade walls should be in accordance with procedures outlined in this report. In addition, backfilled material must be placed within a 1:1 (H:V) backfill geometry, up and away from the base of the structure. If the excavation is steeper than 1H:1V, the Select Fill values outlined above should be used. Consideration has not been given to vertical loads applied to the backfill surfaces during or after construction as a result of traffic and/or other surcharge loads.

5.3. Mechanically Stabilized Earth Wall Recommendations

A Mechanically Stabilized Earth (MSE) retaining wall is anticipated to be constructed adjacent to the proposed North Rim WWTP access road with a maximum height of 10 feet. Design of the retaining wall should consider the lateral earth pressures presented in Section 5.2 above as well as all anticipated surcharge loading including but not limited to construction stockpiles, construction equipment, and all temporary and permanent traffic loading. GEI Recommends the following criteria be used in the design of MSE walls.

1. MSE walls should be founded on shallow foundations consisting of spread footings with design criteria including bearing capacity, settlement, sliding friction, and coefficient of subgrade reaction as outlined in Section 5.1 above. MSE wall foundation should be excavated and prepared in accordance with the recommendations outlined in Section 6.2.4 prior to construction of the retaining wall.
2. Reinforced backfill for MSE walls should consist of free-draining imported material such as an ADOT Class 4 aggregate or equivalent. MSE walls should incorporate a minimum 1-foot zone of free-draining uniform aggregate with a nominal size of 3/4-inch immediately behind the wall face to facilitate drainage. Shear strength parameters for retaining wall materials are provided in Table 5-4 below.

Table 5-4. Retaining Wall Material Shear Strength Parameters

Material Type	Unit Weight (pcf)	Friction Angle (degrees)	Cohesion (psf)
Foundation (Native)	125	28	500
Reinforced Backfill (ADOT Class 4)	130	34	0
Drainage Aggregate (3/4-inch)	130	36	0

3. MSE wall reinforcement should consist of a uniaxial geogrid such as Tensar UX series, Mirafi Miragrid, or equivalent with tensile strength as determined during retaining wall design. Geogrid should be placed between modular blocks at a vertical spacing of approximately 1 to 2 feet with the exact vertical spacing to be determined during retaining wall design. Geogrid should be oriented such that the direction of highest tensile strength is perpendicular to the wall face. Geogrid length should be determined through retaining wall design calculations.

5.4. Frost Protection

In colder climates, certain soils can freeze and can cause heave resulting in larger than expected differential movement. Typically, clays and clean sands or gravels have negligible frost susceptibility whereas silts, silty sands, and very fine sand have high frost susceptibility. Additionally, as moisture contents increase so does the potential for frost heave. We recommend that perimeter shallow foundations and shallow foundations in non-heated enclosed areas have a minimum cover of 30 inches for frost protection per Coconino County Building Codes, Ordinances, and Design Criteria. Frost depth recommendations for buried pipelines should be provided by the pipeline designer.

5.5. Pavement Design

On-site pavement for access roads and parking areas should consist of 6 inches of full-depth asphalt concrete (without base course) or 4 inches of asphalt concrete over 8 inches of moisture conditioned and compacted Arizona DOT Class 2 aggregate base. In high traffic areas and where semi-trucks make tight turns, a minimum of 6 inches of concrete over minimum 8 inches of moisture conditioned and compacted Arizona DOT Class 2 aggregate base should be used. The pavement sections should be constructed atop a minimum 1-foot-thick zone of moisture conditioned and compacted soil consisting of reworked existing on-site soil. The resulting subgrade should be proof-rolled to verify stability and if the proof-roll reveals soft or loose soils, the subgrade should be stabilized. If performed, proof rolling should

utilize a fully loaded 10,000-gallon water truck or equivalent, with acceptable rutting criteria as deemed appropriate by the Owner's onsite geotechnical engineering representative. Stabilization can be accomplished by over excavating existing soil and replacing with at least 12 inches of compacted, moisture conditioned Select Fill. If the subgrade is still not stabilized, geotextile stabilization fabric meeting the requirements of an AASHTO Class 2 nonwoven stabilization fabric should be placed under the 12 inches of Select Fill.

The asphalt concrete pavement shall consist of a bituminous plant mix composed of a mixture of high-quality aggregate and bituminous material, which meets the requirements of a job-mix formula established by a qualified engineer. The asphalt concrete material used should be based on a SuperPave Gyratory Design Revolution of 75. The base lift should be constructed using an asphalt concrete mix designation 3/4-inch (per MAG, 2021 gradation) and asphalt cement binder grade PG 58-22, with a thickness of 2.5 to 4 inches. The top lift should be constructed using an asphalt concrete mix designation 1/2-inch (per MAG, 2021 gradation) and asphalt cement binder grade PG 58-22, with a thickness of 1.5 to 3 inches. An intermediate lift of 1/2-inch asphalt concrete may be utilized if desired.

5.6. Seismic Site Classification

Based on the International Building Code and the American Society of Civil Engineers Minimum Design Loads for Buildings and other Structures, our subsurface investigation, and our experience and knowledge of seismic conditions in the area, a Site Classification of C is recommended for seismic design.

5.7. Surface and Subsurface Drainage

Construction details should ensure that surface water does not penetrate into utility trenches and migrate beneath the structures. Also, the upper few feet of backfill adjacent to structures should consist of low permeability soils to reduce the potential for surface water to migrate into foundation wall backfill. Vegetation should not be located within a distance of 5 feet from buildings. Irrigation systems should be arranged so that water under pressure does not spray within 5 feet of foundation walls.

Current schematic design drawings show a designated area for snow piling within the vicinity of the proposed North Rim WTP and the proposed raw water tanks. Due to the potential for snow melt water to migrate beneath building foundations, we recommend this area is relocated away from the structure footprints. Alternatively, the snow piling area may be paved and proper drainage installed such that snow melt drains away from the building foundations.

5.8. Corrosion

Pipe installation trenches are anticipated to be backfilled with properly compacted granular pipe bedding material meeting the requirements outlined in Section 6.2.3. Depending on the specific pipe type selected, the corrosion potential of soils and water surrounding trench backfill should be assessed. Water soluble sulfate testing was conducted on 12 representative samples with test results ranging from 0.0015 to 0.0081% as shown in Appendix D which would characterize the concrete risk as “Not Applicable” as shown in Table 5-5 below.

Table 5-5. American Concrete Institute Code 318-19 Water Soluble Sulfates

Exposure Class	Water-Soluble Sulfate (SO_4) in Soil, Percent by Mass	Risk
S0	SO_4 <0.10	Not Applicable
S1	0.10< SO_4 <0.20	Moderate
S2	0.20< SO_4 <2.00	Severe
S3	SO_4 >2.00	Very Severe

5.9. Pipeline Recommendations

5.9.1. Differential Movement

Pipelines excavated through various soils and bedrock will be susceptible to differential movements up to 1 inch. Pipe penetrations into structures should be designed to tolerate a total of at least 1 inch of upward or downward relative movement. Existing and proposed structures are expected to function as relatively rigid structures, therefore, flexible joints and/or connections between lines and structures are recommended. Allowance for differential movement should be expected and planned for where pipelines penetrate structures.

5.9.2. Thrust Restraint

Buried, pressurized pipes experience thrust forces at various pipe configurations including tees, reducers, dead ends, valves, bends, and wyes. In order to balance internal transient pressures, thrust blocks or restrained joints are recommended for pipe restraint. For thrust block design, an allowable lateral bearing pressure of 1,500 psf is recommended. A minimum of 1 foot of cover over thrust blocks is recommended and thrust blocks should be oriented such that the passive pressure influence zones do not overlap for adjacent blocks and do not impact other subsurface utilities. If restrained joints are selected, various coefficients of friction for potential pipe materials against anticipated trench backfill material are presented in Table 5-6 below.

Table 5-6. Coefficient of Friction

Pipe Type	Coefficient of Friction
Steel	0.30
PVC	0.25

5.9.3. Modulus of Soil Reaction (E')

The modulus of soil reaction (E') is an empirically derived coefficient which is related to the stiffness of the native soils and the nature and degree of compaction of the pipe bedding and backfill material. Various design guidelines specific to pipe material provide recommended values for E'. For full embedment of pipe in compacted granular pipe bedding (meeting the requirements outlined in Section 6.2.3), where the trench is excavated in soils that are loose/medium stiff or better (typically with blow counts greater than 4 blows per foot), and where the trench width measured at the pipe elevation is at least two times the pipe diameter, an E' value as shown in Table 5-7 below should be used.

Table 5-7. Modulus of Soil Reaction (E')

Pipe Type	E' (psi)
Steel	1,500
PVC	2,000

6. Construction Considerations

The following sections are intended for the Engineer producing specifications and the contractor constructing the proposed project. Construction considerations include temporary excavations, site grading and earthwork, backfill materials, fill placement and compaction, bedrock or oversize material excavation, unstable subgrade conditions, trench plugs, and drilled shaft construction.

6.1. Temporary Excavations

General site safety including temporary excavations are the sole responsibility of the contractor performing construction. GEI is providing temporary excavation information strictly as an informational benefit to the project team, specifically the general contractor. An Occupational Safety and Health Administration (OSHA) defined competent person should be identified by the contractor to be in charge of temporary excavations. In general, the contractor's competent person should have experience or training in determining soil types, benching and shoring, and have the ability to detect potential temporary slope stability and protective system issues.

OHSA defines an excavation as a man-made cut, trench, or depression formed by the removal of earth. A trench is a specific type of narrow excavation with a geometry including a greater depth than width and a width of 15 feet or less. Trenches 5 feet deep or greater should be sloped, retained with shoring, or shielded appropriately. Shielding most commonly includes trench boxes. In general, shoring can include inclined, horizontal, or vertical systems depending on the excavation geometry and availability of retention alternatives. Sloping and benching should be in accordance with OSHA recommendations. Benching should include a maximum 4-foot vertical face for each bench and an overall excavation geometry less than or equal to the OSHA defined slope.

A professional engineer registered in the State of Arizona should approve the contractor's approach for trenches greater than or equal to 20 feet in depth. In addition, shoring for trench excavations should be designed by a professional engineer or be based on tabulated data prepared or approved by a registered professional engineer in accordance with OSHA 1926.652(b) and (c).

GEI has evaluated observed soil conditions likely to be penetrated by the proposed construction. Based on the OSHA determined soil Types, Table 6-1 below presents maximum recommended temporary excavation slopes to be utilized during construction.

Table 6-1. OSHA Temporary Excavation Slopes

Backfill Material Type	OSHA Classification	Maximum Recommended Slope (H:V)[1,2]
Existing Fill	Type C	1 1/2:1
Residuum	Type B	1:1
Bedrock	Type A	3/4:1

Notes:
1. H:V is an abbreviation for Horizontal:Vertical.
2. Valid for trench excavations less than 20 feet in depth.

During construction, heavy equipment, excavated material stockpiles, or construction material stockpiles should be kept away from excavation edges to the extent possible. It is recommended that heavy equipment or excavated material stockpiles be kept at least 10 feet from the edges of excavations. Existing underground utilities should be fully understood and documented prior to initiating excavations including information regarding location, depth, type, and size. Finally, the contractor's competent person should inspect trenches and excavations routinely for signs of instability including sliding, toppling, subsidence, bulging, heaving, squeezing, boiling, and/or other visual concerns.

6.1.1. Shored Temporary Excavations

The contractor may elect to use a temporary shoring system to limit excavation extents, particularly if other site infrastructure may be impacted. For excavations greater than 20 feet deep, the contractor will be required to submit a design prepared by a professional engineer registered in the state of Arizona to demonstrate that the shoring system maintains both local and global slope stability for the duration of the open excavation.

Typically trench boxes are used to support lateral loads during excavation and construction in soils that have sufficient standup time to excavate the trench, install the trench box, complete installation and remove the trench box without full contact of the soils. Tight shoring typically involves full contact of the soil with the shoring system. Typical tight shoring systems include sheet pile walls, soldier pile and lagging walls, secant pile walls, soil nail walls, and slide rail systems.

6.1.2. Permanent Slopes

GEI recommends a maximum slope of 3:1 for permanent cuts and fills. Erosion control may be required for slope stabilization. There are several options regarding erosion control including erosion control blankets, fiber rolls, reinforcement mats, and vegetation.

6.2. Site Grading and Earthwork

Appropriate site preparation, material placement and compaction, and backfill selection can reduce the risk of post-construction settlement and potential issues related to lateral earth pressures. General site grading and earthwork considerations are presented in the following sections.

6.2.1. Site Preparation

Areas supporting backfill and pipe bedding should be properly prepared. Once the rough grade has been established in excavated areas, and prior to placement of backfill, the exposed subgrade should be carefully inspected via proofrolling, probing and testing, as determined necessary by the geotechnical engineer and as outlined in Section 6.2.4. If performed, proofrolling should utilize a fully loaded 10,000-gallon water truck or equivalent, with acceptable rutting criteria as deemed appropriate by the Owner's onsite geotechnical engineering representative. At the time of such observation, it may be necessary to make hand auger borings and/or use a hand penetration device in the base of the foundation excavation to confirm that the soils below the base are satisfactory for foundation support. The necessary depth of penetration will be established during observation. Frozen, wet, soft, or loose soil, as well as any other undesirable material should be removed. Once suitable soil conditions are achieved, the subgrade should be scarified and compacted prior to fill placement. The suitable exposed soil materials should be scarified and moistened or dried, as necessary, to a minimum depth of 8 inches below the proposed construction. Scarified material should be compacted to the minimum specifications defined in Section 6.2.4.

6.2.2. Structure and Trench Backfill

Select Fill should be used for backfill adjacent to structures, backfill beneath spread footings and mat foundations, trench backfill, and support of structure floor slabs. Select Fill should consist of imported material or processed site-generated material having a maximum particle size of 3 inches, a maximum of 60 percent fines (passing No. 200 sieve), a minimum of 30 percent passing the No. 4 sieve, a maximum liquid limit of 40, and a maximum Plasticity Index of 20. Select Fill should not contain any frozen debris, trash, cobbles greater than 3 inches in diameter, boulders, organic material, wood or plant debris, or other deleterious materials. It is anticipated that Select Fill may be generated from on-site soil, provided it meets the criteria presented above. GEI anticipates some of the native soil will require blending and processing for use as Select Fill. If another material is considered for any of the presented applications, GEI should be contacted to review submitted particle size distribution and plasticity testing results.

ADOT Class 4 aggregate may be considered for imported structural backfill against foundation walls in order to use the lower earth pressures presented in Section 5.2. ADOT Class 4 Aggregate also meets requirements for Select Fill. For ADOT Class 4 aggregate, the maximum PI is 5 and the gradation is as shown in Table 6-2 below:

Table 6-2. ADOT Class 4 Aggregate Gradation

Sieve Size	Percentage Passing
3 inch	100
1 1/2 inch	-
1 inch	-
3/4 inch	-
1/4 inch	35-70
#8	-
#200	0-10

6.2.3. Pipe Bedding

Piping for the proposed pipelines is anticipated to be within residuum deposits. Piping should be bedded in a clean granular material from a depth of 6 inches below the pipe to 6 inches above the pipe. To achieve the design E' values, the contractor must adequately compact the pipe bedding, especially under the haunches of the pipe. Bedding can consist of ASTM C33 No. 67 coarse aggregate with less than 5 percent fines. It is not recommended that the on-site soil be used for pipe bedding.

6.2.4. Fill Placement and Compaction

GEI recommends fill placement occurs in maximum 8-inch loose lifts for fill beneath structures and backfill adjacent to structures. Lift thickness for pipe bedding should not exceed 6 inches. Minimum recommended compaction specifications are outlined in Table 6-3 below.

Table 6-3. Fill Placement and Compaction

Backfill Material Type	Moisture and Compaction Specifications		
	Moisture Content[1]	Dry Density[2]	Relative Density[3]
Select Fill and Native Soil	-2% to +2%	≥ 95%	-
Imported ADOT Class 4 Aggregate	-2% to +2%	≥ 95%	-
Granular Pipe Bedding	-	-	≥ 65%

Notes:
1. Moisture content relative to the optimum moisture content as determined by ASTM D698.
2. Dry density relative to the maximum dry density as determined by ASTM D698.
3. Relative Density as determined by ASTM D4263 and D4254.

Mechanical compaction is required for all materials placed as backfill during construction. Compaction of cohesive materials is best accomplished with equipment such as a jumping jack or padfoot roller. Non-cohesive soils are best compacted with a vibratory plate or vibratory smooth-drum roller. Compaction utilizing any flooding type technique should not be allowed. Care should be taken when compacting fill adjacent to structures. Generally, we recommend operating only light-weight compaction equipment such as jumping jacks and vibratory plates immediately adjacent to structures.

Quality assurance of backfill material and backfill placement will be necessary to reduce potential for long-term differential settlements. Inspection of subgrade materials prior to placing or forming and

casting structural elements is also critical to project success. GEI recommends a qualified testing agency is retained to provide quality assurance services during the backfill process.

6.3. Bedrock or Oversized Material Excavation

Based on the subsurface investigation, bedrock at the site was classified as predominately sandy limestone with some sandy dolomite and chert. Bedrock was encountered at depths of between 5 and 10 feet in borings GB 14, GB 29, and the Ballfield Boring. Therefore, bedrock is likely to be encountered during excavation of the pipeline trench in the southern portion of the alignment near the proposed bore location. Based on our subsurface investigation and understanding of the project, GEI does not anticipate bedrock will be encountered during excavation activities at the WTP and WWTP sites.

Due to the high compressive strength of the bedrock, it is anticipated that specialized rock trenching equipment may be required to excavate the pipeline trench in the southern portion of the alignment. Pipeline trenchers such as the Vermeer T955III Pipeline Trencher have been successful in excavating similar bedrock on the South Rim. Alternatively, the use of a heavy-duty excavator with an impact hammer attachment is common for excavations into bedrock in the area. Utilizing a minimal width bucket to maximize tooth pressure should also be considered. If blasting is used, the contractor must adhere to all applicable project specifications including, but not limited to, safe blasting vibration criteria for existing infrastructure and existing slopes, take all necessary precautions for site safety, and observe all applicable laws and guidelines for storage and use of explosives.

GEI anticipates the onsite fill and residuum can be excavated with heavy duty excavators in good operating condition. Occasional hard limestone boulders and cobbles should be anticipated during excavations into the fill and residuum.

6.4. Unstable Subgrade Conditions

If soft/loose pockets or undesirable materials are encountered in the excavations, the proposed excavations may be re-established by backfilling after the material has been removed. The backfill should consist of coarse granular material such as ADOT Class 1 or 2 base course. If the base of the overexcavation is still soft and further overexcavation is not practical, then a stabilization product meeting the minimum requirements for AASHTO Class 4A Woven Stabilization fabric should be placed in the base of the overexcavation per the manufacturer's recommendations. The backfill should be placed and compacted according to specifications outlined in this report. Field observations by a geotechnical engineer and testing of the open excavations should be performed prior to concrete placement to verify the subgrade conditions and/or to make corrective recommendations, as necessary.

6.5. Trench Plugs

Though the pipeline will be above groundwater, spring snowmelt can create a temporary condition where large volumes of surface water may infiltrate into the pipe bedding. To prevent the accelerated flow of water through the bedding, trench plugs are recommended. Trench plugs should be constructed from impervious material and should be keyed into the in-situ soils or bedrock along the trench bottom and sidewalls, extending at least 6 inches to 12 inches above the top of the bedding elevation. Trench

plugs are recommended to be installed at a spacing not to exceed 1,000 feet and on either side of structures. Trench plugs may be constructed from compacted clay soils with an USCS classification of CL or with at least 60 percent fines and a plasticity index of 15 or greater. Trench plugs constructed from onsite lean clay soils should have a minimum thickness of 2 feet as measured parallel to the pipe. Trench plugs may also be constructed from reinforced concrete, lean concrete, controlled low-strength material (CLSM), or site soils processed with bentonite.

7. Limitations

This study was conducted in accordance with generally accepted geotechnical engineering and engineering geologic practices and principles; no warranty, express or implied is made. The subsurface conditions described in this report were based on data obtained from widely spaced exploratory borings, geotechnical laboratory testing, information provided by the client, engineering judgement, and our experience with similar subsurface conditions and projects. The boring logs presented in this report only depict the subsurface conditions at the actual boring locations. Subsurface conditions are typically variable, both laterally and vertically, and the nature and extent of the subsurface variations across the site may not become evident until construction. The boundaries between different soil types and bedrock presented in this report are approximate and, in some cases, may be more abrupt or gradual than described herein. Groundwater levels may vary with time, adjacent surface water levels, precipitation, and changes to the hydrogeological conditions at or surrounding the project site.

This report has been prepared exclusively for our client for design purposes for the subject project. GEI is not responsible for technical interpretations by others of the data presented in this report or use of this report by others for the subject project or other projects. If differing site conditions are encountered during further evaluation of the subsurface conditions by others or during construction, GEI should be notified immediately to determine if any changes to our recommendations presented in this report are warranted.

The recommendations presented in this report are only intended for the proposed design and construction as understood by GEI at the time of issuing this report. If the proposed design and construction changes, GEI should be notified immediately and given the opportunity to review the proposed changes and if necessary, modify our recommendations presented herein.

An environmental assessment was not included in GEI's scope of work for this project. Any statements regarding the absence or presence of hazardous and/or toxic substances presented herein are only intended for informational purposes. If the client is concerned about the environmental conditions at the site, GEI recommends the client and/or owner retain a qualified environmental firm to conduct an environmental site assessment.

8. References

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Figures

Figure 1. Site Vicinity and Figure Location Map

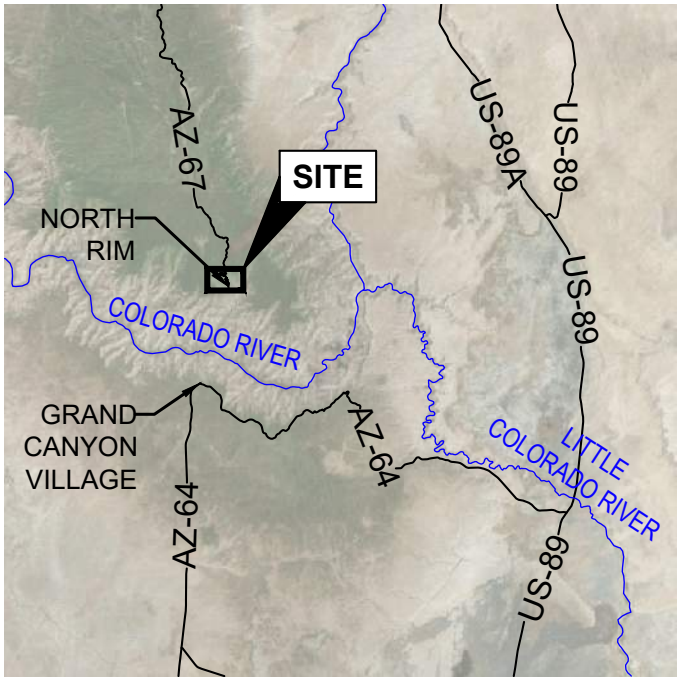
Figure 2. Boring/Improvement Locations

Figure 3. Boring/Improvement Locations (cont.)

Figure 4. Boring/Improvement Locations (cont.)

Figure 5. Boring/Improvement Locations (cont.)

Figure 6. Boring/Improvement Locations (cont.)



SITE VICINITY

SCALE: 1" = 20 MI

LEGEND:

- – – – PROPOSED PIPELINE ALIGNMENT
- ——— MAJOR ROADWAYS

NOTES:

1. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

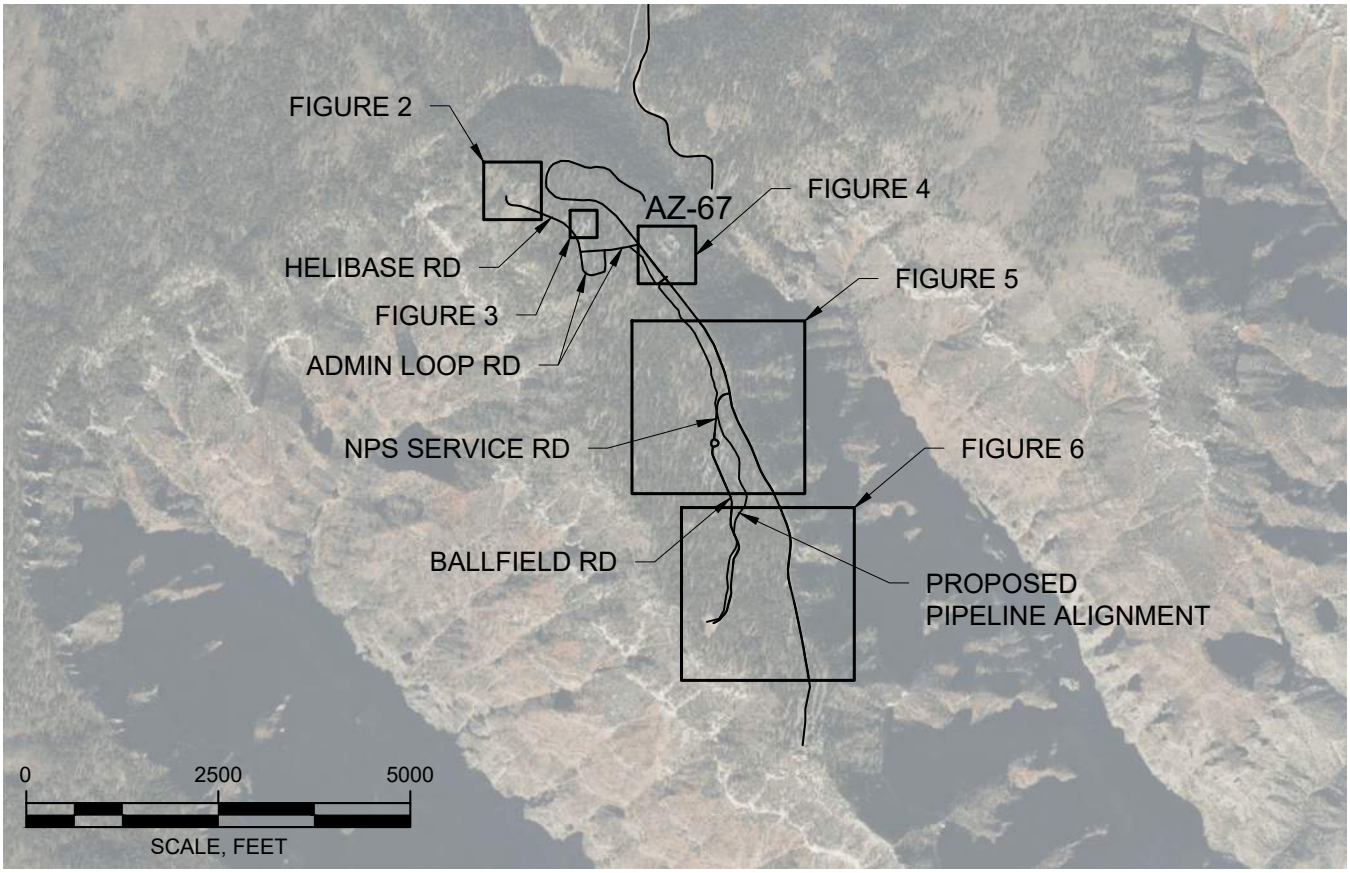
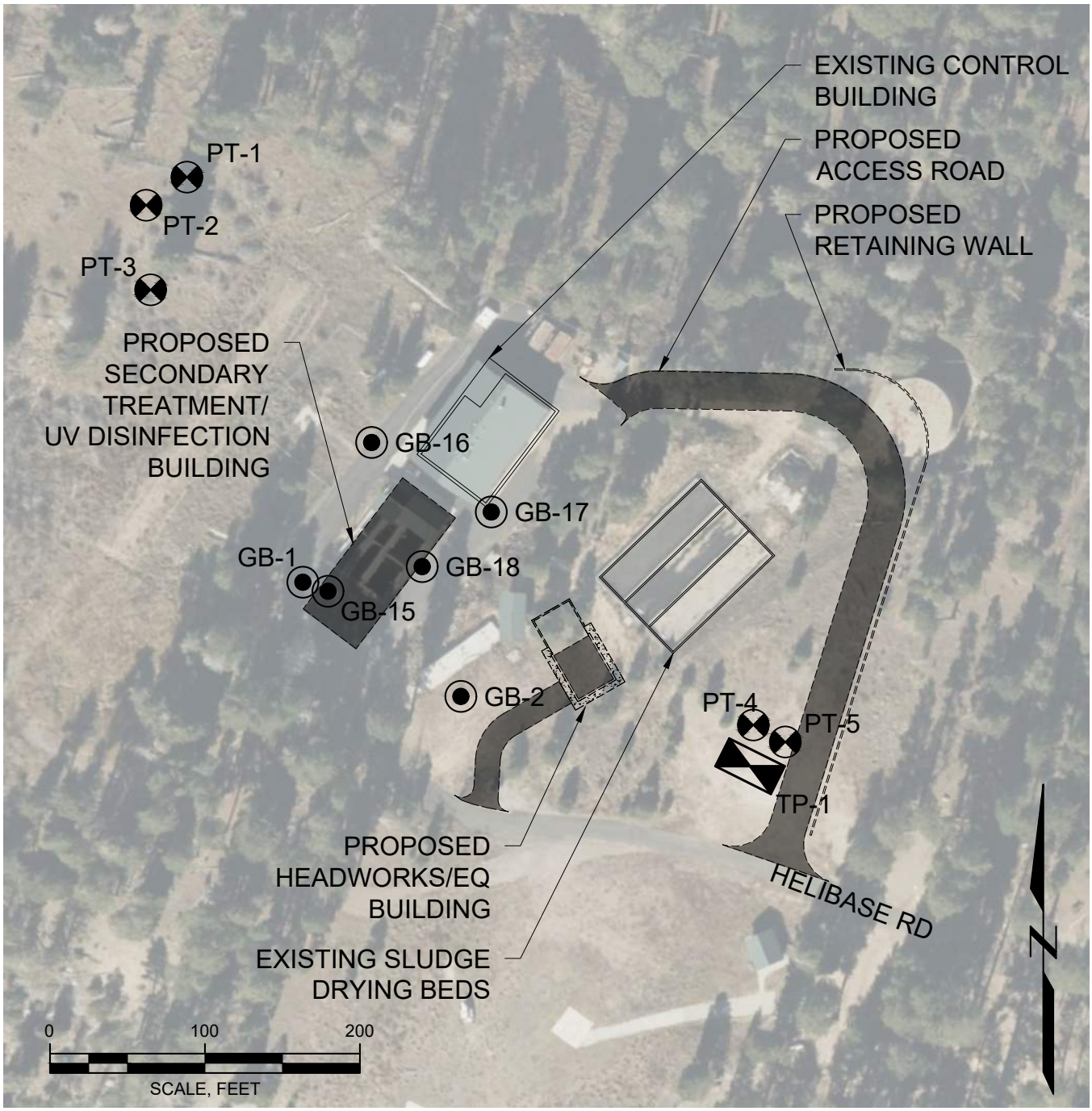


FIGURE LOCATIONS

SCALE: 1" = 2500'

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	SITE VICINITY AND FIGURE LOCATION MAP
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 1



BORING/IMPROVEMENT LOCATIONS

SCALE: 1" = 100'

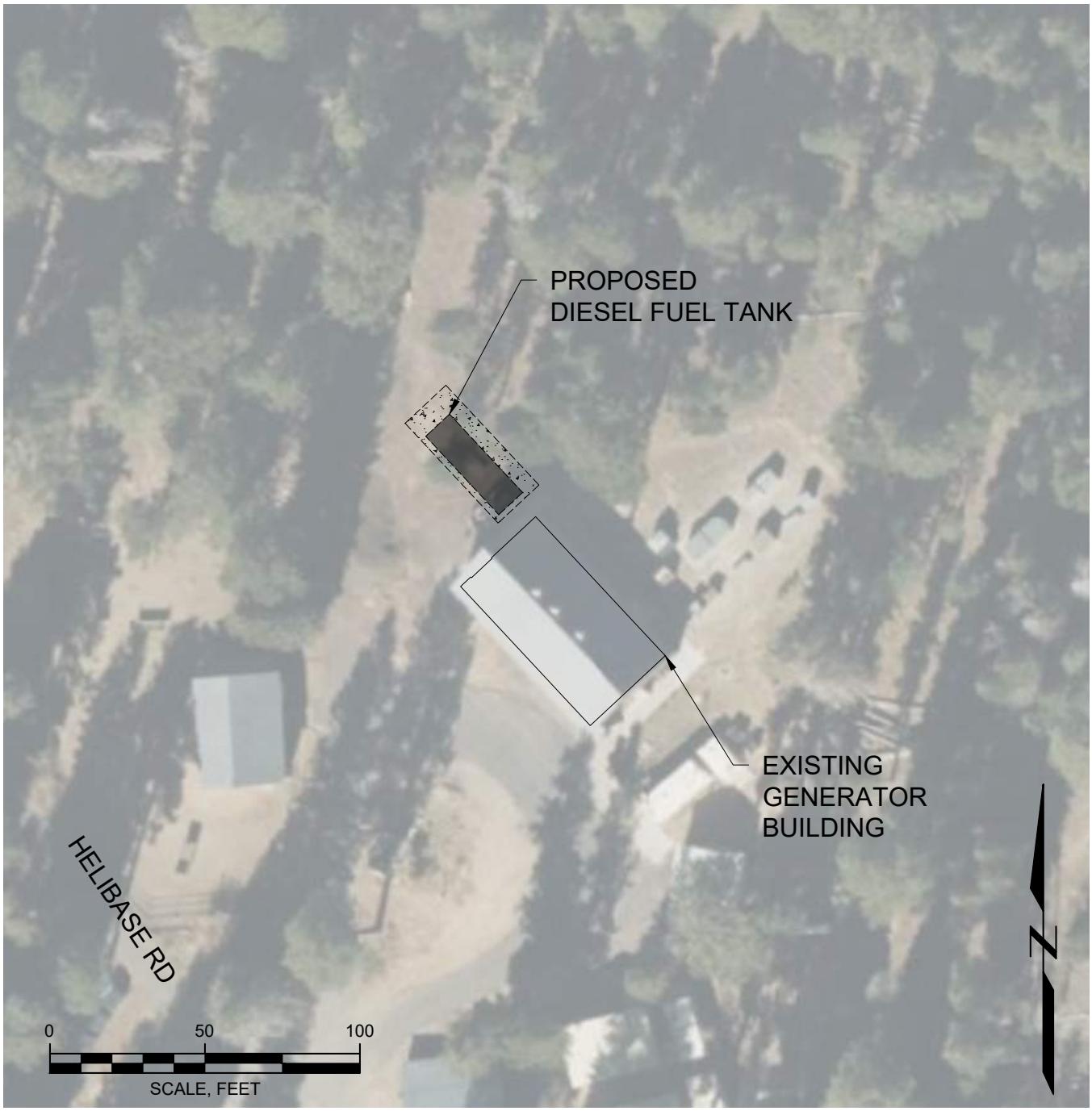
LEGEND:

- BORING LOCATION
- PERCOLATION TEST LOCATION
- TEST PIT LOCATION
- PROPOSED STRUCTURE
- EXISTING STRUCTURE

NOTES:

1. BORING, PERCOLATION TEST, AND TEST PIT LOCATIONS ARE APPROXIMATE.
2. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	BORING/IMPROVEMENT LOCATIONS
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 2



BORING/IMPROVEMENT LOCATIONS

SCALE: 1" = 50'

LEGEND:

PROPOSED STRUCTURE

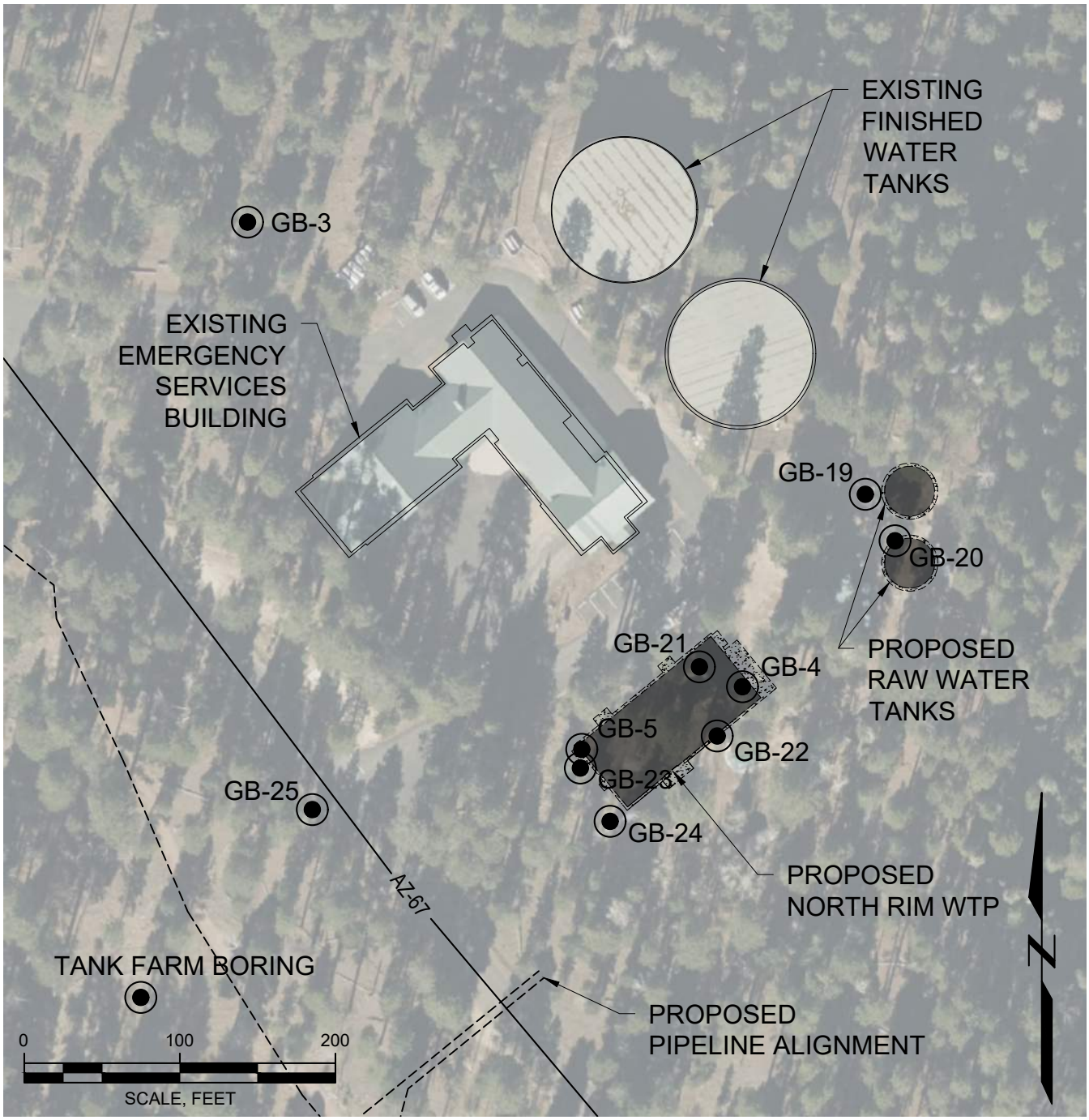
EXISTING STRUCTURE

NOTES:

1. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	BORING/IMPROVEMENT LOCATIONS
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 3

JOHANSON, ROBERT B:\Working\HDR\2305987 21186 GRCA North Rim Water Supply LR\Design\Dwgs\North Rim GIR Figures\Boring_Location_Map.dwg - 5/27/2025



BORING/IMPROVEMENT LOCATIONS

SCALE: 1" = 100'

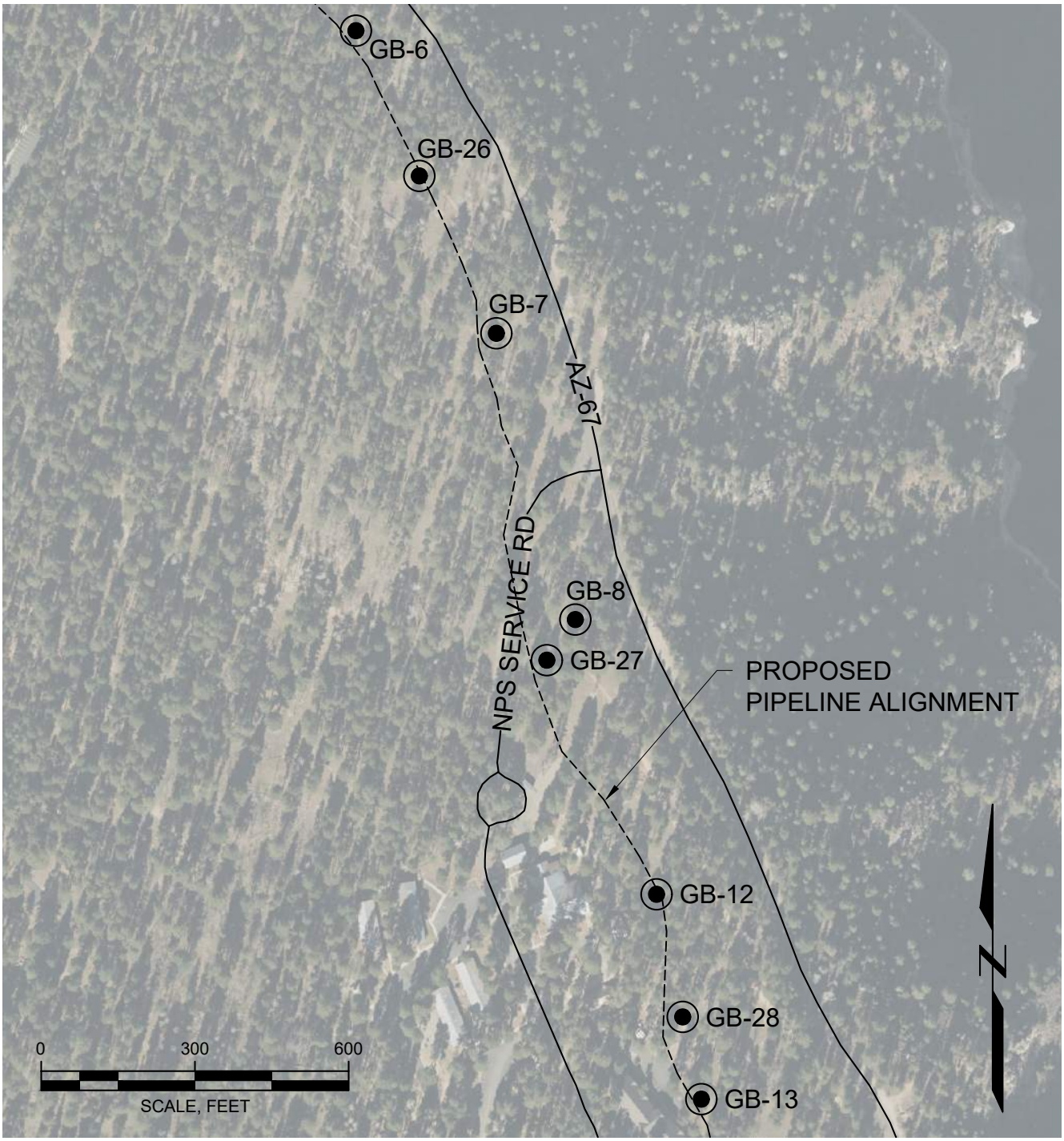
LEGEND:

- BORING LOCATION
- PROPOSED RAW WATERLINE
- EXISTING ROAD
- PROPOSED STRUCTURE
- EXISTING STRUCTURE

NOTES:

1. BORING LOCATIONS ARE APPROXIMATE.
2. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	BORING/IMPROVEMENT LOCATIONS (CONT.)
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 4



BORING/IMPROVEMENT LOCATIONS

SCALE: 1" = 300'

LEGEND:

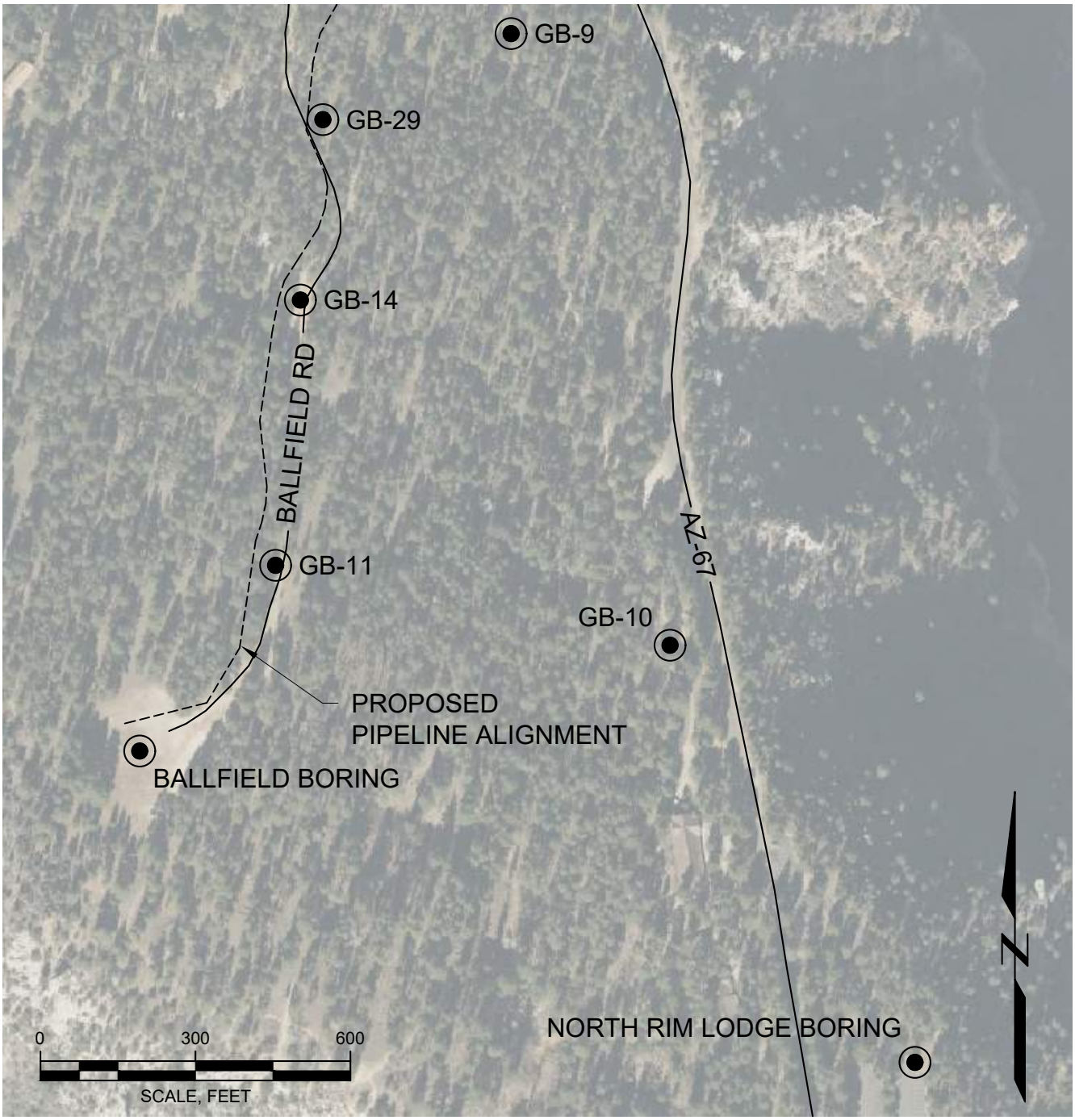
- BORING LOCATION
- PROPOSED RAW WATERLINE
- EXISTING ROAD

NOTES:

1. BORING LOCATIONS ARE APPROXIMATE.
2. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	BORING/IMPROVEMENT LOCATIONS (CONT.)
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 5

JOHANSON, ROBERT B:\Working\HDR\2305987 21186 GRCA North Rim Water Supply LR\Design\Dwgs\North Rim GIR Figures\Boring_Location_Map.dwg - 5/27/2025



BORING/IMPROVEMENT LOCATIONS

SCALE: 1" = 300'

LEGEND:

- BORING LOCATION
- PROPOSED RAW WATERLINE
- EXISTING ROAD

NOTES:

1. BORING LOCATIONS ARE APPROXIMATE.
2. STRUCTURE LOCATIONS ARE APPROXIMATE AND ARE SHOWN FOR INFORMATIONAL PURPOSES ONLY. SEE CONTRACT DRAWINGS FOR EXACT LOCATIONS.

Geotechnical Investigation Report GRCA 318719 Rehabilitate North Rim Utility Infrastructure Grand Canyon North Rim, Arizona	GEI Consultants	BORING/IMPROVEMENT LOCATIONS (CONT.)
HDR Engineering, Inc. Fort Collins, Colorado	Project 2305987	August 2025 Fig. 6

JOHANSON, ROBERT B:\Working\HDR\2305987 21186 GRCA North Rim Water Supply LR\Design\Dwgs\North Rim GIR Figures\Boring_Location_Map.dwg - 5/27/2025

Appendix A Standard Geotechnical Boring Log Key and Boring Logs

BORING LOG KEY

STANDARD GEOTECHNICAL DRILLING

Soil Classifications:

Clear Square Sieve Openings — U.S. Standard Series Sieve Sizes

12"	3"	3/4"	4	10	40	200

Boulders	Cobbles	Gravel		Sand			Silts and Clays
		Coarse	Fine	Coarse	Medium	Fine	

300mm	75mm	19mm	4.75mm	2.0mm	0.42mm	0.075mm

Gradation Estimates by Field Observation	
Description	**Quantity (%)**
Trace	<5
Few	5 to 10
Little	15 to 25
Some	30 to 45
Mostly	> 50

Relative Density or Consistency of Non-cohesive and Cohesive Soils			
Non-cohesive Soils		**Cohesive Soils**	
Classification	**Blows per 12 in**	**Classification**	**Blows per 12 in**
Very Loose	0 to 4	Very Soft	0 to 2
Loose	5 to 10	Soft	3 to 4
Medium Dense	11-30	Medium Stiff	5 to 8
		Stiff	9 to 15
Dense	31 to 50	Very Stiff	16 to 30
Very Dense	>50	Hard	>30

Color: Sample colors are in general accordance with basic brown, red, yellow, and gray combinations

Description of Moisture	
Description	**Criteria**
Dry	Absence of moisture, dusty, dry to the touch
Moist	Damp but no visible water
Wet	Visible free water, usually soil below the groundwater table

Plasticity	
Description	**Criteria**
Nonplastic	A $\frac{1}{8}$" diameter thread cannot be rolled
Low	A $\frac{1}{8}$" in diameter thread can be rolled with difficulty; a lump cannot be formed at a moisture lower than the plastic limit
Medium	A $\frac{1}{8}$" in diameter thread can be rolled easily; a crumbly lump can be formed at a moisture lower than the plastic limit
High	A $\frac{1}{8}$" in diameter thread can be rolled very easily; a lump can be formed at a moisture lower than the plastic limit

Description of Odor	
Description	**Criteria**
No Organic Odor	Organic odor is not present
Trace Organic Odor	Mild organic odor; mixture of soil and organics
Strong Organic Odor	Prominent organic odor; sample is primarily organic

Cementation	
Description	Criteria
Weak	Crumbles with light finger pressure
Moderate	Crumbles with considerable finger pressure
Strong	Will not crumble with finger pressure

Rock Descriptions:

Weathering	
Description	**Criteria**
Fresh	No visible sign of rock material weathering: perhaps slight discoloration on major discontinuity surfaces.
Slightly Weathered	Discoloration of rock material on discontinuity surfaces.
Moderately Weathered	Less than half of the rock material is decomposed and/or disintegrated to soil. Fresh or discolored rock is present either as a continuous framework or as corestones.
Highly Weathered	More than half of the rock material is decompsed and/or disintegrated to a soil. Fresh or discolored rock is present either as a discontinuous framework or as corestones
Completely Weathered	All rock material is decomposed and/or disintegrated to soil. The original mass structure is still largely intact.

Texture	
Description	**Criteria**
Very Fine Grained	Grains not individually visible to the unaided eye
Fine Grained	Grains barely visible to the unaided eye, up to $\frac{1}{16}$" diameter
Medium Grained	Grain diameter between $\frac{1}{16}$" and $\frac{3}{16}$"
Coarse Grained	Grains diameter between $\frac{3}{16}$" and $\frac{1}{4}$"
Very Coarse Grained	Grains larger than $\frac{1}{4}$" in diameter

Field Hardness	
Description	**Criteria**
Very Hard	Cannot be scratched with a knife or sharp pick.
Hard	Can be scratched with a knife or pick only with difficulty
Medium	Can be gouged $\frac{1}{16}$" deep by firm pressure on knife or pick point
Soft	Can be grooved or gouged readily with knife or pick point
Very Soft	Can be carved with knife and scratched readily by fingernail

Geologic Interpretation:

A ***Geologic Interpretation*** of encountered soil and bedrock units is provided for each specific ***Visual Material Description***. Examples of geologic interpretations for soil that may be presented include: FILL, ALLUVIUM, AEOLIAN, AND GLACIAL TILL, AND RESIDUUM. Rock geologic interpretations are referenced based on a combination of field classifications and applicable geologic maps.

Sample Graphics and Descriptions:

California Barrel Sampler: Barrel sampler loaded with sample liners and driven to collect a relatively representative and intact specimen of soil or weak rock.

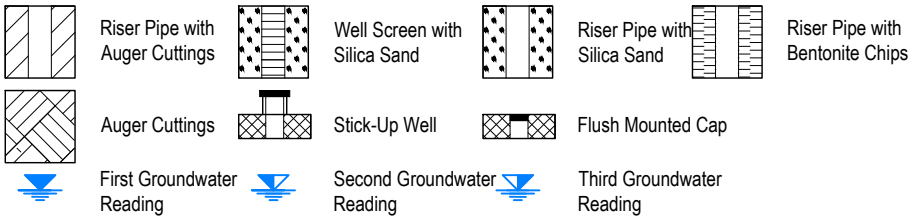
Split-Spoon Sampler: Split-barrel sampler driven in accordance with ASTM D1586 used to provide visual material descriptions and collect a disturbed specimen.

Shelby Tube Sampler: Thin wall tube hydraulically pushed into the subsurface to collect a representative and intact specimen of soil.

Bulk Sample: Bulk or bagged sample taken from auger cuttings.

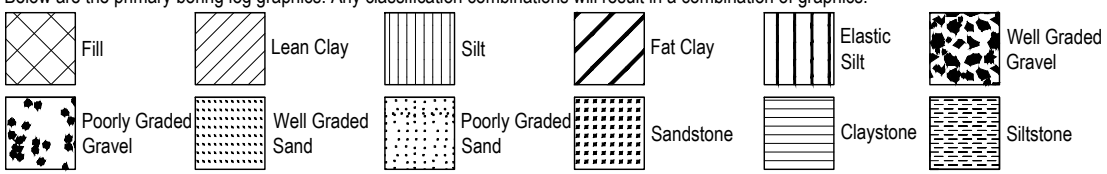
Rock Core

Groundwater Monitoring Well Graphics:



Boring Graphics:

Below are the primary boring log graphics. Any classification combinations will result in a combination of graphics.



GEI Consultants

BORING: GB 1

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/7/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21733, -112.06603
Boring Elevation: 8289 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**ASPHALT - 2"**												
							0.2 ft.												
							FILL												
	8285		5 9 8	9			CL - LEAN CLAY with sand, few gravel, very stiff, olive brown to reddish brown with gray gravel, dry, topsoil in uppermost inch of sample.												
5			4 6	0			CL - Sandy LEAN CLAY with gravel, stiff, dark brown, moist, from bulk sample after no recovery in Mod Cal.												
	8280																		
10			4 13	8			10.0 ft.		16.9	127.2				37	50	31		1250	0.5
							RESIDUUM GC - Clayey GRAVEL with sand, medium dense, redish brown, moist.												
	8275																		
15			3 4 6	12			CL - Sandy LEAN CLAY with gravel, stiff, reddish brown, moist.												
	8270																		
20			2 2 3	18			CL - As above except medium stiff.							57	47	27			
	8265																		
25			3 2 3	2			CL - As above, gravel in split spoon shoe limiting sample recovery				0.004								
	8260																		
30			2 5 4	3			CL - LEAN CLAY with sand, few gravel, stiff, reddish brown to brown, moist.												

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 1

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/7/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21733, -112.06603
Boring Elevation: 8289 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Sampling Data							Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Laboratory Testing Results											
									In-Situ States			INDEX DATA					Strength & Compressibility		
Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic			Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35	8255																			
40	8250		1 2 7	18			CL - LEAN CLAY with sand, stiff, reddish brown with some grey, moist. Bottom 2" is poorly graded sand with little clay and few gravel, light brownish gray, dry.													
45	8245						41.5 ft. **END OF EXPLORATION**													
50	8240																			
55	8235																			
60	8230																			
65	8225																			

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 2

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/5/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21712, -112.06571
Boring Elevation: 8312 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8310		5 7 13	18			**RESIDUUM** CL - Sandy LEAN CLAY, few gravel, very stiff, olive brown, dry.												
5	8305		5 11	8			SC - Clayey SAND with gravel, pale olive, medium dense, dry, fragmented pieces of carbonate rock present.		6.6		0.005			19					
10	8300		18 50/4"	10			SC - As above except very dense. Gravel is angular and up to 2" diam.												
15	8295		8 8	10			CL - LEAN CLAY with sand, few gravel, very stiff, reddish brown, moist.											105	-0.5
20	8290		7 28	16			CL - Gravelly LEAN CLAY with sand, hard, reddish brown, moist.											0	-0.2
25	8285		5 10 11	16			CL - As above except very stiff.												
30	8280		3 11 11	18			CL - Gravelly LEAN CLAY with sand, very stiff, reddish brown, moist.												

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 2

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/5/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21712, -112.06571
Boring Elevation: 8312 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35	8275		50/5"	8			CL - Sandy LEAN CLAY, reddish brown, moist, bottom 3" is poorly graded sand with clay, yellow olive, dry.		18.8	101.4				54	42	24			
40	8270		50/1"	0.5			40.0 ft. **KAIBAB LIMESTONE** Sandy LIMESTONE, medium hard, fresh, light brownish gray to white, very fine grained. 40.1 ft. **END OF EXPLORATION**												
45	8265																		
50	8260																		
55	8255																		
60	8250																		
65	8245																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 3

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/6/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21567, -112.05942
Boring Elevation: 8329 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Visual Material Description	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	UCS (psi)
0	8329	16 20 41	15		**RESIDUUM** SC - Clayey SAND, few gravel, very dense, pale olive, dry.	0.006				
5	8325	15 20 18	11		SC - As above except dense.					
10	8320	26 50/3"	7		SC - Clayey SAND with gravel, very dense, olive yellow with some reddish brown, dry.		21	35	44	
15	8315		16 0%	7.0	14.0 ft. **KAIBAB LIMESTONE** Sandy LIMESTONE, hard, fresh, gray to light brownish gray, very fine grained, close to moderate joint spacing.					
20	8310		14 0%	1.8	As above except very close joint spacing, rock mostly reduced to gravel sized pieces. Considerable sand found in the circulation tank accounting for poor recovery.					
25	8305		35 37%	1.8	As above. 21.5' to 23' is hard with extremely close joint spacing (broken down to gravel) while bottom 23' to 26.5' is medium hardness with close to moderate joint spacing. 3-4" void encountered at about 24' deep which caused loss of circulation.					
30	8300		60 67%	3.8	LIMESTONE, medium to hard, fresh, light brownish gray to gray, very fine grained, very close fracture spacing for first 16" and moderate fracture spacing for remaining 44".					6327
					As above except wide fracture spacing. 2-3" fracture encountered around 34' deep.					

Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
		0.006								
			21	35	44			6327		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 3

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/6/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21567, -112.05942
Boring Elevation: 8329 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Visual Material Description
35	8295	53 / 70%	1.8	As above except close to moderate fracture spacing.
40	8290	42 / 47%	1.6	
				41.5 ft.
				END OF EXPLORATION
45	8285			
50	8280			
55	8275			
60	8270			
65	8265			

Sampling Data: Sample Identification; Blow Count/6 in; Geologic Graphic

Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
								8451		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 4

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/5/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21485, -112.05834
Boring Elevation: 8327 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8325		9 11 16	7			**RESIDUUM** CL - Sandy LEAN CLAY, few gravel, very stiff, pale olive, dry, moderate to low plasticity.													
5	8320		9 46 42	13			SP-SC - Poorly Graded SAND with clay and gravel, very dense, light brownish gray to white, dry, angular grains, possibly sampled through cobble(s).													
10	8315		10 50/2"	10			SC - Clayey SAND with gravel, very dense, reddish brown clay with gray gravel and sand, dry.		13.2	97.3	0.002	25	27	48	32	9		303	0
15	8310		21 50/4"	15			SC - Clayey SAND with gravel, very dense, pale olive, dry.							18					
20	8305		3 0 0	0			20.0 ft. **VOID** No Recovery - SPT advances 0.5" through limestone before dropping 2'. 22.0 ft. **RESIDUUM**												
25	8300		2 3 3	3			SC - Clayey SAND with gravel, loose, brown with gray gravel, dry.													
30	8295		3 1 1	14			GC - Clayey GRAVEL with sand, very loose, reddish brown with light grey carbonate rock gravel, moist.							48	40	21			

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 4

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/5/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21485, -112.05834
Boring Elevation: 8327 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
35	8290		2 1 3	8			CL - LEAN CLAY with gravel, few sand, soft, reddish-brown, moist.	
40	8285		8 9	6			GC - Clayey GRAVEL with sand, mostly fine to coarse gravel, little fine to coarse sand, few to little clay, medium dense, white to reddish brown, dry to moist.	
							41.0 ft. **END OF EXPLORATION**	
45	8280							
50	8275							
55	8270							
60	8265							
65	8260							

Laboratory Testing Results

Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 5

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/4/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.21474, -112.05869
Boring Elevation: 8330 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
0	8330						**RESIDUUM**	
			37 50/4"	11			SC - Clayey SAND with gravel, very dense, light brownish gray, dry.	
5	8325		15 50/5"	7			SC - As above except light gray.	
10	8320		50/5"	5			SC - As above. 10.4 ft. **END OF EXPLORATION**	
15	8315							
20	8310							
25	8305							
30	8300							

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
					27					

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 6

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.21250, -112.05758
Boring Elevation: 8329 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**RESIDUUM**												
			20 35 47	12			CL - Sandy LEAN CLAY, few gravel, hard, pale olive, dry.												
	8325																		
5			50/5"	4			SC - Clayey SAND with gravel, very dense, pale olive, dry.												
	8320																		
10			15 30 28	18			SM - Silty SAND with gravel, very dense, pale olive, dry, angular sand and gravel grains.					24	42	34	33	7			
	8315																		
15			50/3"	7			SM - As above except max particle size to 2" diam.												
							15.3 ft.												
							END OF EXPLORATION												
20	8310																		
	8305																		
25																			
	8300																		
30																			

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 7

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.21084, -112.05661
Boring Elevation: 8325 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8325						**RESIDUUM**												
			13 35 40	18			SC - Clayey SAND with gravel, very dense, pale olive, dry, sand and gravel grains are angular.												
5	8320		18 40 50/5"	16			SC - As above except for some thin layers (1- 2") of high dry strength reddish brown clay.					25	30	45					
10	8315		33 50/3"	8			SC - Clayey SAND with gravel, very dense, pale olive, dry, some thin layers of high dry strength reddish brown clay.												
							10.8 ft.												
							END OF EXPLORATION												
15	8310																		
20	8305																		
25	8300																		
30	8295																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 8

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.20936, -112.05617
Boring Elevation: 8306 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8305						**RESIDUUM**												
			10 16 38	14			CL - Sandy LEAN CLAY with gravel, hard, light brownish grey, dry, gravel 0.5" to 1.5" diam.												
5	8300		30 32 35	18			SC - Clayey SAND with gravel, very dense, pale olive with some reddish brown, dry.					26	33	41					
10	8295		29 50/1"	7			SC - Clayey SAND with gravel, very dense, light brownish grey, dry, angular gravel and sand.												
15	8290		50/2"	3			SC - As above except few gravel. 15.2 ft. **END OF EXPLORATION**												
20	8285																		
25	8280																		
30	8275																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 9

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/2/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20567, -112.05447
Boring Elevation: 8233 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Blow Count/6 in	Recovery (in) / RQD (%)	Visual Material Description
0	8230	9 9 11	0	**RESIDUUM** SC - Clayey SAND with gravel, medium dense, olive grey, dry, collected bulk sample after no recovery in split spoon sampler.
5	8225	3 5 7	8	CL - Sandy LEAN CLAY, trace gravel, stiff, reddish brown, dry.
10	8220	7 14 17	16	CL - As above except hard.
				11.5 ft.
				END OF EXPLORATION
15	8215			
20	8210			
25	8205			
30	8200			

Sampling Data: Sample Identification; Drilling Rate (min./ft.); Geologic Graphic
Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
					60	46	27			

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 10

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/2/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20241, -112.05344
Boring Elevation: 8213 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8213						**RESIDUUM**												
	8210		4 9 10	8			CL - Sandy LEAN CLAY with gravel, very stiff, olive brown, dry.												
5			14 25 19	12			SP-SC - Poorly Graded SAND with clay and gravel, dense, grey, dry, freshly broken angular grains.												
	8205																		
10			13 15 23	14			CL - LEAN CLAY, few sand, few gravel, hard, reddish brown, dry, very high dry strength, high toughness.												
	8200																		
15			2 16	10			GC - Clayey GRAVEL, few sand, medium dense, reddish brown with light grey gravel, dry, bottom 3 inches of sample is fat clay with some gravel.							47	51	34			
							16.0 ft.												
							END OF EXPLORATION												
	8195																		
20																			
	8190																		
25																			
	8185																		
30																			
	8180																		

Sampling Data | **Laboratory Testing Results** | In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 11

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/2/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20284, -112.05601
Boring Elevation: 8255 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8255						**RESIDUUM**												
			14 50/5"	12			SM - Silty SAND with gravel, very dense, olive yellow, dry, low dry strength.					23	44	33	NV	NP			
5	8250		31 37 50/1"	9			SC - Clayey SAND with gravel, very dense, light brownish gray, dry.												
10	8245		50/5"	6			SC - As above except few gravel. 10.4 ft. **END OF EXPLORATION**												
15	8240																		
20	8235																		
25	8230																		
30	8225																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 12

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.20789, -112.05563
Boring Elevation: 8303 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
0	8303						**RESIDUUM**	
	8300		26 42 43	16			SC - Clayey SAND with gravel, very dense, light brownish gray, dry.	
5			47 50/5"	14			SC - As above.	
	8295							
10			24 40 50/4"	15			SM - Silty SAND with gravel, very dense, light brownish gray, dry.	
	8290						11.3 ft.	
15							**END OF EXPLORATION**	
	8285							
20								
	8280							
25								
	8275							
30								
	8270							

Laboratory Testing Results

Depth (ft)	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psi)	Swell Percent (%)
10				27	39	34	NV	NP			

In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 13

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.20679, -112.05539
Boring Elevation: 8295 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8295						**RESIDUUM**												
			50/5"	10			CL - Sandy LEAN CLAY with gravel, hard, light brownish gray, dry, moderate plasticity.												
5	8290		50/2"	5			CL - As Above.												
10	8285		19 28 50/5"	10			SC - Clayey SAND with gravel, very dense, pale olive, dry.												
15	8280		18 36 50/5"	12			SC - Clayey SAND with gravel, very dense, light yellowish brown with light gray fine grained rock chips, dry.					36	47	17					
							16.4 ft. **END OF EXPLORATION**												
20	8275																		
25	8270																		
30	8265																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 14

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/4/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20425, -112.05585
Boring Elevation: 8259 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0						**RESIDUUM**												
			29 19 36	18		SC - Clayey SAND, few gravel, very dense, light brownish gray, dry.												
	8255																	
5			16 50/2"	12		GM - Silty GRAVEL with sand, very dense, light brownish gray, dry.					44	35	21	NV	NP			
						8.0 ft.												
	8250					**KAIBAB LIMESTONE** Sandy LIMESTONE, medium hard, fresh, light brownish gray to pale olive, very fine grained, very close joint spacing.												
10				26 39%	3.3	As above, very low recovery, substantial amounts of sand found in circulation tank.												
	8245			6 10%	3.2													
15						Sandy LIMESTONE, hard, fresh, olive yellow to white, very fine grained, very close joint spacing, highly fragmented.													
	8240			23 0%	3.4													
20						As above.												
				13 0%	3.5													
						23.0 ft.												
	8235					**END OF EXPLORATION**												
25																		
	8230																	
30																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 15

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/8/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217305, -112.066001
Boring Elevation: 8310 ft
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8310						**ASPHALT** 0.4 ft.												
							FILL CL - Cuttings consist of LEAN CLAY with sand, mostly clay, little fine sand, reddish brown to reddish yellow, moist.													
5	8305		4 5	8			5.0 ft. **RESIDUUM** GC - Clayey GRAVEL, mostly fine to coarse gravel, little to some clay, few fine to coarse sand, loose, reddish yellow to white, moist, limestone fragments present.					63	8	29	47	26			
10	8300		8 15 16	20			CL - Sandy LEAN CLAY with gravel, mostly clay, some fine to coarse sand, little fine to medium gravel, hard, reddish yellow to white, moist, limestone and chert fragments present.													
15	8295		3 4	10			CL - As above except medium stiff.		18.2					53					
20	8290		5 4 9	26			CL - Sandy LEAN CLAY, mostly clay, some fine sand, few fine to medium gravel, stiff, reddish yellow, moist, few limestone fragments present.													
25	8285		8 7				SC - Clayey SAND with gravel, mostly fine to coarse sand, little to some clay, little fine to coarse gravel, medium dense, reddish yellow to white, moist, limestone and chert fragments present.													
30	8280		3 6 42	18			SC - Clayey SAND, mostly fine sand, little clay, few fine to coarse gravel, dense, white to reddish yellow, dry to moist. 31.0 ft. **KAIBAB LIMESTONE** Sandy LIMESTONE, medium hard, fresh, white to yellow, fine grained, thin to medium bedding thickness, chert nodules present,													
			46 91%		4.6														

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 15

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/8/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217305, -112.066001
Boring Elevation: 8310 ft
Notes: Location and elevation approximated using Google Earth.

Sampling Data

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic
35	8275					
				42 / 76%	3.9	
40	8270					
45	8265					
50	8260					
55	8255					
60	8250					
65	8245					

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

joints are horizontal, partially open, slightly rough, irregular, unweathered.
As above except slightly weathered at 38 ft, mostly mechanical breaks, quartz infill present throughout core run.

No recovery 38' - 39', likely completely weathered limestone to clay
Sandy LIMESTONE, medium hard, fresh, white to yellow, fine grained, thin to medium bedding thickness.

40.0 ft.

END OF EXPLORATION

Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 16

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/7/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217569, -112.065906
Boring Elevation: 8308 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
0	8308						**ASPHALT - 2"**	
							0.2 ft.	
							FILL	
	8305							
5			1 0 1	3			CL - LEAN CLAY with sand, mostly clay, little fine to coarse sand, few fine gravel, very soft, brown to dark brown, moist, plant debris present.	
	8300							
10			3 4	12			CL - As above except medium stiff, reddish yellow to white.	
	8295							
15			4 6 10	12			SC - Clayey SAND, some fine to coarse sand, some clay, few fine gravel, medium dense, brown to dark brown, moist, high plasticity, chert fragments present.	
							16.5 ft.	
	8290						**RESIDUUM**	
20			10 8	12			CL - Sandy LEAN CLAY, mostly clay, some fine to coarse sand, few fine to coarse gravel, very stiff, reddish yellow to white, moist, chert and limestone fragments present.	
	8285							
25			50/5"	5			25.0 ft.	
							KAIBAB LIMESTONE At 25', SPT sample is Sandy LIMESTONE, soft to medium hard, slightly to moderately weathered, white to pink, very fine grained.	
	8280			42 66%	4.2		At 25.5', core is as above except medium hard, fresh, fine to medium grained, thin to medium bedding thickness, joints are low angle, open, slightly rough, irregular, slightly weathered.	
30							No recovery 28' - 29', drillers note loss of circulation, likely partially infilled fracture.	
							At 30', as above except joints are horizontal to moderately dipping, open to partially open, slightly rough, irregular, unweathered, quartz infill present, chert nodlules present throughout core run	
	8275			50 73%	4.0			

Laboratory Testing Results

Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
					47	61	39			
			7	27	66	48	28			
								7506		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 16

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/7/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217569, -112.065906
Boring Elevation: 8308 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
35							No recovery 34' - 37', likely completely weathered limestone to clay.	
	8270			45 53%	4.3		Sandy LIMESTONE, medium hard, fresh, white to reddish yellow, fine to medium grained, joints are moderately dipping, partially open, smooth, irregular, unweathered, quartz infill, clay present at top of core run.	
40							40.0 ft.	
							END OF EXPLORATION	
	8265							
45								
	8260							
50								
	8255							
55								
	8250							
60								
	8245							
65								

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 17

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/9/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217446, -112.065644
Boring Elevation: 8314 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**ASPHALT - 2"** 0.2 ft.												
							FILL SC - Cuttings consist of clayey SAND with gravel, some fine to coarse sand, little to some clay, little fine gravel, dark brown, moist. 3.0 ft.					25	47	28	27	9			
	8310						**RESIDUUM**												
5			4 5	12			GC - Clayey GRAVEL with sand, some fine to coarse gravel, some clay, little fine to coarse sand, loose, white to reddish yellow, moist, limestone and chert fragments present.		22.1	91.0		49	17	34	39	20		0.0	0.0
	8305																		
10			5 3 4	14			SC - Clayey SAND with gravel, mostly fine to coarse sand, some clay, little fine to coarse gravel, loose, white to reddish yellow, moist, limestone and chert fragments present. At 12'-13', drill rig chatter, cobble/gravels may be present.												
	8300																		
15			3 5	8			CL - Sandy LEAN CLAY, mostly clay, some fine sand, medium stiff, dark reddish brown, moist. 16.0 ft.							62	47	27			
							KAIBAB LIMESTONE Sandy LIMESTONE, medium hard, fresh, white to pink, fine grained, medium bedding thickness, chert nodules present throughout. As above except iron oxide staining present, joints are horizontal, partially open, slightly rough, irregular, unweathered to moderately weathered. As above except joints are horizontal, open, rough, irregular, moderately weathered, clay infill.										8086		
	8295			49 90%	5.3														
20																			
				56 90%	3.4														
	8290																		
25							As above except joints are horizontal, partially open, slightly rough, planar, slightly weathered, clay infill.												
				61 100%	3.4														
	8285																		
30							As above except joints are horizontal to moderately dipping, partially open to open, smooth to rough, planar to irregular, unweathered to slightly weathered, clay infill present.												
				54 90%	3.6														

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 17

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/9/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217446, -112.065644
Boring Elevation: 8314 feet
Notes: Location and elevation approximated using Google Earth.

Sampling Data

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic
35	8280					
40	8275					
45	8270					
50	8265					
55	8260					
60	8255					
65	8250					

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

35.0 ft.

END OF EXPLORATION

Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 18

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/13/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217349, -112.065795
Boring Elevation: 8311 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psi)	Swell Percent (%)
0	8310						**ASPHALT - 1.5"** — 0.1 ft. **FILL** SC - Clayey SAND with gravel, mostly fine to coarse sand, little fine to coarse gravel, little clay, dark brown, moist. — 2.0 ft. **RESIDUUM**												
5	8305		4 5 7	11			SC - Clayey SAND, some fine to coarse sand, some clay, few fine to coarse gravel, medium dense, reddish yellow to white, moist, high plasticity, limestone and chert fragments present.							44	55	35			
10	8300		6 44	12			SC - As above except mostly fine to coarse sand, dense.							31	58	37			
15	8295		3 4 5	26			CL - Sandy LEAN CLAY, mostly clay, some fine sand, trace fine gravel, stiff, reddish yellow, moist, low to medium plasticity.				0.003								
20	8290		23 50/1"	12			— 20.0 ft. **KAIBAB LIMESTONE** Sandy LIMESTONE, very soft to medium hard, moderately to highly weathered, white to reddish yellow, fine grained. As above except medium hard, fresh, white to pink, thin to medium bedding, joints are horizontal to low angle, partially open, smooth, planar, slightly weathered.												
				48 90%	4.8														
25	8285						As above except slightly to moderately weathered, highly fractured, joints are moderately dipping, tight, slightly rough, irregular, unweathered, clay present. As above except joints are horizontal to moderately dipping, partially open, smooth to slightly rough, planar to irregular, slightly weathered, clay infill.											6497		
				54 83%	4.4														
30	8280						As above.												
				59 98%	4.4		As above except slightly weathered, pitted with quartz infill, joints are horizontal, partially open to open, slightly rough, irregular,												

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 18

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/13/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.217349, -112.065795
Boring Elevation: 8311 feet
Notes: Location and elevation approximated using Google Earth.

Sampling Data

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic
35	8275					
40	8270			57 86%	4.7	
45	8265					
50	8260					
55	8255					
60	8250					
65	8245					

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

slightly weathered, clay infill.
As above except fresh.
As above.

As above except slightly weathered, fractured, clay present.
As above except fresh.

40.0 ft.

END OF EXPLORATION

Groundwater Depth / Monitoring Well Configuration

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 19

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/12/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.215190, -112.058073
Boring Elevation: 8367 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8365						**RESIDUUM** SM - Cuttings consist of silty SAND with gravel, some fine sand, some silt, little fine gravel, olive, brown, dry.													
5	8360		22 27 25	12			SM - Silty SAND with gravel, some fine sand, some gravel, little to some silt, very dense, white to pink, dry, limestone fragments present.					33	39	28	NV	NP			
10	8355		50/5"	8			SC - Clayey SAND with gravel, mostly fine to coarse sand, little clay, little gravel, very dense, reddish yellow, dry to moist; limestone fragments present.							41					
15	8350		50/4"	4			SC - As above.												
20	8345		2 4	10			ML - Sandy SILT, mostly silt, some fine sand, trace fine gravel, medium stiff, reddish yellow, moist, low plasticity.		19.8	93.0				67	31	8	10		
25	8340		3 2 3	10			CL - Sandy LEAN CLAY with gravel, mostly clay, some fine to coarse sand, little fine to coarse gravel, medium stiff, yellow-reddish yellow, moist.								21	12			
							27.5 ft.												
30	8335			29 34%	4.4		**KAIBAB LIMESTONE** Sandy LIMESTONE, soft to medium hard, slight to moderately weathered, white to pink, fine grained, thick bedding, highly fractured at 27.7' and 28.5', joints are horizontal, tight, partially open, smooth to slightly rough, planar, slightly weathered. As above except soft, moderately weathered, highly fractured. As above except very soft, highly weathered, clay present. As above except soft, moderately weathered, highly fractured. As above except medium hard, slightly weathered, joints are												
				59 49%	3.3														

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 19

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/12/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.215190, -112.058073
Boring Elevation: 8367 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35	8330			36 52%	3.7		horizontal, low angle, partially open, slightly rough, planar to irregular, slightly weathered. As above. As above except moderately weathered, highly fractured. As above except slightly weathered, joints are horizontal, partially open, slightly rough, planar, slightly weathered. No recovery 38' - 40', likely completely weathered limestone to clay.											7685		
40	8325						40.0 ft. **END OF EXPLORATION**													
45	8320																		
50	8315																		
55	8310																		
60	8305																		
65	8300																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 20

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/12/2023, 11/13/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.215108, -112.058008
Boring Elevation: 8364 feet
Notes: Location and elevation approximated using Google Earth.

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**RESIDUUM** SM - Cuttings consist of silty SAND with gravel, some fine to coarse sand, some silt, little fine to coarse gravel, reddish yellow, dry.													
	8360																		
5			21 20	12			GM - Silty GRAVEL with sand, mostly fine to coarse gravel, some fine to coarse sand, little silt, dense, white to pink, dry, limestone fragments present.				0.008									
	8355																		
10			14 15 4	12			GM - As above except medium dense, reddish yellow to white.					50	30	20	NV	NP			
	8350																		
15			50/6"	8			GM - As above except very dense.												
	8345																		
20			6 14 14	14			SC-SM - Silty, clayey SAND with gravel, mostly fine to coarse sand, little silty clay, little gravel, medium dense, reddish yellow to white, dry, limestone fragments present.												
							22.5 ft.												
	8340			23 91%	5.5		**KAIBAB LIMESTONE** Sandy LIMESTONE, medium, fresh to slightly weathered, white to pink, fine grained, medium bedding, pitted, highly fractured from 23'-23.2'.												
25							As above except slightly to moderately weathered, highly fractured.												
				58 82%	3.4		As above except slightly weathered, joints are low angle to moderately dipping, partially open to open, smooth to slightly rough, planar to irregular, slightly to moderately weathered.												
	8335																		
30							As above except fresh.										7474		
				60 72%	3.0		As above except soft, moderately weathered, highly fractured with												

Sampling Data | **Visual Material Description** | **Laboratory Testing Results** (In-Situ States; INDEX DATA; Strength & Compressibility)

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 20

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/12/2023, 11/13/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.215108, -112.058008
Boring Elevation: 8364 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
35	8330			60 100%	2.9		clay infill, joints are horizontal to moderately dipping, smooth to slightly rough, planar to irregular, slightly weathered. As above except joints are horizontal, open, slightly rough, undulating, unweathered, quartz infill.	
40	8325						40.0 ft. **END OF EXPLORATION**	
45	8320							
50	8315							
55	8310							
60	8305							
65	8300							

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 21

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/11/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214885, -112.058434
Boring Elevation: 8368 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8365						**RESIDUUM**												
5	8360		22 50/4"	10			SC - Clayey SAND with gravel, some fine to coarse sand, some gravel, little to some clay, very dense, white to reddish yellow, dry, chert fragments present.					35	36	29					
10	8355		22 50/4"	10			SC - Clayey SAND, mostly fine to coarse sand, little clay, trace fine to medium gravel, very dense, reddish brown to pink, dry.												
15	8350		20 18	12			GM - Silty GRAVEL with sand, mostly fine to coarse gravel, little fine to coarse sand, little silt, dense, white to pink, dry.		5.8			45	36	19					
20	8345		50/3"	4			GM - As above except very dense, white to reddish yellow.												
25	8340		6 12	12			GM - As above except some fine to coarse gravel, some silt, little fine to coarse sand, medium dense, reddish yellow to white, dry to moist.					43	16	41	30	3			
30	8335		7 2 17	10			CL-ML - Sandy SILTY CLAY with gravel, mostly silty clay, little fine sand, little fine to medium gravel, very stiff, olive yellow to white, moist.		14.7						22	7			
							32.5 ft. **KAIBAB LIMESTONE**												

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 21

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/11/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214885, -112.058434
Boring Elevation: 8368 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35				29 77%	5.5		Sandy LIMESTONE, soft to medium hard, slightly weathered, white to pink, fine grained, medium bedding, pitted, iron oxidation present, joints are low angle to moderately dipping, partially open, smooth, planar, slightly weathered.												
	8330						As above except abundant chert nodules.												
40				57.5 68%	3.4		As above except moderately weathered, highly fractured, joints are low angle to moderately dipping, open, smooth to slightly rough, planar to irregular, slightly to moderately weathered, clay infill.											1460		
	8325						40.0 ft.												
45							**END OF EXPLORATION**												
	8320																		
50																			
	8315																		
55																			
	8310																		
60																			
	8305																		
65																			

Laboratory Testing Results
In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 22

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214763, -112.058395
Boring Elevation: 8370 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8370						**RESIDUUM** Drillers note drill chatter, gravel/cobbles may be present.												
5	8365		36 50/5"	10			GM - Silty GRAVEL with Sand, mostly fine to coarse gravel, little fine to coarse sand, little silt, very dense, white to reddish yellow, dry.					54	22	24	NV	NP			
10	8360		46 34	10			GM - As above, chert fragments present.												
15	8355		13 8 17	18			GW-GC - Well-graded GRAVEL with clay and sand, mostly fine to coarse gravel, little fine to coarse sand, few clay, medium dense, white to reddish brown, dry to moist.												
20	8350		6 8	10			GW-GC - As above except some fine to coarse sand, dry.												
25	8345		2 1 2	16			GM - Silty GRAVEL, mostly fine to coarse gravel, little silt, few to little fine to coarse sand, very loose, reddish brown to white, moist.					68	12	20	NV	NP			
30	8340		1 1 2	4			GM - As above.												

Sampling Data | **Laboratory Testing Results** | In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 22

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214763, -112.058395
Boring Elevation: 8370 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35	8335		33 50/1"	12			CL - Sandy LEAN CLAY, mostly clay, little fine sand, hard, yellowish brown, moist. SP - At 35.5', poorly graded SAND, mostly fine sand, very dense, white, dry.													
40	8330		9 12 8	12			SC - Clayey SAND with gravel, some fine sand, some fine gravel, some clay, medium dense, white, dry.					30	31	39					
							41.5 ft. **END OF EXPLORATION**												
45	8325																		
50	8320																		
55	8315																		
60	8310																		
65	8305																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 23

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/10/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214706, -112.058693
Boring Elevation: 8368 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8365						**RESIDUUM**												
5	8360		14 19 33	22			SC - Clayey SAND with gravel, mostly fine to coarse sand, little clay, little fine to medium gravel, very dense, reddish brown to white, dry, limestone and chert fragments present.				0.003									
10	8355		26 39	12			GC - Clayey GRAVEL with Sand, some fine to coarse gravel, some clay, little fine to coarse sand, very dense, white to reddish yellow, dry, limestone and chert fragments present.					31	23	46	35	13				
15	8350		11 22 7	19			GC - As above except medium dense, moist in tip of sampler.													
20	8345		7 50/5"	10			GC - As above except very dense.													
25	8340		2 1 9	26			CL - Sandy LEAN CLAY with gravel, mostly clay, some fine to coarse sand, little fine gravel, stiff, reddish brown to reddish yellow, moist.							52	35	12				
30	8335		50/3"	4			SM - Silty SAND, mostly sand, few fine gravel, little silt, very dense, white, dry.													
							30.5 ft.													
				53.5 87%	4.4		**KAIBAB LIMESTONE** Sandy LIMESTONE, medium hard, fresh, white to pink, fine grained, thin to medium bedding thickness, pitted, quartz infill													

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 23

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/10/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214706, -112.058693
Boring Elevation: 8368 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35							throughout, joints are horizontal, open, slightly rough, planar, unweathered.										9273		
	8330			62 90%	3.0		As above except moderately weathered, highly fractured, iron oxide staining present.												
40							As above except joints are horizontal to moderately dipping, tight to open, smooth to slightly rough, planar to unudlating, unweathered to slightly weathered.												
							40.0 ft.												
							END OF EXPLORATION												
	8325																		
45																			
	8320																		
50																			
	8315																		
55																			
	8310																		
60																			
	8305																		
65																			

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 24

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/10/2023, 11/11/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214612, -112.058628
Boring Elevation: 8369 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**RESIDUUM**												
	8365																		
5			20 50/3"	12			CL - Gravelly LEAN CLAY, mostly clay, little to some fine to coarse gravel, few to little fine to coarse sand, hard, reddish yellow to white, dry, limestone fragments present.					28	11	61	42	21			
	8360																		
10			50/6"	8			SM - Silty SAND, mostly fine sand, little silt, trace fine gravel, very dense, white, dry.												
	8355																		
15			50/3"	3			SM - As above except white to reddish yellow.												
	8350																		
20			50/3"	6			GC-GM - Silty, Clayey GRAVEL with Sand, some fine to medium gravel, some fine to medium sand, little silty clay, very dense, reddish yellow to white, dry.					46	39	15	26	6			
	8345																		
25			16 11	10			GC-GM - As above except medium dense. SW - Well-graded SAND with gravel, mostly fine to coarse sand, little fine to coarse gravel, very dense, white, dry.												
							27.0 ft.												
	8340			40 68%	5.7		**KAIBAB LIMESTONE** Sandy LIMESTONE, medium hard, slightly weathered, white to red, fine grained, thin to medium bedding, pitted throughout with quartz infill, joints are low angle, partially open, slightly rough, planar, unweathered.												
30							At 28.5', as above except moderately weathered, highly fractured with clay infill. At 29', as above except slightly weathered. At 30', as above except fresh, joints are low angle, partially open, slightly rough, irregular, slightly weathered.										5326		
				61 74%	3.0														

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 24

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/10/2023, 11/11/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214612, -112.058628
Boring Elevation: 8369 feet
Notes: Location and elevation approximated using Google Earth.

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
35	8335			35 38%	3.3		At 32', as above except slightly weathered, highly fractured. At 35', as above except slightly to moderately weathered, increasing iron oxide staining, joints are horizontal, partially open, rough, irregular, moderately weathered, slight clay infill. At 36.3', as above except moderately weathered, highly fractured. At 36.7', as above except slightly to moderately weathered. At 37.5', as above except moderately weathered, highly fractured, very pitted. At 38' to 40', no recovery, likely completely weathered limestone to clay.												
40	8330						40.0 ft. **END OF EXPLORATION**												
45	8325																		
50	8320																		
55	8315																		
60	8310																		
65	8305																		

Sampling Data

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Laboratory Testing Results

In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 25

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/7/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.214632, -112.059277
Boring Elevation: 8365 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8365						**ASPHALT - 4.5"**												
							0.4 ft.												
							RESIDUUM SM - Silty SAND with gravel, mostly fine sand, little fine to coarse gravel, little silt, very dense, white to reddish yellow, dry to moist, limestone and chert fragments present. CL - LEAN CLAY with sand, mostly clay, little fine to coarse sand, trace fine gravel, hard, reddish yellow, dry to moist.													
			30 50/4"	10															
5	8360		13 14 50/4"	14			SW - Well-graded SAND with gravel, mostly fine to coarse sand, little fine to coarse gravel, very dense, white to reddish yellow, dry.					2	17	81	37	16			
10	8355		50/4"	4			SW-SM - Well-graded SAND with silt, mostly fine to coarse sand, few silt, few fine gravel, very dense, reddish yellow, dry to moist.												
15	8350		50/3"	NR			No recovery.												
							15.3 ft.												
							END OF EXPLORATION												
20	8345																		
25	8340																		
30	8335																		

Sampling Data

Laboratory Testing Results
In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 26

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.211738, -112.057206
Boring Elevation: 8345 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8345						**RESIDUUM**												
			12 12 17	18			CL - LEAN CLAY with sand, mostly clay, little fine to coarse sand, trace fine gravel, very stiff, brown, dry.				0.003								
5	8340		10 23	10			CL - As above except hard, roots present.		10.5	125.6				82	29	14	76		
10	8335		50/5"	6			SM - Silty SAND with gravel, mostly fine sand, little fine to coarse gravel, little silt, very dense, reddish yellow to white, dry, chert fragments present.												
15	8330		10 40	10			GW - Well-graded GRAVEL with sand, mostly fine to coarse gravel, some fine to coarse sand, dense, white to reddish yellow, dry to moist, chert fragments present. 16.0 ft. **END OF EXPLORATION**												
20	8325																		
25	8320																		
30	8315																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 27

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.209143, -112.056358
Boring Elevation: 8323 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
0	8323						**RESIDUUM**	
	8320		10 17	9			GW - Well-graded GRAVEL with sand, mostly fine to coarse gravel, little fine to coarse sand, trace silt, medium dense, white to reddish yellow, dry, chert fragments present.	
5			8 50/5"	12			CL - Sandy LEAN CLAY with gravel, mostly clay, some fine sand, little fine to coarse gravel, hard, pink, dry, chert fragments present.	
	8315							
10			8 25	10			GC - Clayey GRAVEL with sand, some fine to coarse gravel, some fine to coarse sand, little clay, dense, reddish yellow to white, dry to moist, chert fragments present.	
	8310							
15			30 50/3"	12			SW - Well-graded SAND with gravel, mostly fine to coarse sand, little fine to coarse gravel, very dense, white to reddish yellow, dry, chert fragments present.	
							15.8 ft.	
							END OF EXPLORATION	
	8305							
20								
	8300							
25								
	8295							
30								
	8290							

Laboratory Testing Results

Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psi)	Swell Percent (%)
In-Situ States			INDEX DATA					Strength & Compressibility		
					59					

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 28

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.207232, -112.055456
Boring Elevation: 8309 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**RESIDUUM**												
			13 17 32	16			GM - Silty GRAVEL with sand, some fine to coarse gravel, some fine to coarse sand, little to some silt, dense, light brown to pink, dry, chert fragments present.					39	35	26	NV	NP			
	8305																		
5			14 34	8			GM - As above except pink to white.												
	8300																		
10			5 2 4	14			SW-SC - Well-graded SAND with clay and gravel, mostly fine to coarse sand, some fine to medium gravel, few clay, loose, reddish yellow, dry.												
	8295																		
15			5 5	10			SC - Clayey SAND with gravel, mostly fine to coarse sand, little fine to medium gravel, little clay, loose, reddish yellow to white, moist.												
							16.0 ft. **END OF EXPLORATION**												
	8290																		
20																			
	8285																		
25																			
	8280																		
30																			

Sampling Data | **Laboratory Testing Results** | In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: GB 29

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Rob Johanson
Date(s) of Drilling: 11/6/2023

Drilling and Sampling Methods

Drill Make and Model: CME-85
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon / Mod Cal
Sampler Diameter(s): 1.4" / 2.0"

GEI Consultants

Boring Location: 36.205210, -112.055704
Boring Elevation: 8291 feet
Notes: Location and elevation approximated using Google Earth.

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration
0	8290						**RESIDUUM**	
			3 10	10			CL - LEAN CLAY with gravel, mostly clay, little gravel, few sand, stiff, reddish yellow to yellowish brown, moist, limestone fragments present.	
5	8285		50/1"	1			5.0 ft. **KAIBAB LIMESTONE** LIMESTONE, medium hard, fresh to slightly weathered, white to brown, very fine to fine grained.	
10	8280		50/4"	4			As above except fresh.	
			50/1"	1			As above except light brown. 13.1 ft. **END OF EXPLORATION** Auger refusal at 13.1 feet bgs.	
15	8275							
20	8270							
25	8265							
30	8260							

Laboratory Testing Results

In-Situ States			INDEX DATA					Strength & Compressibility		
Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
		0.002								

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: Ballfield Boring

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/2/2022 - 11/4/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger, Coring
Bit Type: Auger Cutting Bit, HQ Core
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20185, -112.05690
Boring Elevation: 8235 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8235						**RESIDUUM**												
			8 9 8	10			CL - Gravelly LEAN CLAY, very stiff, olive brown with light brownish gray gravel, dry.				0.008								
5	8230		8 12 50/3"	18			CL - Gravelly LEAN CLAY, few sand, hard, dark brown to light brownish gray, dry.							58	39	23			
10	8225			24 63%	4.0		10.0 ft. **KAIBAB LIMESTONE** Sandy LIMESTONE, medium hardness, fresh, light gray, fine to very fine grained, close to moderately spaced joints.											5892		
							As above except moderate joint spacing from 12' to 14' followed by close to moderate joint spacing in remainder. High quantities of chert in lower half											3249		
15	8220			48 42%	2.4														
							Sandy LIMESTONE, medium hardness, fresh, light gray, very fine to fine grained, close to moderate joint spacing with layers of very close joint spacing and highly fractured rock.												
20	8215			30 25%	2.5														
							As above except very close joint spacing, highly fractured rock.												
				15 0%	4.0														
25	8210						25.0 ft. **END OF EXPLORATION**												
30	8205																		

Sampling Data | **Laboratory Testing Results** | In-Situ States | INDEX DATA | Strength & Compressibility

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: North Rim Lodge Boring

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/02/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon, Mod Cal
Sampler Diameter(s): 1.4", 2.0"

GEI Consultants

Boring Location: 36.20020, -112.05181
Boring Elevation: 8269 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0							**RESIDUUM**												
			15 35 50/2"	14			SP-SC - Poorly Graded SAND with clay and gravel, very dense, light gray, dry, angular grains.				0.006								
	8265																		
5			50/5"	5			SC - Clayey SAND with gravel, very dense, light gray, dry, freshly fractured angular grains.												
	8260																		
10			27 50/2"	9			SC - As above.							24.4					
	8255																		
15			50/3"	3			SC - As above.												
							15.3 ft.												
							END OF EXPLORATION												
20	8250																		
25	8245																		
30	8240																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

BORING: Tank Farm Boring

Project Name: GRCA 318719
Project Number: 21186
Client's Name: HDR
Owner's Name: National Park Service
Drilling Subcontractor: GSI
GEI Representative: Adam Malsam
Date(s) of Drilling: 11/1/2022

Drilling and Sampling Methods

Drill Make and Model: CME-55 Truck Mounted
Drilling Method: Hollow Stem Auger
Bit Type: Auger Cutting Bit
Casing Description: Hollow Stem Auger
Hammer Weight (lbs)/Fall (in): 140 lbs. / 30 in.
Sampler Type(s): Split Spoon
Sampler Diameter(s): 1.4"

GEI Consultants

Boring Location: 36.21430, -112.05965
Boring Elevation: 8340 feet
Notes: Location from CalTOPO, elevation from USGS 3D Elevation Program

Visual Material Description

Soil:
-GEOLOGIC INTERPRETATION-
USCS Classification (group symbol), particle sizes, density or consistency, color, moisture, odor, other descriptions

Rock:
-GEOLOGIC INTERPRETATION-
Bedrock Classification, hardness, weather, color, texture, joint size, other descriptions

Depth (ft)	Elevation (ft)	Sample Identification	Blow Count/6 in	Recovery (in) / RQD (%)	Drilling Rate (min./ft.)	Geologic Graphic	Visual Material Description	Groundwater Depth / Monitoring Well Configuration	Moisture Content (%)	Dry Unit Weight (pcf)	Water Soluble Sulfates (%)	Gravel Content (%)	Sand Content (%)	Fines Content (%)	Liquid Limit (%)	Plasticity Index (%)	UCS (psi)	Swell Pressure (psf)	Swell Percent (%)
0	8340						**RESIDUUM**												
							CL - LEAN CLAY with sand, brown, dry, medium plasticity.												
5	8335		13 43 50/4"	19			SM - Silty SAND with gravel, very dense, yellowish brown to gray, dry.					20.4	33.8	45.8	25	3			
10	8330		12 23 37	18			CL - Sandy LEAN CLAY with gravel, hard, reddish brown with white to gray gravel and sand, dry.				0.004								
15	8325		22 10 20	18			SP-SC - Poorly Graded SAND with clay and gravel, medium dense, light brownish gray, sand and gravel are highly angular.												
							16.5 ft.												
							END OF EXPLORATION												
20	8320																		
25	8315																		
30	8310																		

General Notes:
1) Soil classificaitons are in general accordance with ASTM D2487
2) The maximum particle size identified in the material description is dependent on sampler dimensions.
3) Additional information is provided on the Boring Log Key.
4) Groundwater measurements for monitoring wells present water levels at the time of drilling, highest level, and lowest level. Refer to the respective report for a complete history of groundwater values.

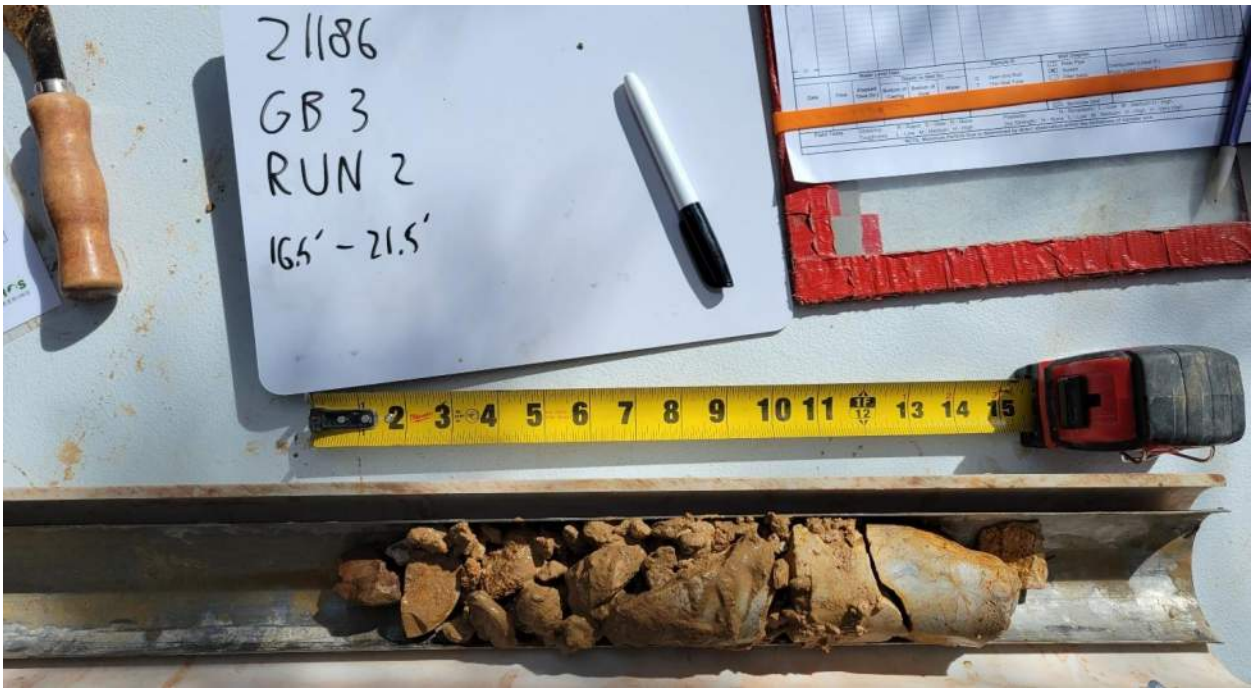
Groundwater Data:

Date:	Elapsed Time:	Depth to Groundwater:
		DNE

Appendix B Rock Core Photographs



GB3, Run 1, 14’ – 16.5’



GB3, Run 2, 16.5’ – 21.5’



GB3, Run 3, 21.5’ – 26.5’



GB3, Run 4, 26.5’ – 31.5’



GB3, Run 5, 31.5’ – 36.5’



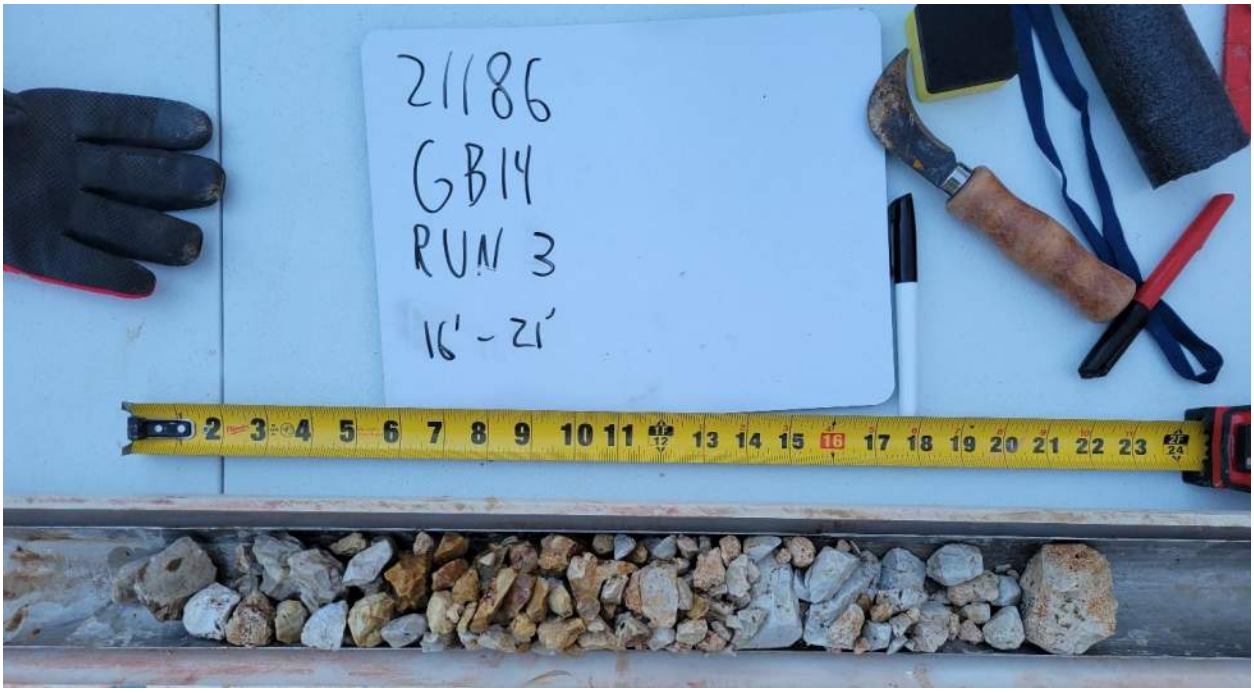
GB3, Run 6, 36.5’ – 41.5’



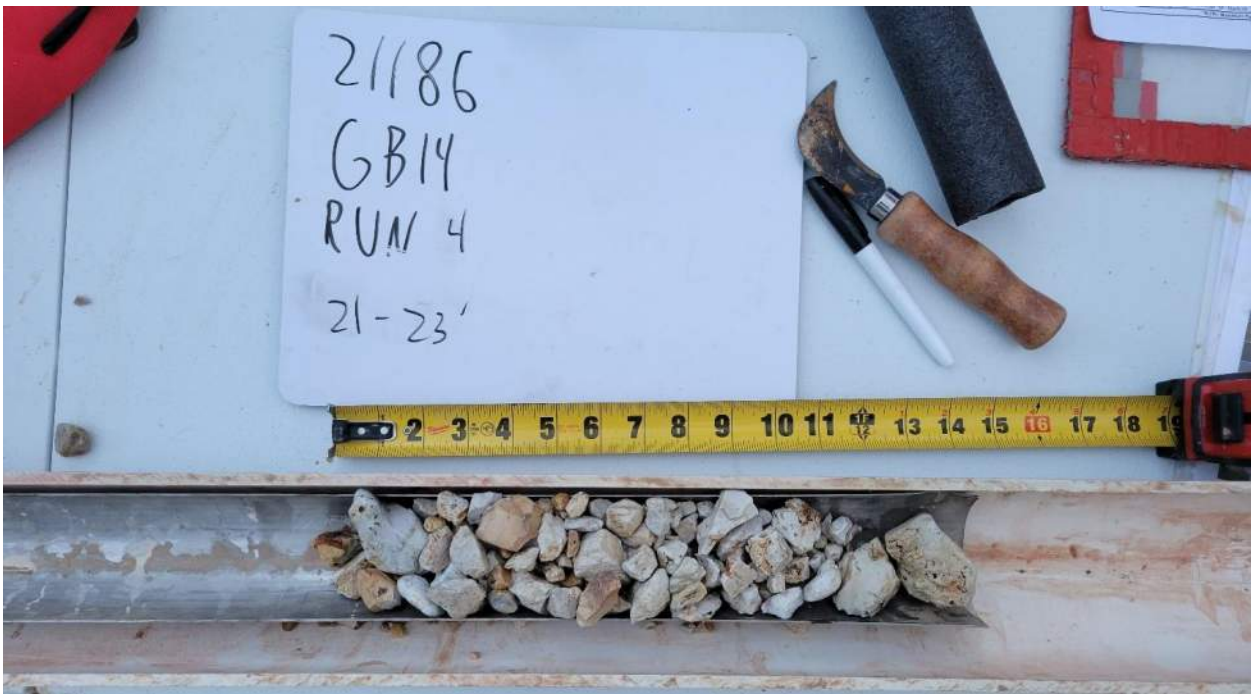
GB 14, Run 1, 8’ – 11’



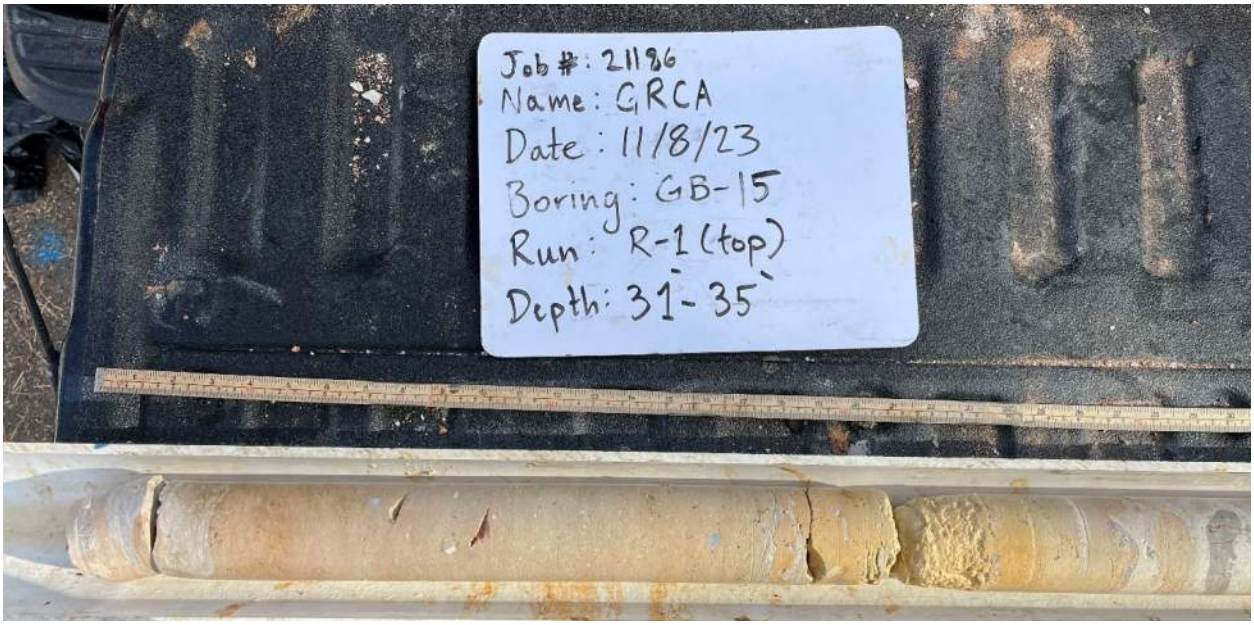
GB 14, Run 2, 11’ – 16’



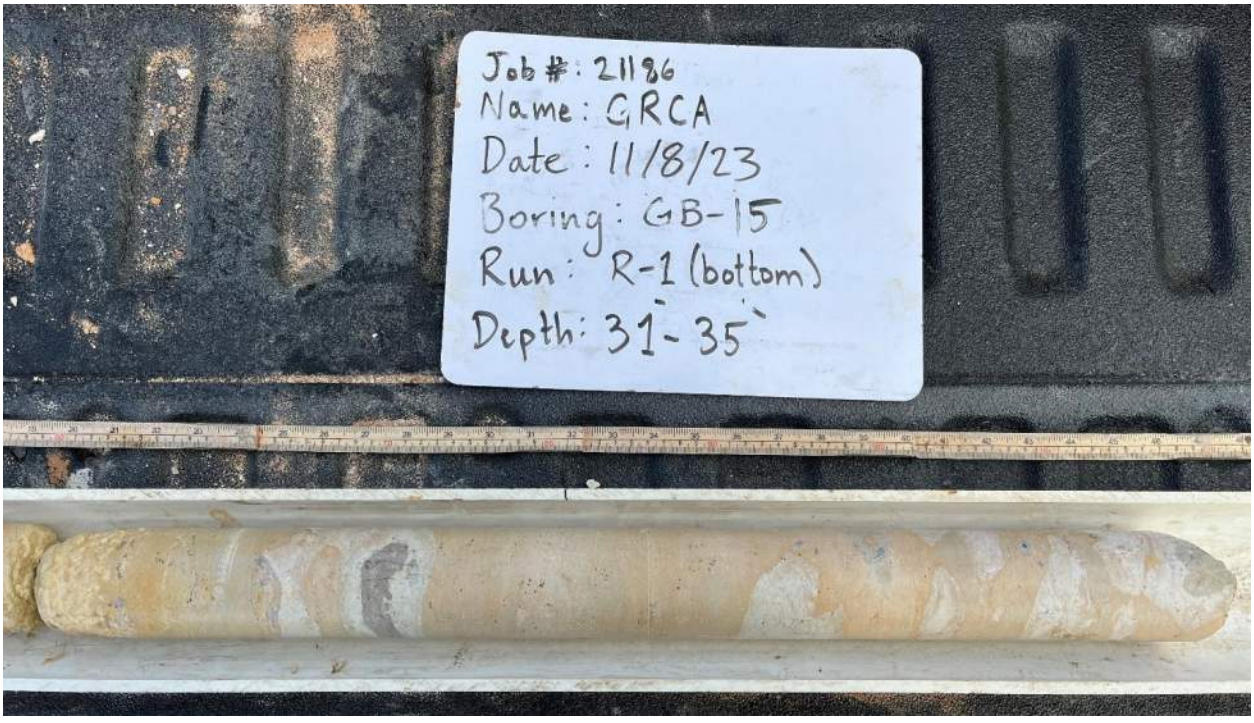
GB 14, Run 3, 16’ – 21’



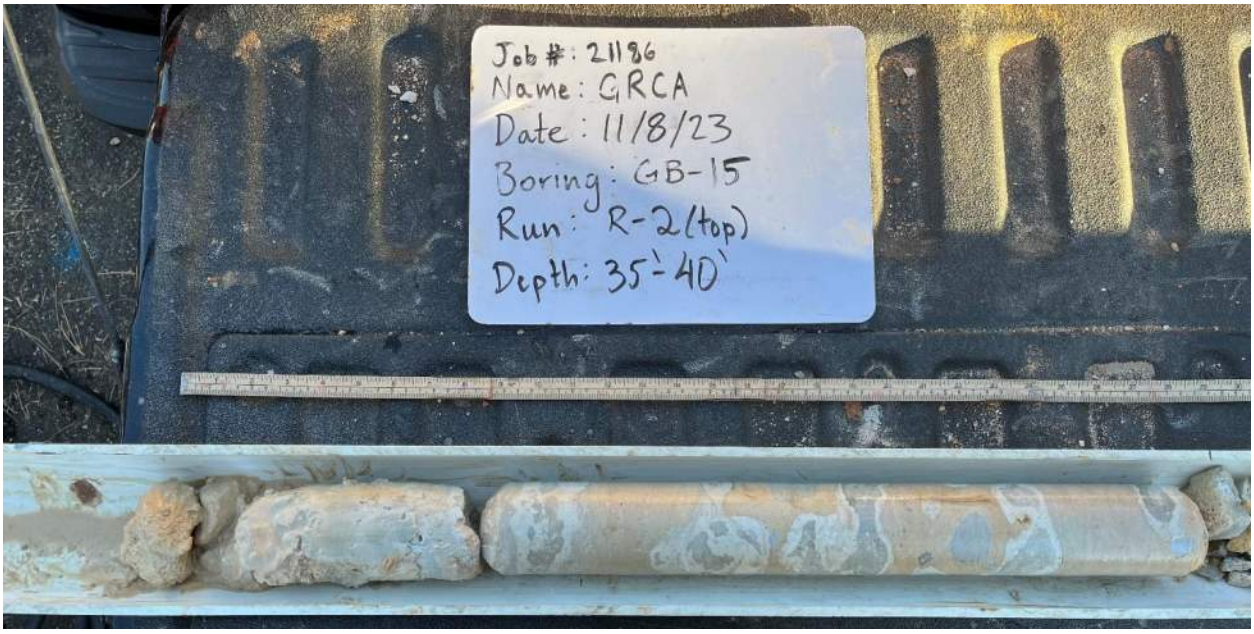
GB 14, Run 4, 21’ – 23’



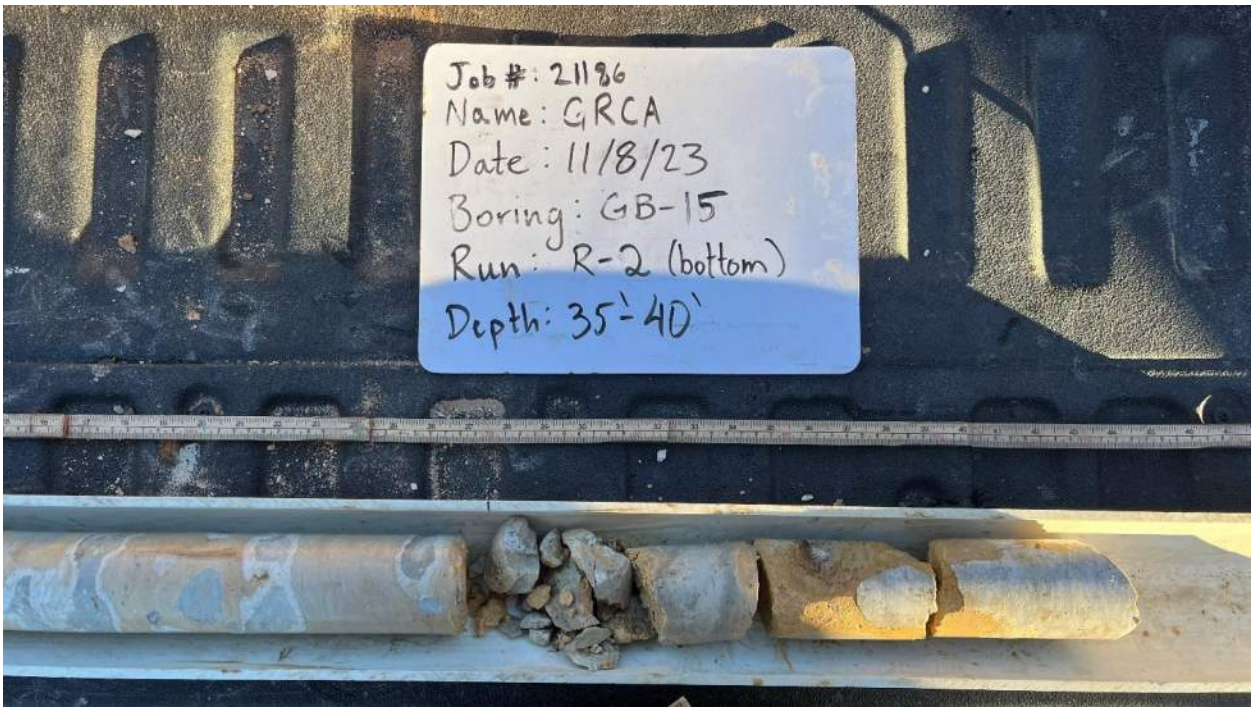
GB 15, Run 1 (top), 31' – 35'



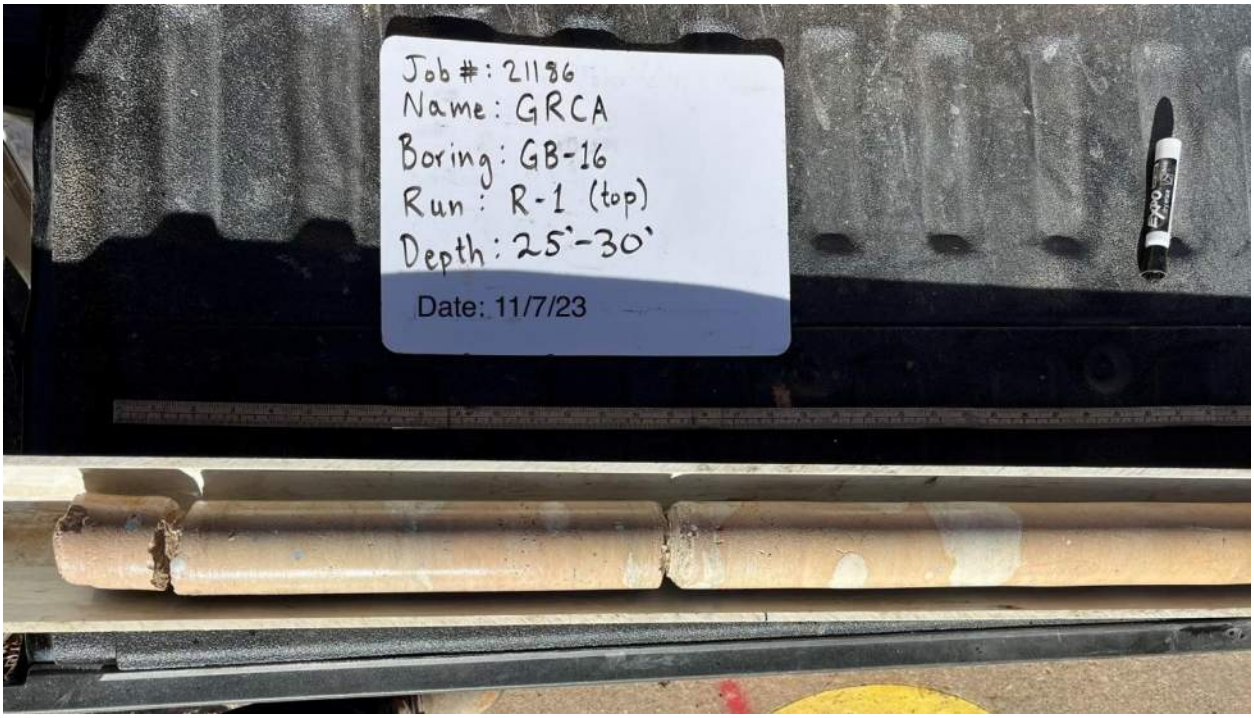
GB 15, Run 1 (bottom), 31' – 35'



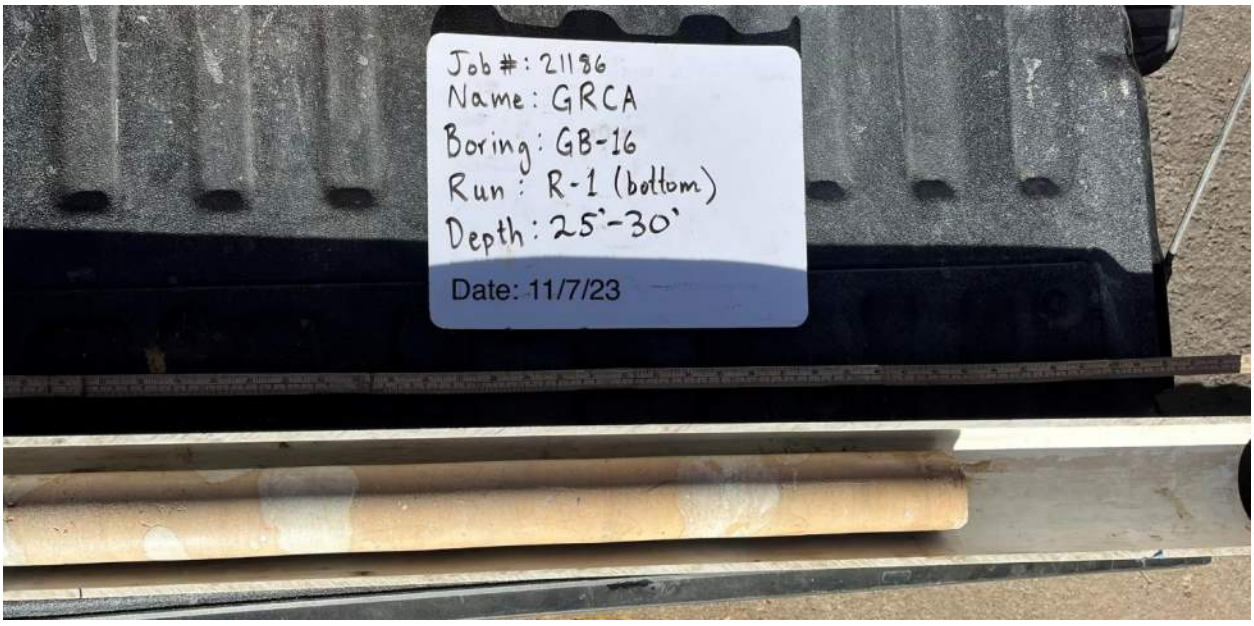
GB 15, Run 2 (top), 35’ – 40’



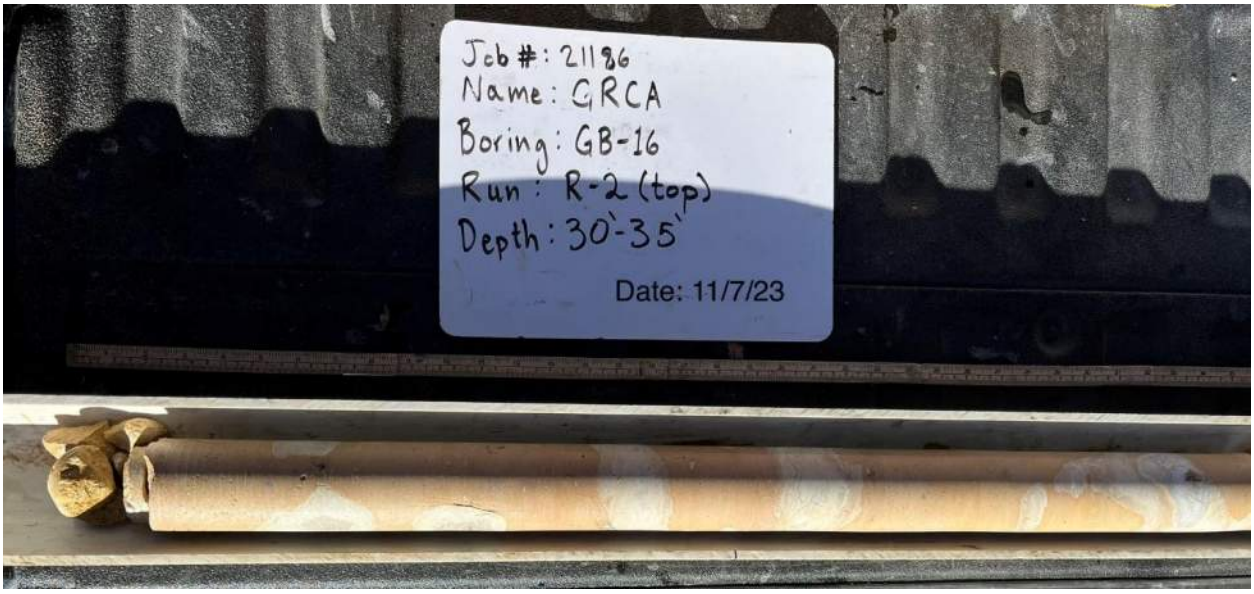
GB 15, Run 2 (bottom), 35’ – 40’



GB 16, Run 1 (top), 25’ – 30’



GB 16, Run 1 (bottom), 25’ – 30’



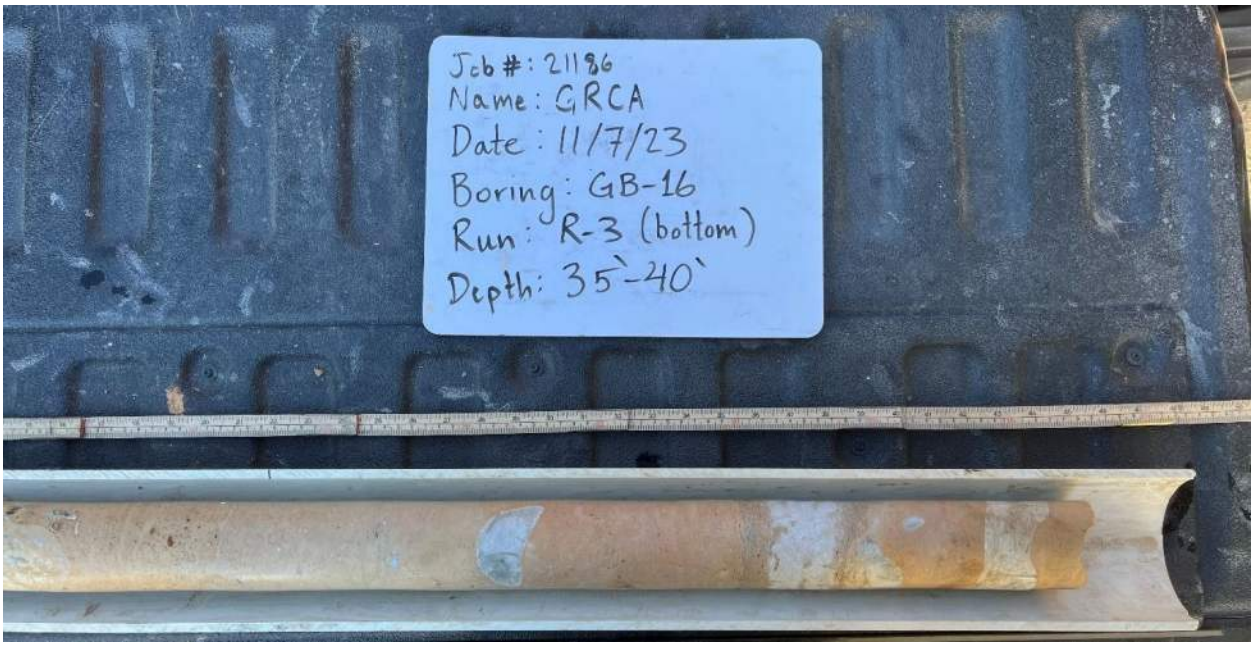
GB 16, Run 2 (top), 30’ – 35’



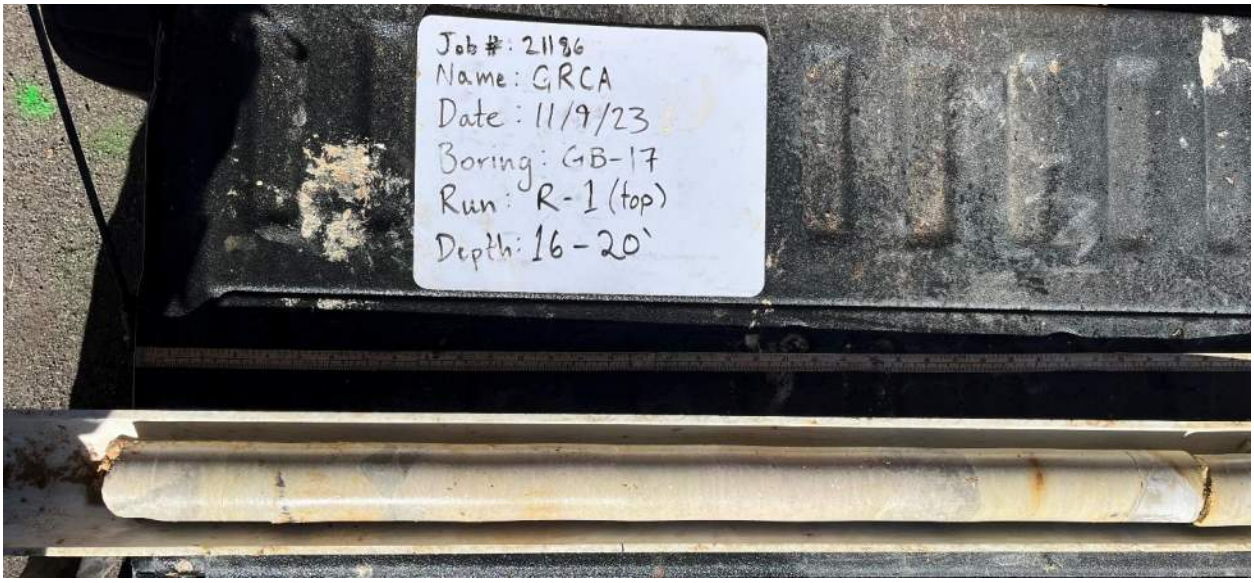
GB 16, Run 2 (bottom), 30’ – 35’



GB 16, Run 3 (top), 35’ – 40’



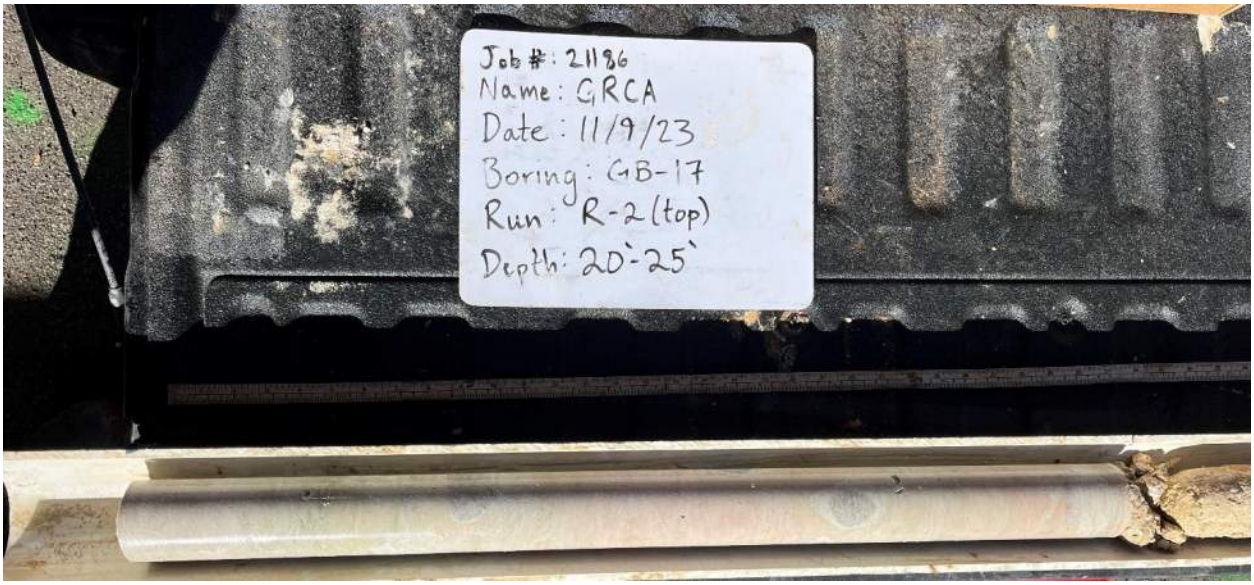
GB 16, Run 3 (bottom), 35’ – 40’



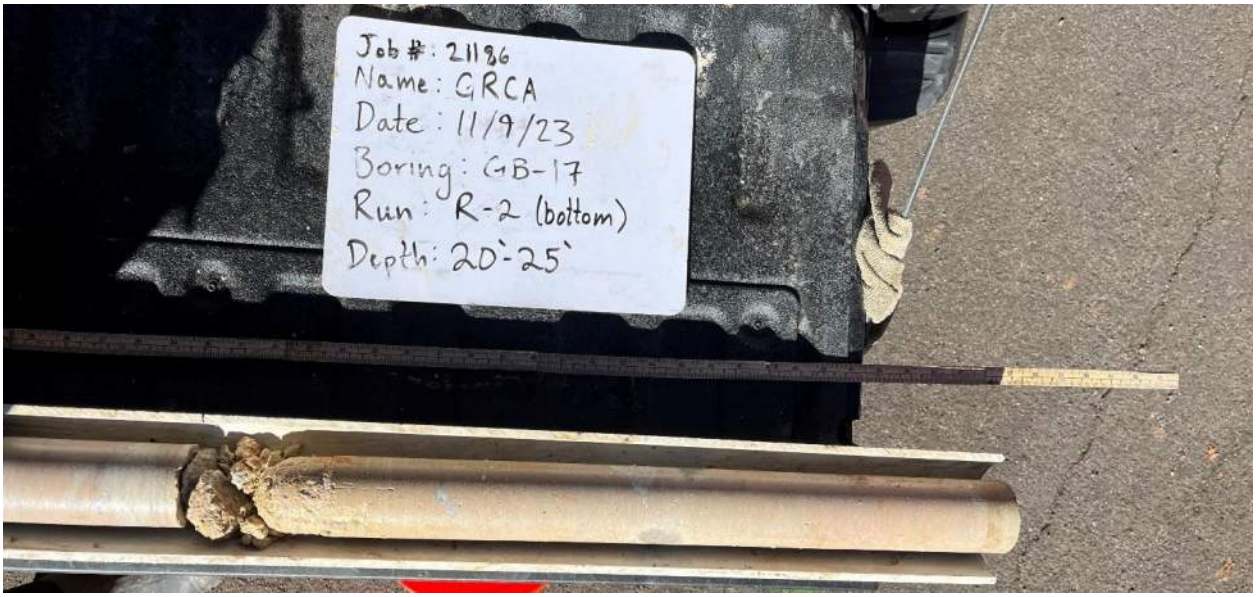
GB 17, Run 1 (top), 16’ – 20’



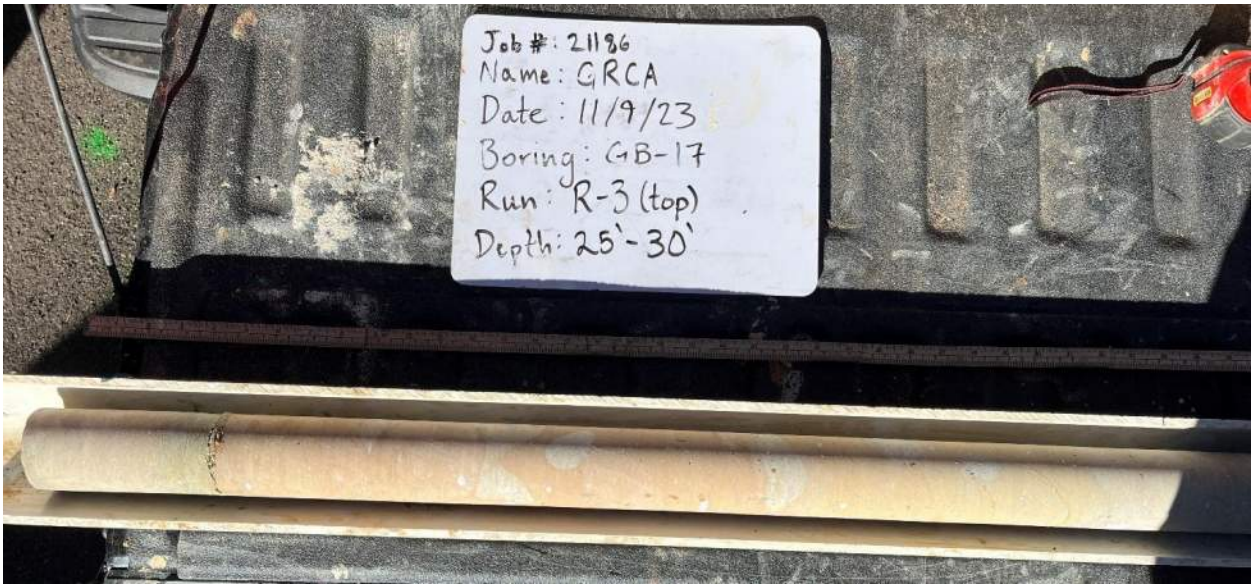
GB 17, Run 1 (bottom), 16’ – 20’



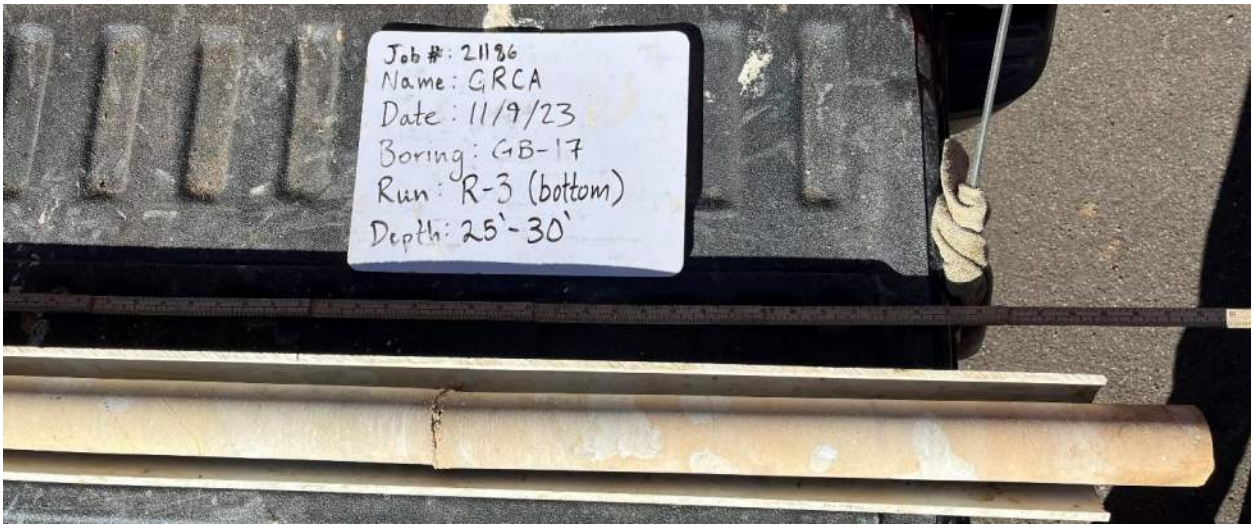
GB 17, Run 2 (top), 20' – 25'



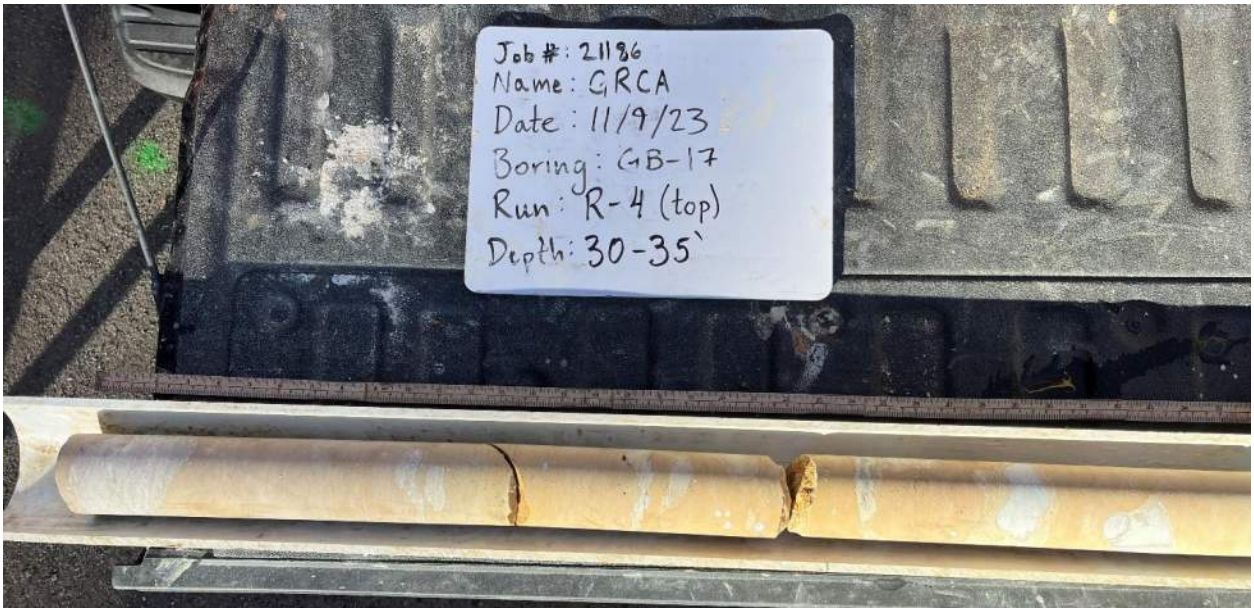
GB 17, Run 2 (bottom), 20' – 25'



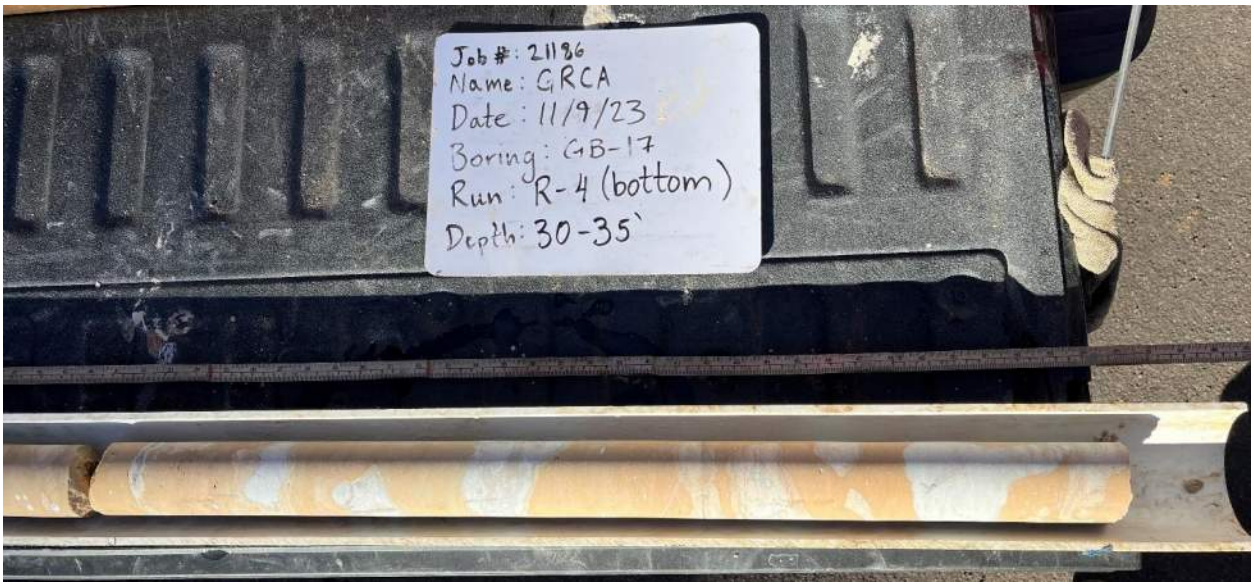
GB 17, Run 3 (top), 25’ – 30’



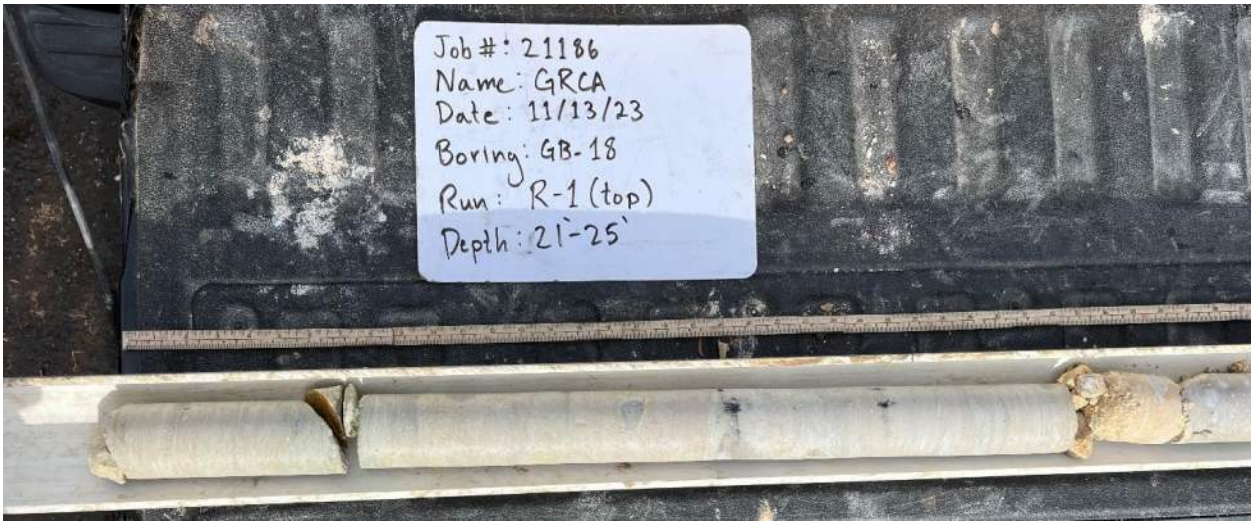
GB 17, Run 3 (bottom), 25’ – 30’



GB 17, Run 4 (top), 30’ – 35’



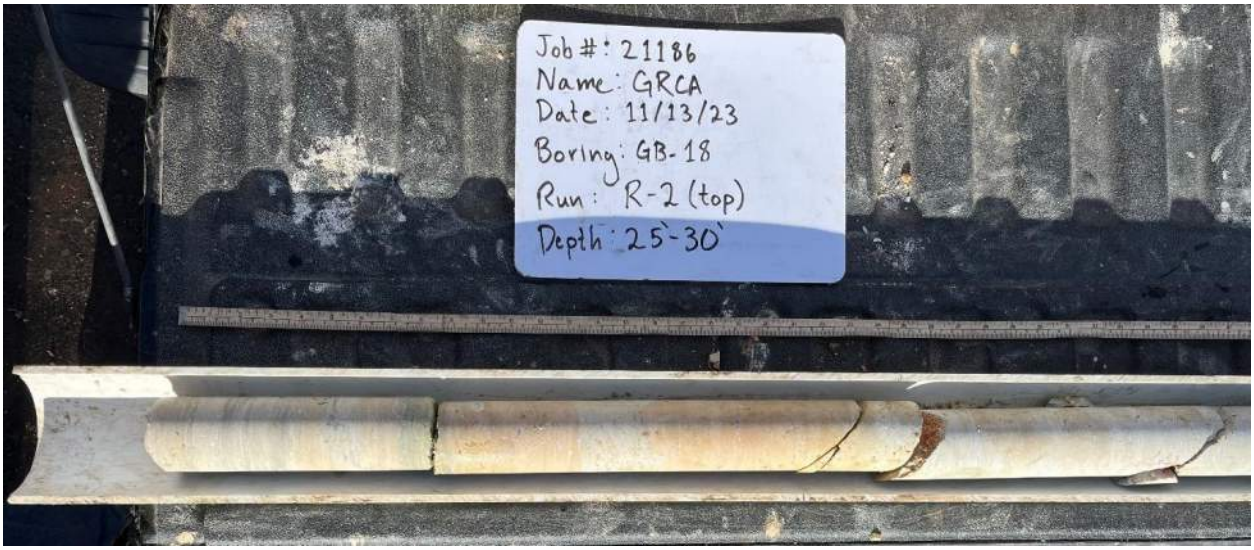
GB 17, Run 4 (bottom), 30’ – 35’



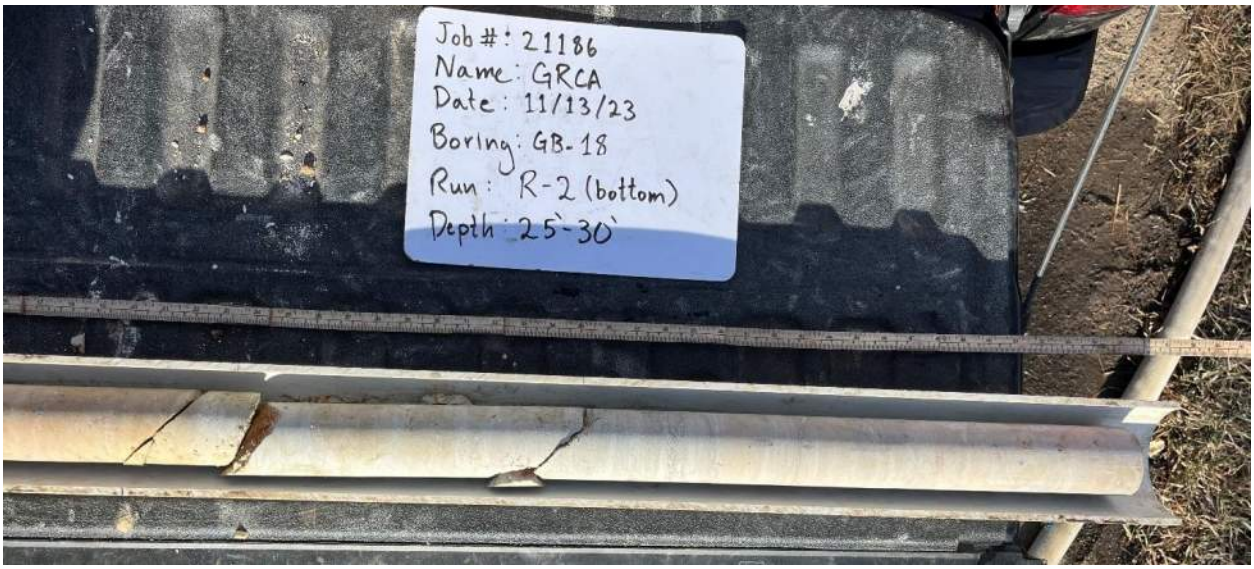
GB 18, Run 1 (top), 21’ – 25’



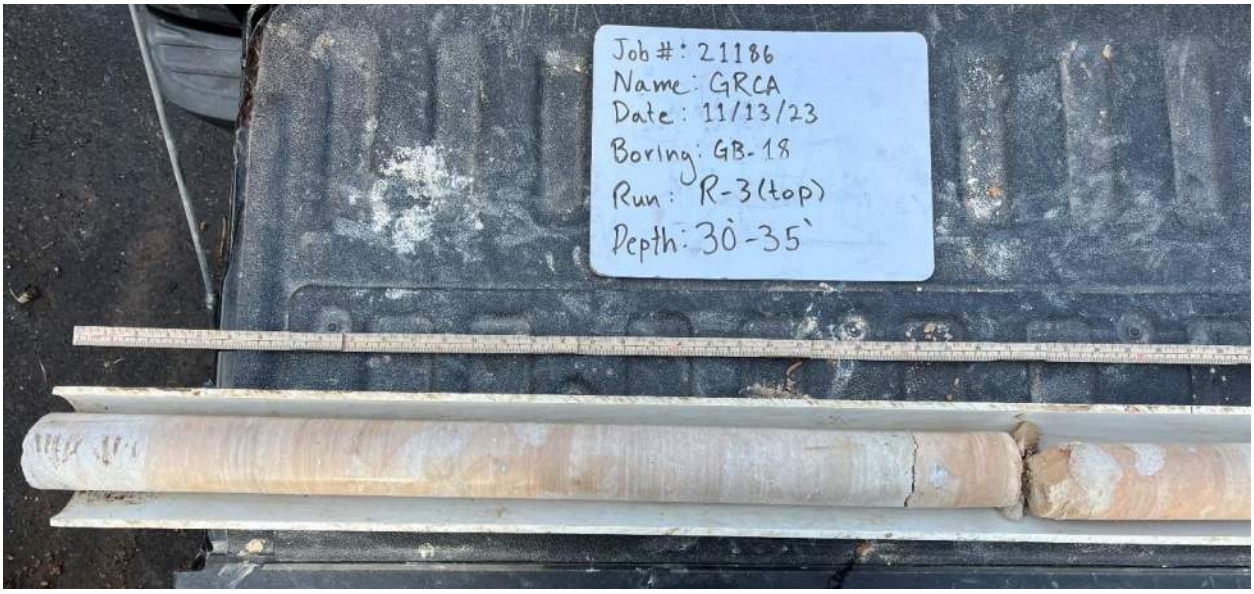
GB 18, Run 1 (bottom), 21’ – 25’



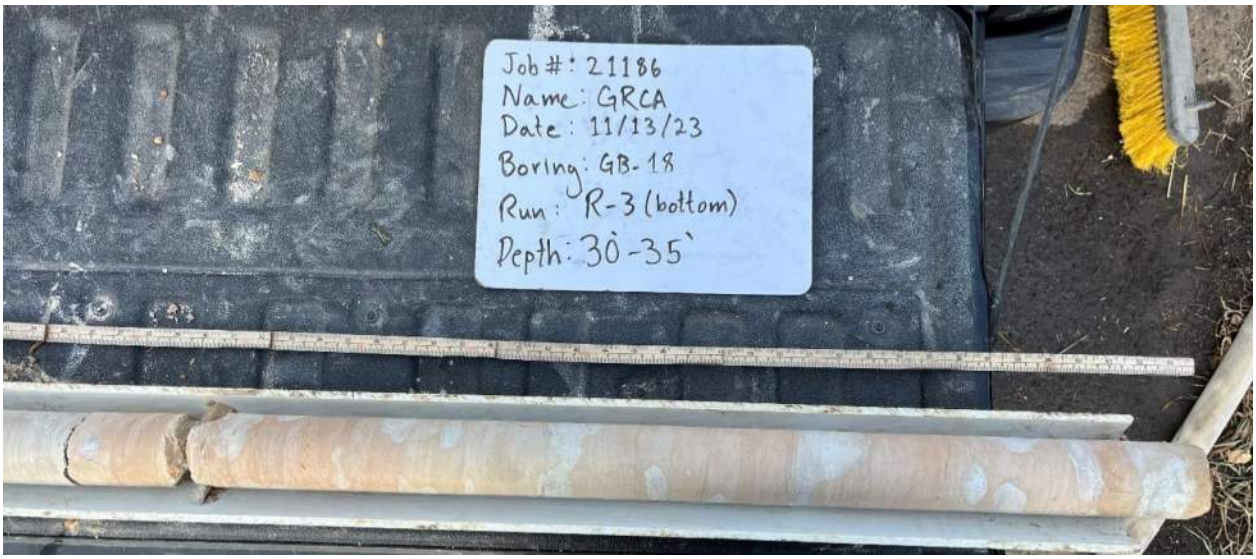
GB 18, Run 2 (top), 25’ – 30’



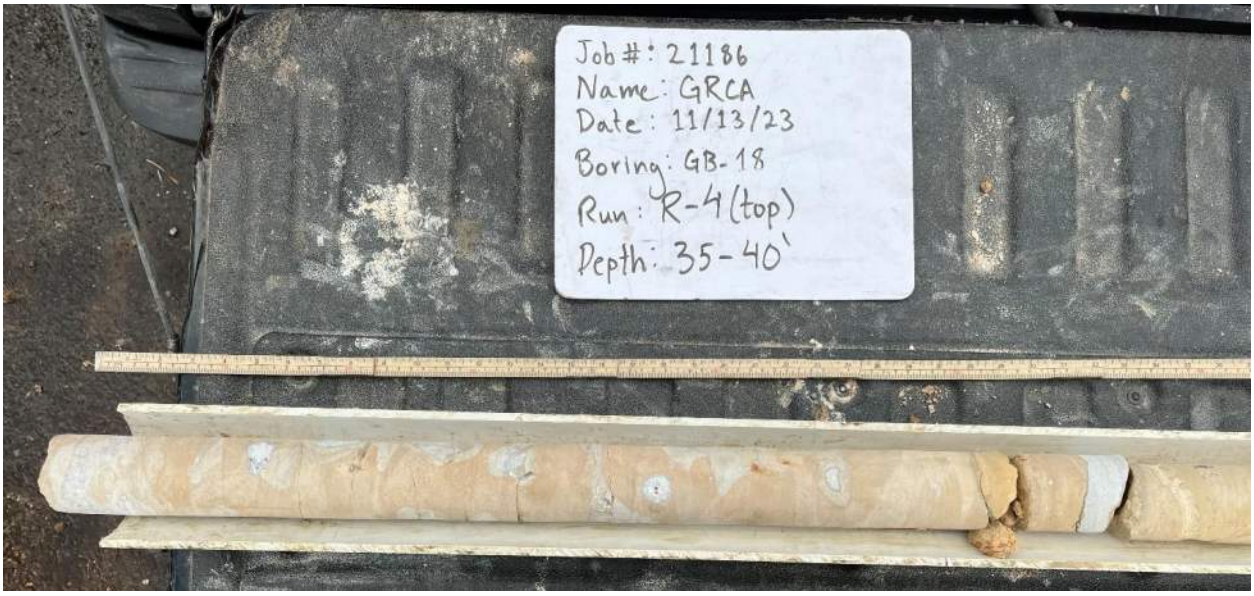
GB 18, Run 2 (bottom), 25’ – 30’



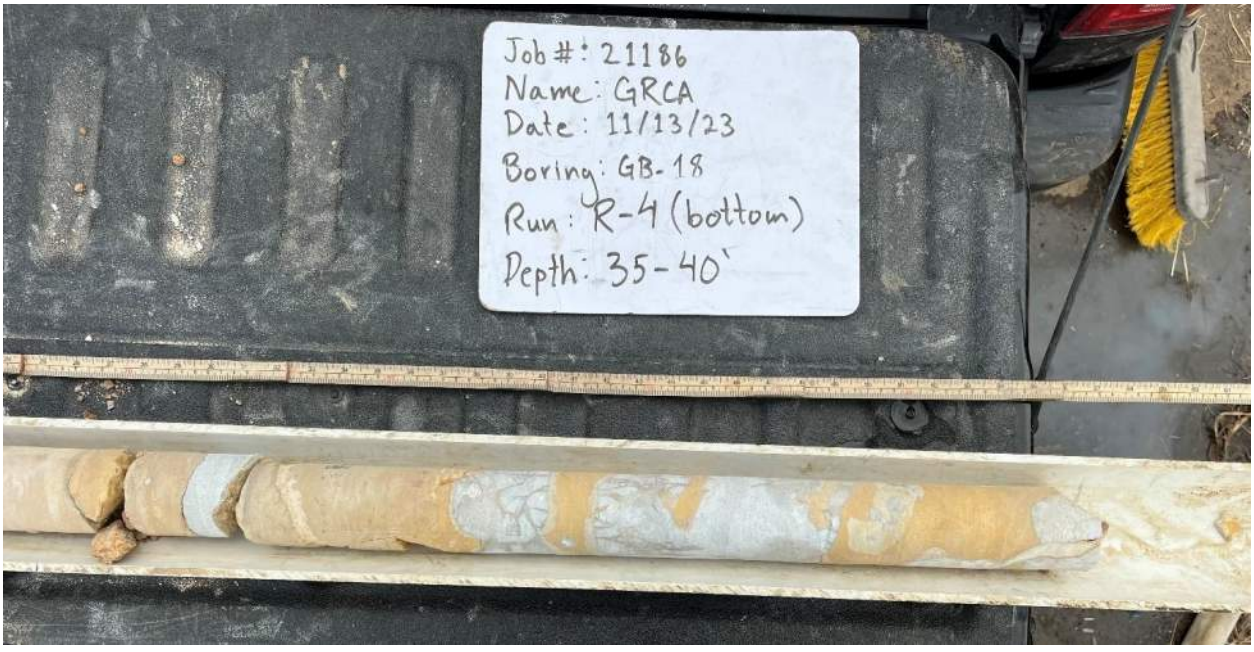
GB 18, Run 3 (top), 30' – 35'



GB 18, Run 3 (bottom), 30' – 35'



GB 18, Run 4 (top), 35’ – 40’



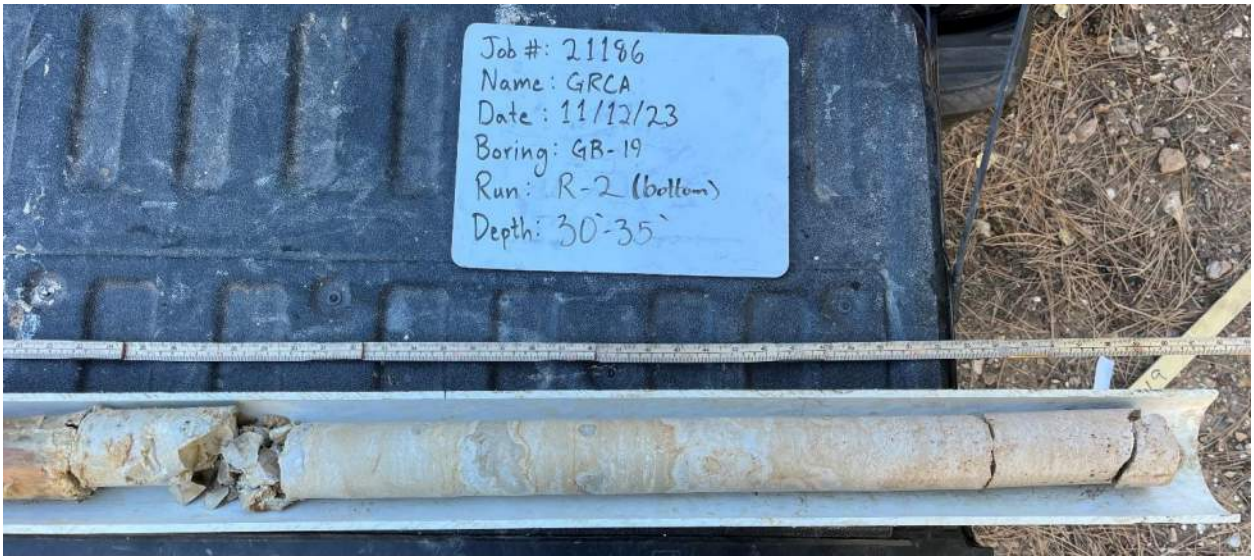
GB 18, Run 4 (bottom), 35’ – 40’



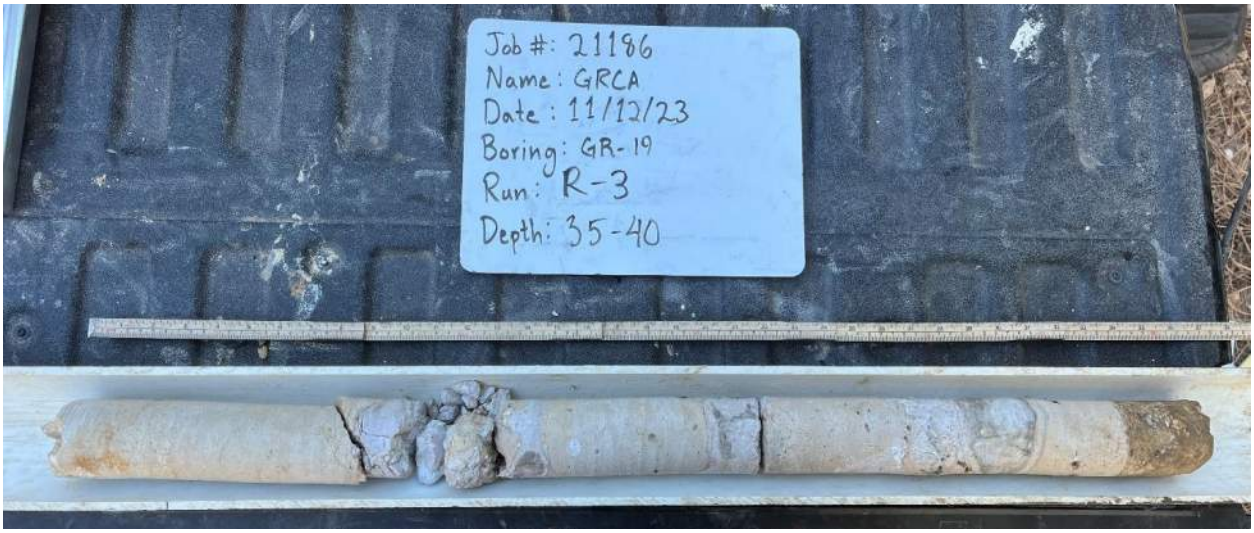
GB 19, Run 1, 27.5’ – 30’



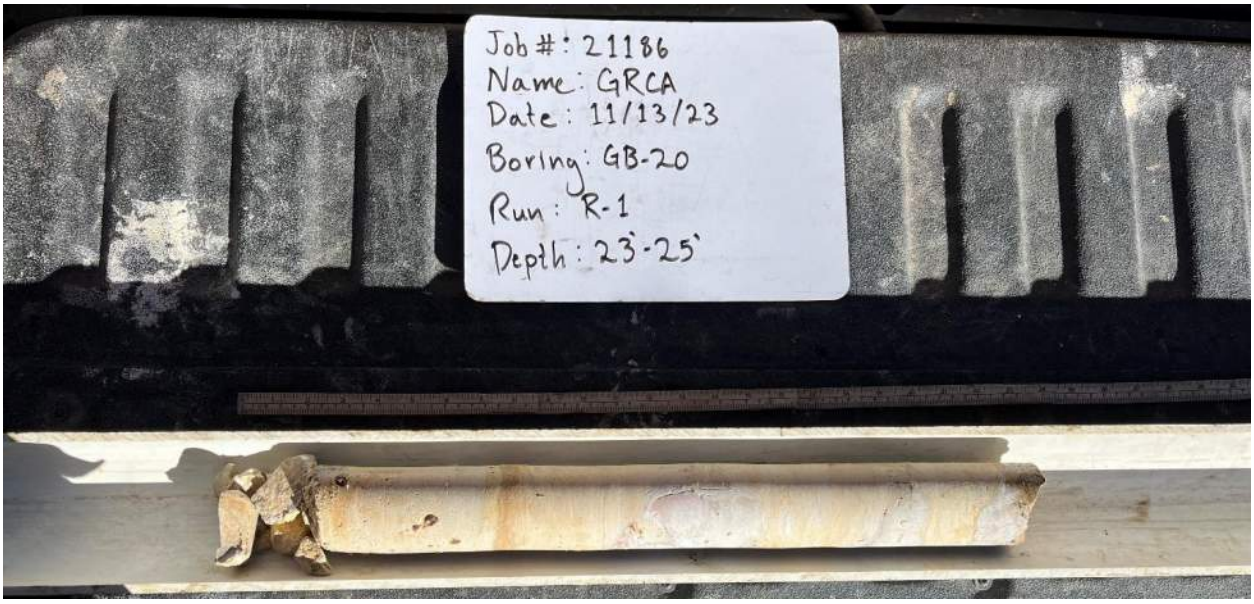
GB 19, Run 2 (top), 30’ – 35’



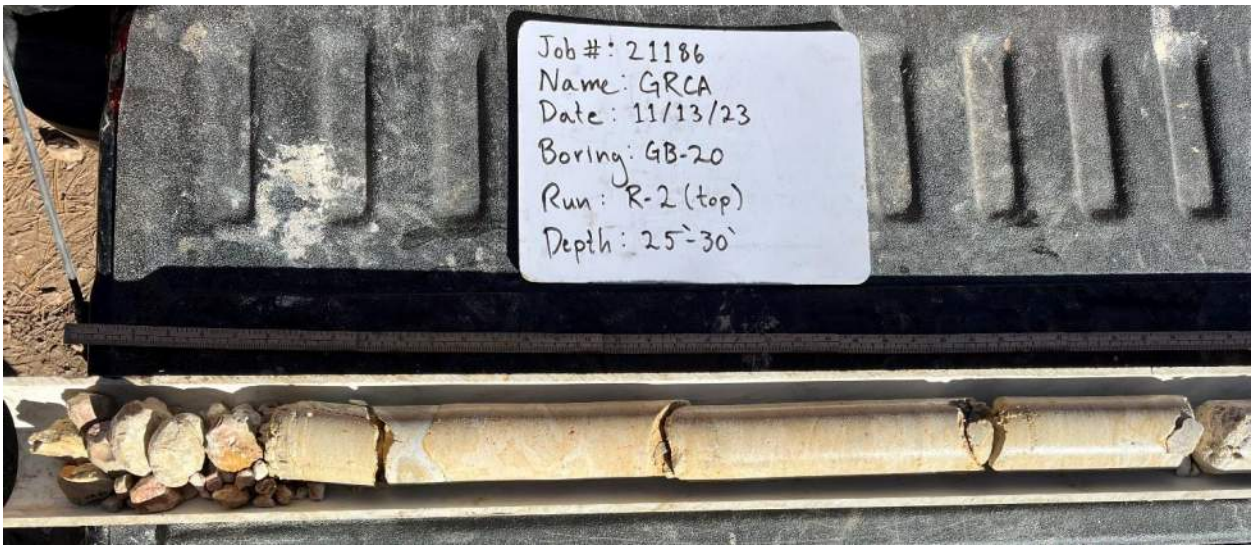
GB 19, Run 2 (bottom), 30’ – 35’



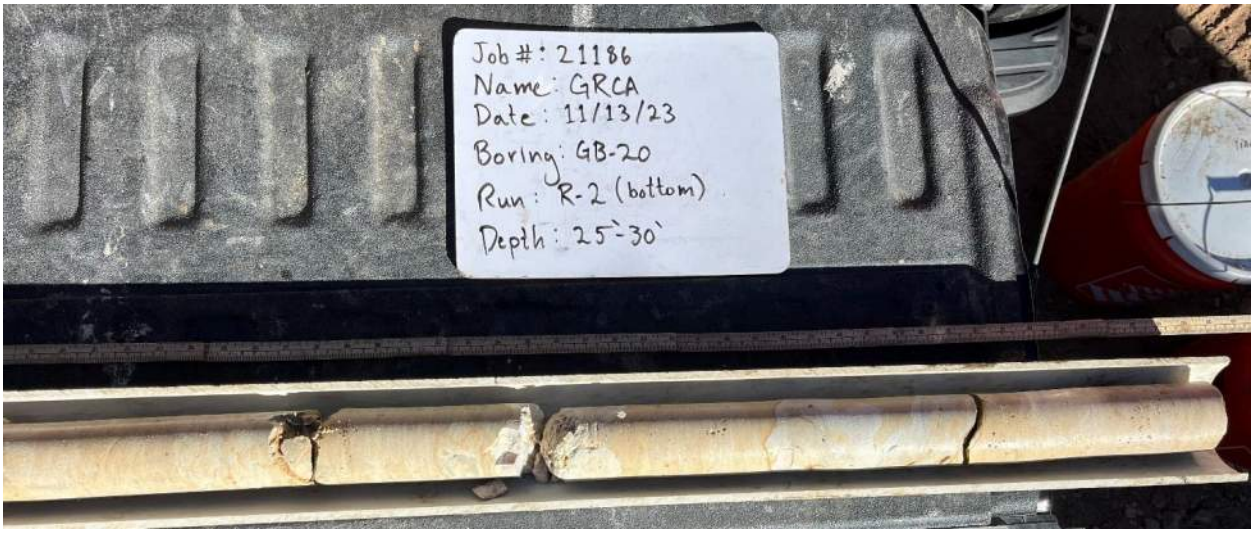
GB 19, Run 3, 35’ – 40’



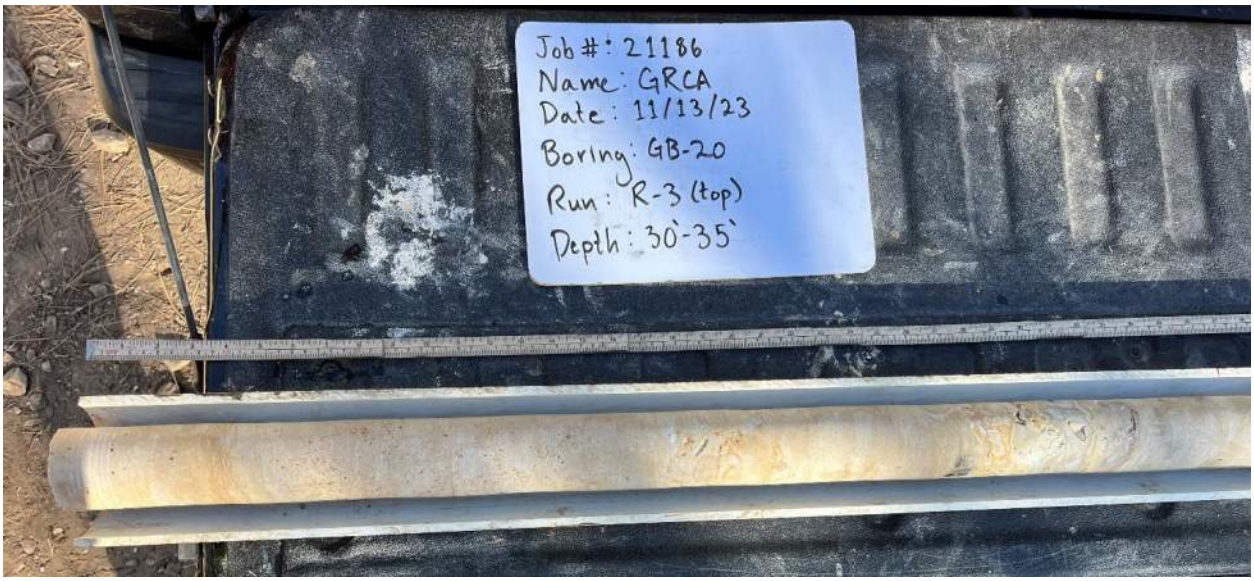
GB 20, Run 1, 23’ – 25’



GB 20, Run 2 (top), 25’ – 30’



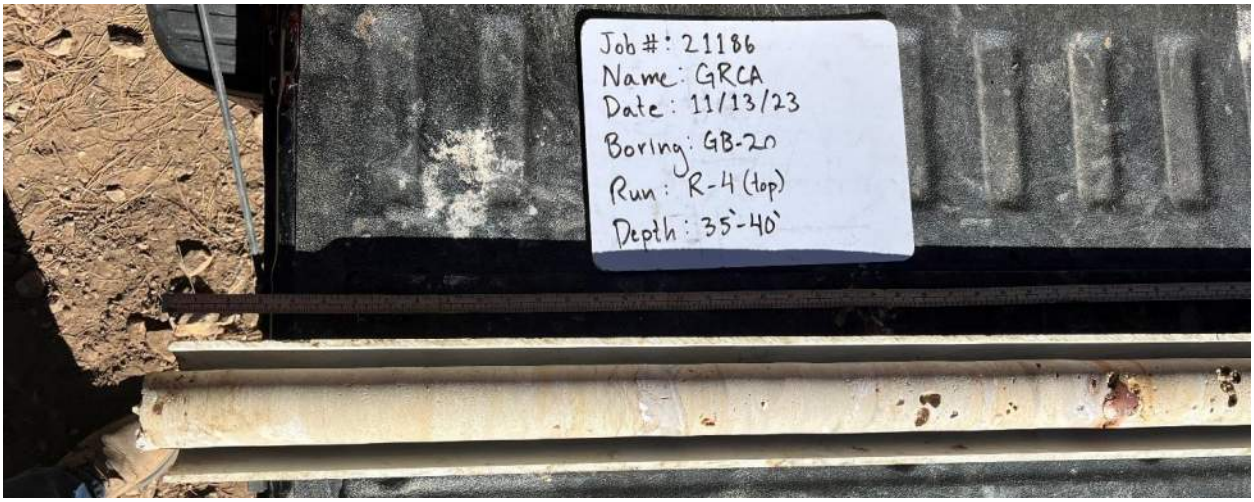
GB 20, Run 2 (bottom), 25’ – 30’



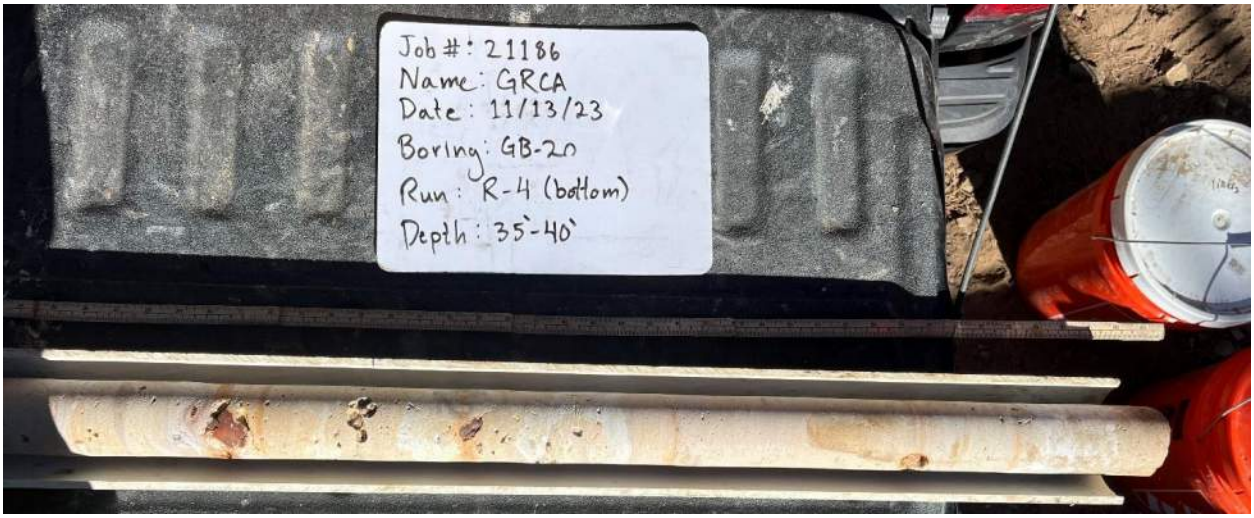
GB 20, Run 3 (top), 30’ – 35’



GB 20, Run 3 (bottom), 30’ – 35’



GB 20, Run 4 (top), 35’ – 40’



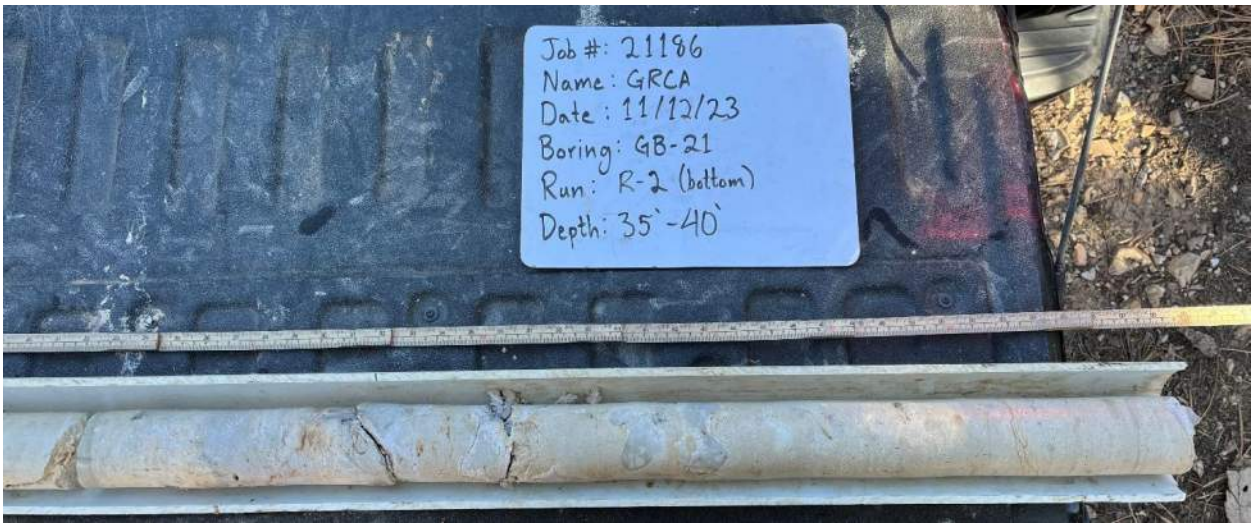
GB 20, Run 4 (bottom), 35’ – 40’



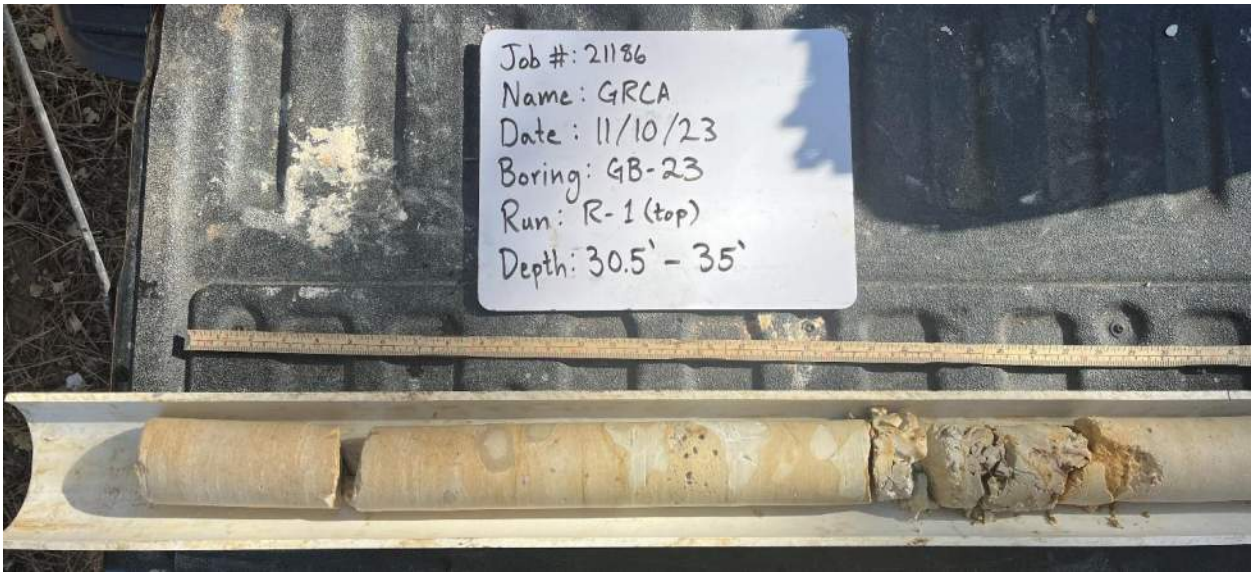
GB 21, Run 1, 32.5’ – 35’



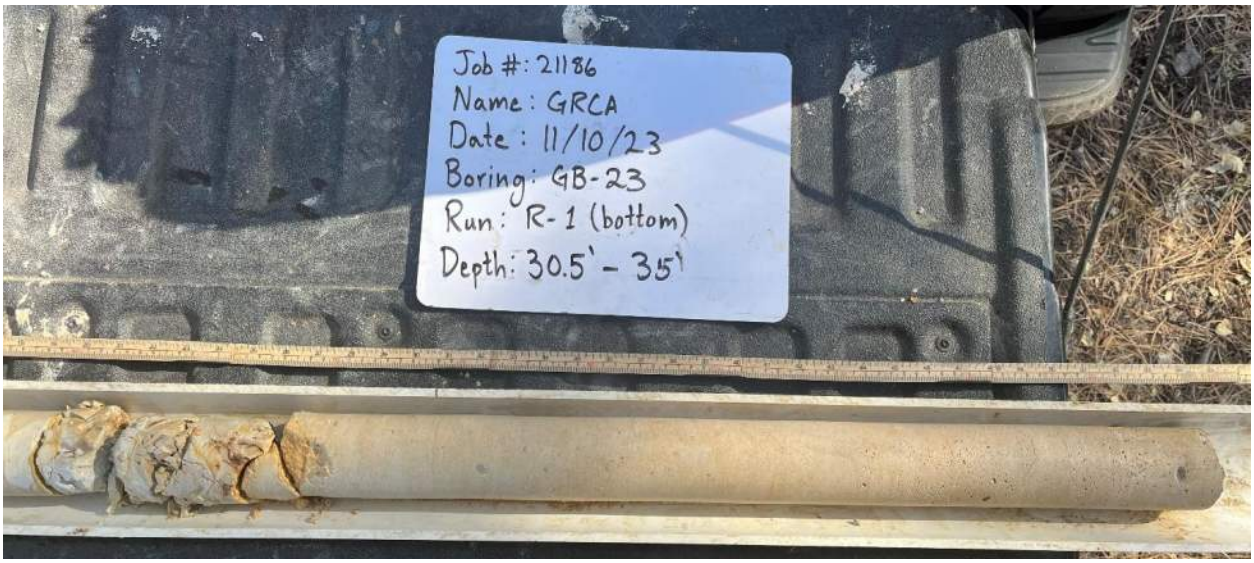
GB 21, Run 2 (top), 35’ – 40’



GB 21, Run 2 (bottom), 35’ – 40’



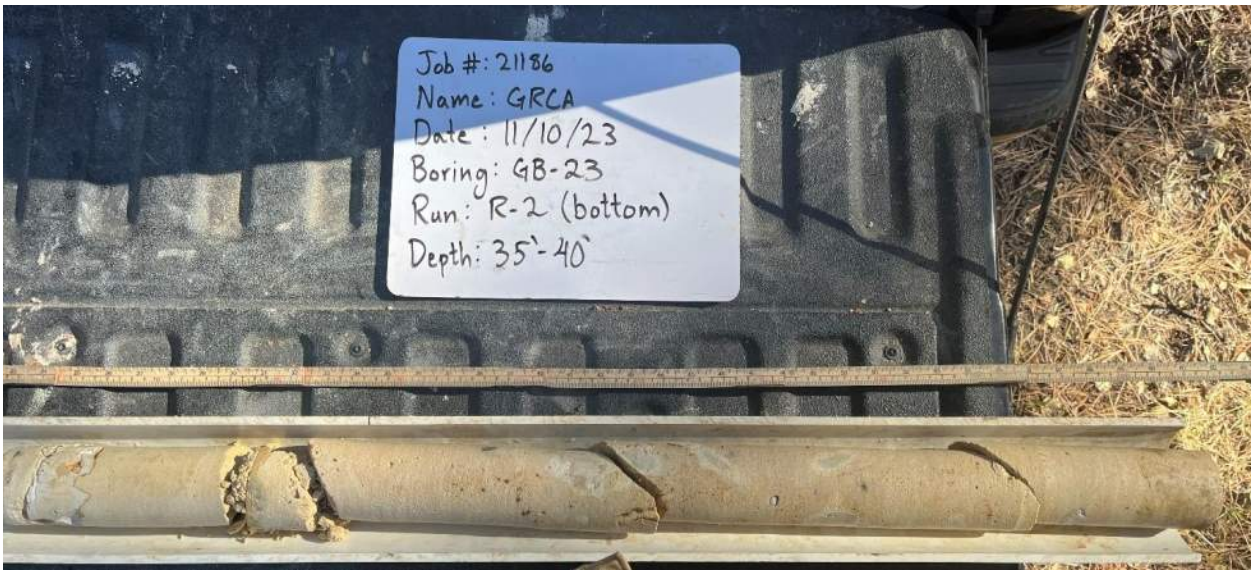
GB 23, Run 1 (top), 30.5’ – 35’



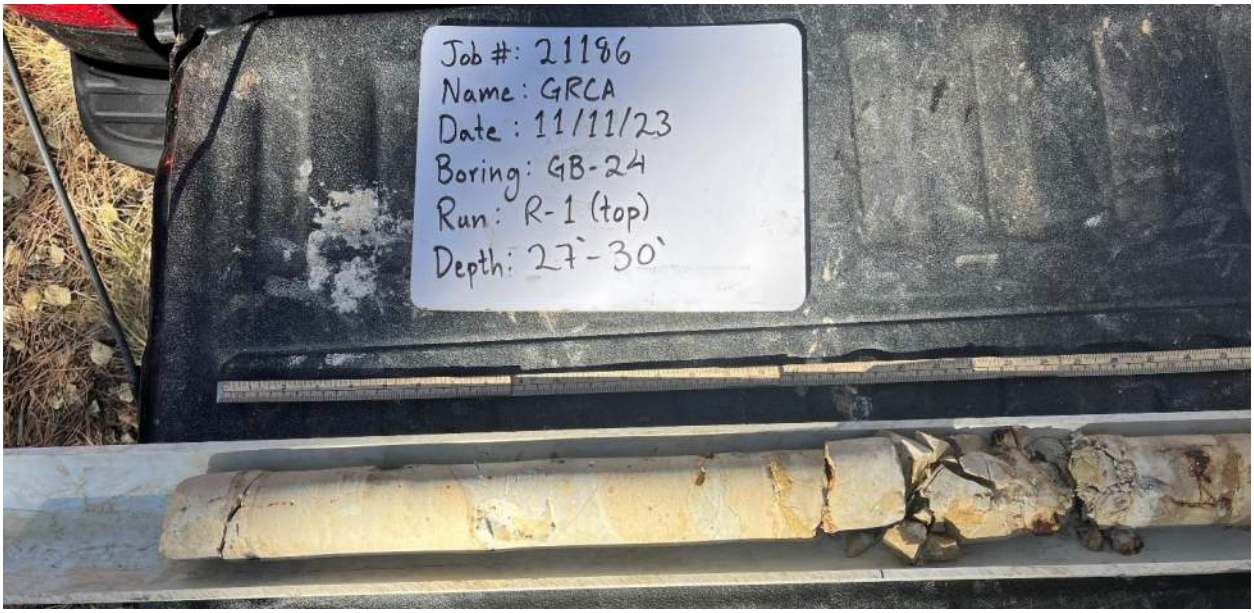
GB 23, Run 1 (bottom), 30.5’ – 35’



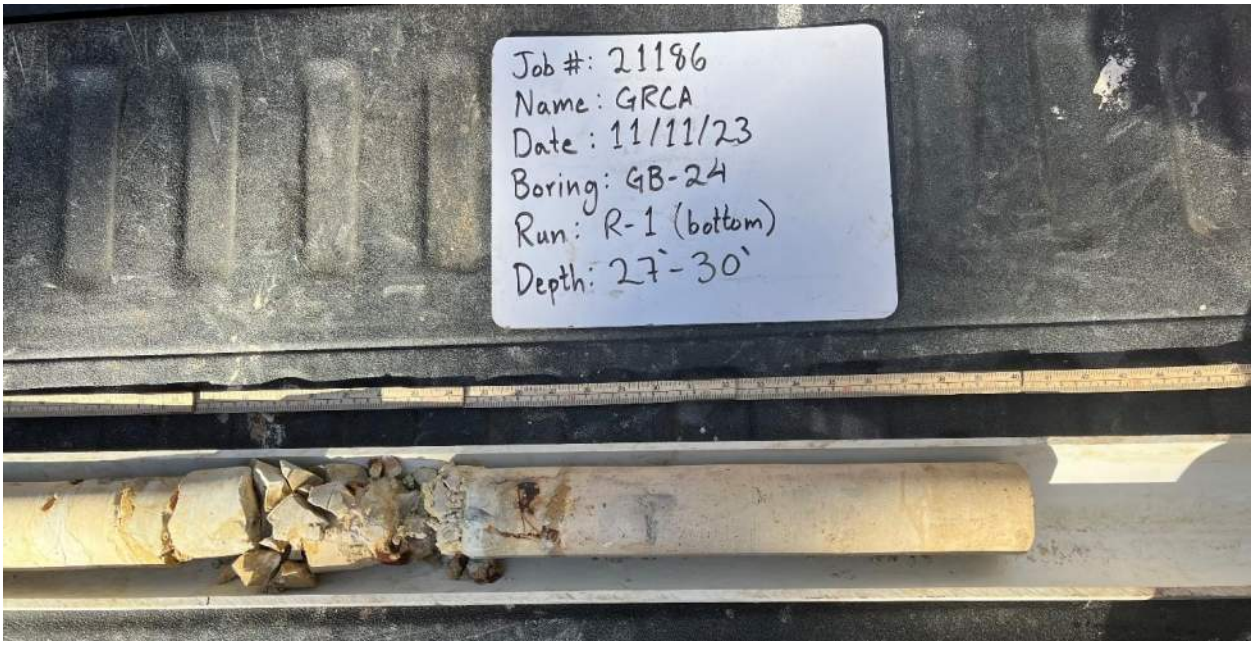
GB 23, Run 2 (top), 35’ – 40’



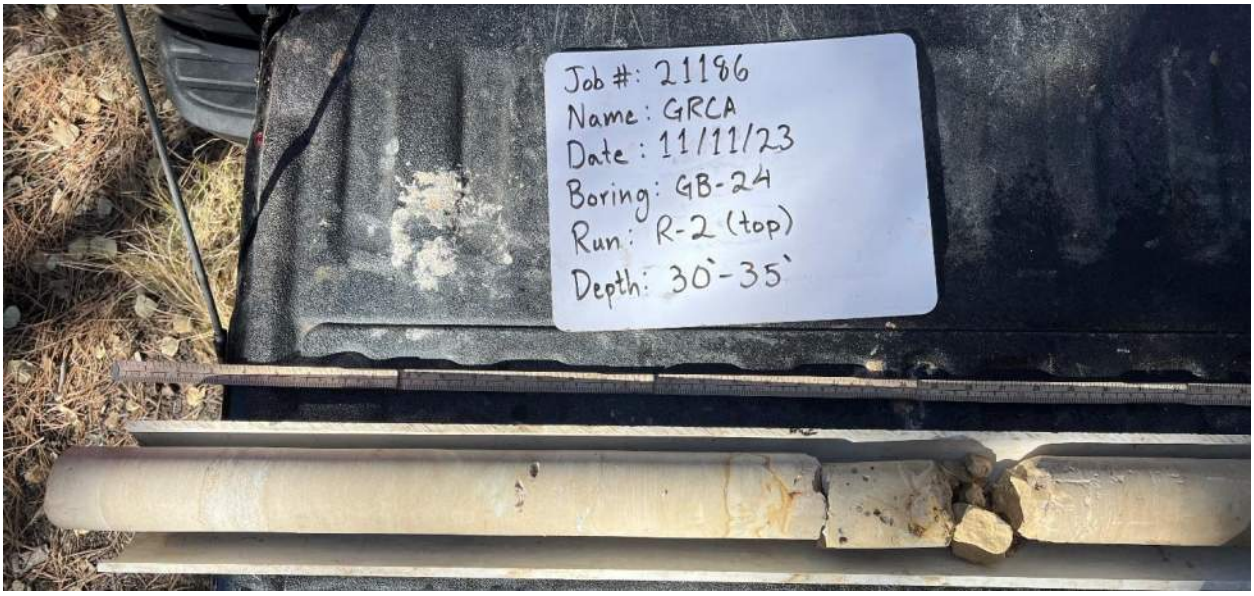
GB 23, Run 2 (bottom), 35’ – 40’



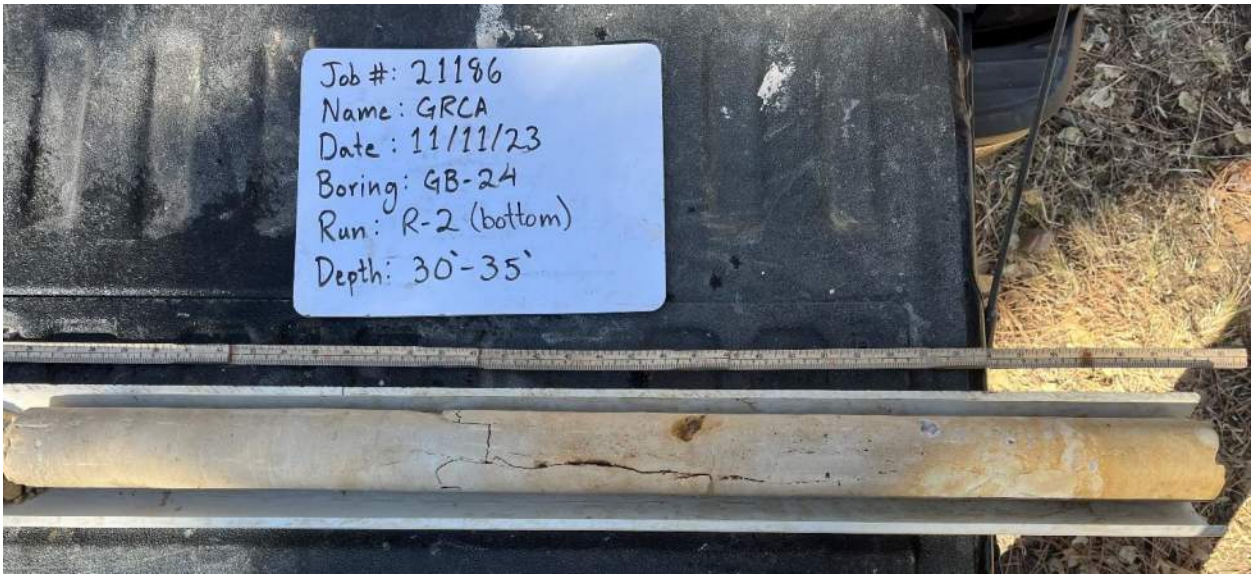
GB 24, Run 1 (top), 27’ – 30’



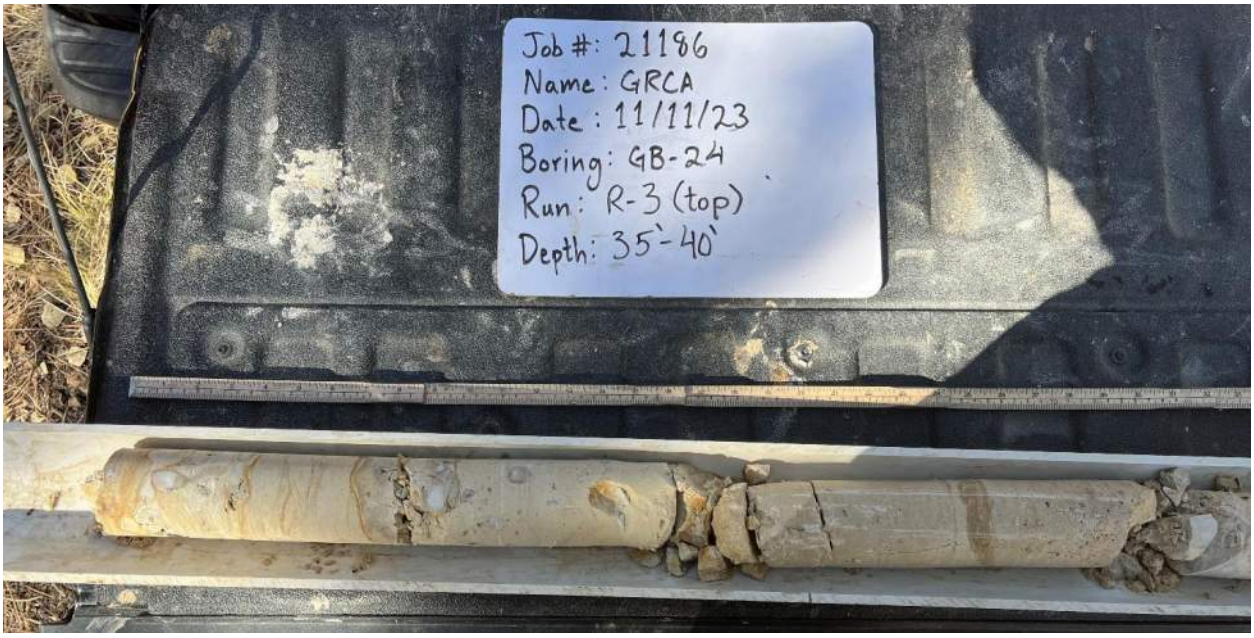
GB 24, Run 1 (bottom), 27’ – 30’



GB 24, Run 2 (top), 30’ – 35’



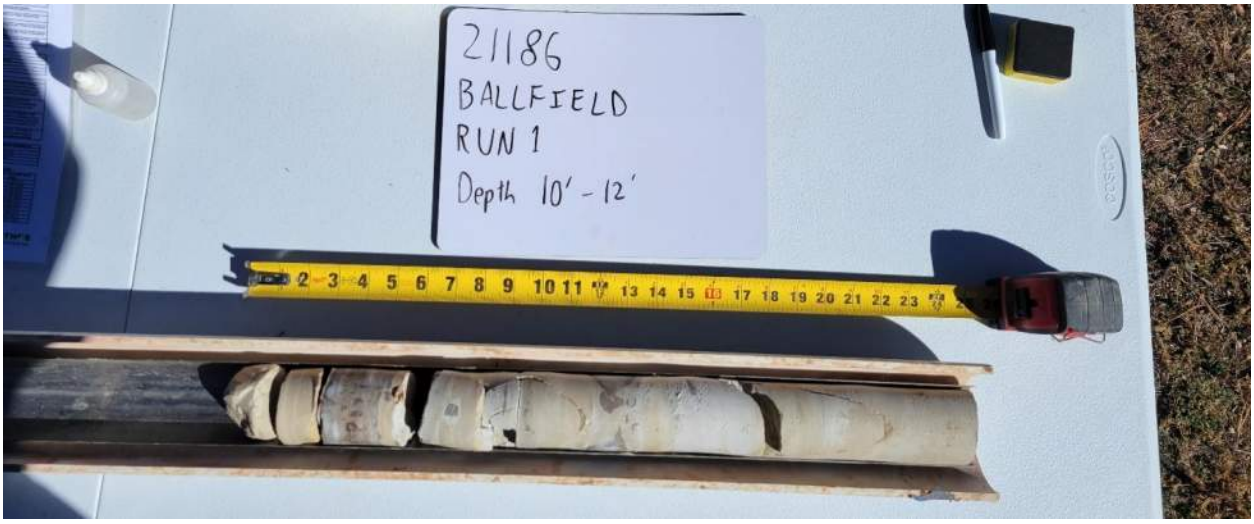
GB 24, Run 2 (bottom), 30’ – 35’



GB 24, Run 3 (top), 35’ – 40’



GB 24, Run 3 (bottom), 35’ – 40’



Ballfield Boring, Run 1, Depth 10’ – 12’



Ballfield Boring, Run 2, Depth 12’ – 17’



Ballfield Boring, Run 3, Depth 17’ – 23’



Ballfield Boring, Run 4, Depth 22’ – 25’

Appendix C Test Pit Logs

TEST PIT REPORT

Project: GRCA 318719
Client: HDR
Excavating Contractor: National Park Service
Excavator Make and Model: N/A
Cotractor Equipment/Procedures: Excavated with backhoe bucket
Field Rep.: Adam Malsam Excavator: N/A
Start Date: 10/19/2022 Finish Date: 10/19/2022

GEI Consultants

TEST PIT NO. Test Pit 1

Project No.: 21186
Elevation: 8323 Datum:
Notes:

Depth (ft)	Elevation (ft)	ID	Stratigraphy	Visual-Manual Identification and Description Density/consistency, color, GROUP NAME, max. particle size, structure, odor, moisture,optional descriptions, geologic interpretation	Groundwater Depth	Laboratory Results: Moisture (%)	Gravel (%)	Sand (%)	Fines (%)	Liquid Limit (%)	Plasticity Index (%)	Sulfates
0	8322			**RESIDUUM** Cobbles 6" to 12" with a clayey sand matrix, few gravels, trace boulders up to 18", olive grey to light brownish grey, dry. Matrix material quickly ravels out until stable, clast supported cobble walls are obtained. Cobbles and boulders are angular and comprised of sandy limestone								
2	8320											
4	8318											
5.5 ft.				**REFUSAL REACHED**								
6	8316											
8	8314											
10	8312											
12	8310											
14	8308											
16												

Note: Soil and rock identification based on visual-manual methods of ASTM D2487 and ASTM D2488.

Appendix D Geotechnical Laboratory Testing Results

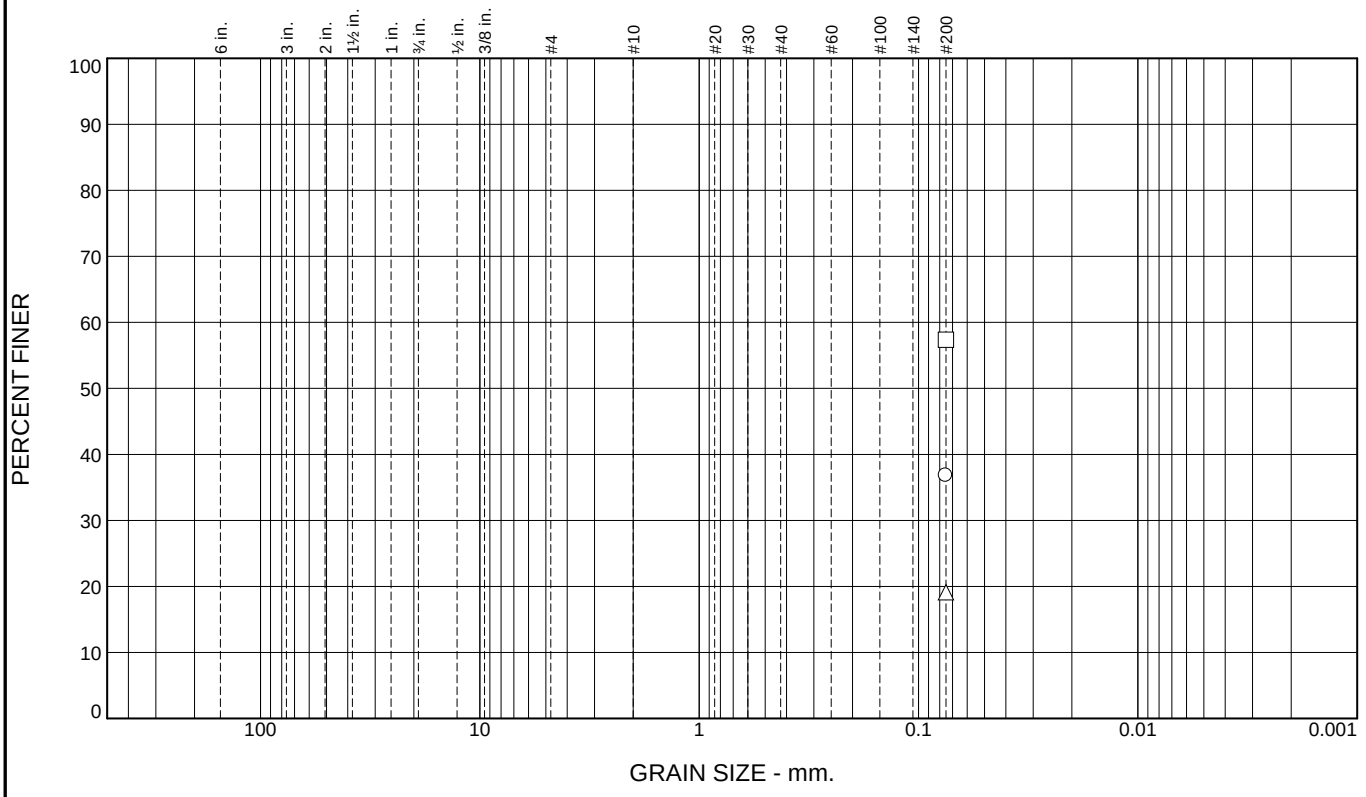
Sample Identification		In-Place States		Corrosion	Particle Size			Atterberg Limits		Strength		Swell		Soil/Rock Classification[1]	
Boring	Sample Depth (ft)	Moisture Content (%)	Dry Density (pcf)	Water Soluble Sulfates (%)	Gravel (%)	Sand (%)	Fines (%)	Liquid Limit (%)	Plasticity Index (%)	Unconfined Compressive Strength (psi)	Unconfined Compressive Strength (psf)	Swell Pressure (psf)	Swell Percent (%)	USCS Symbol	Description
GB1	10	16.9	108.8				37	50	31			1,250	0.5	GC	Clayey GRAVEL with Sand
GB1	20						57	47	27					CL	Sandy LEAN CLAY with Gravel
GB1	25			0.004										CL	Sandy LEAN CLAY with Gravel
GB2	5	6.6		0.005			19							SC	Clayey SAND with Gravel
GB2	15											105	-0.5	CL	LEAN CLAY with Sand
GB2	20											0	-0.2	CL	Gravelly LEAN CLAY with Sand
GB2	35	18.8	101.4				54	42	24					CL	Sandy LEAN CLAY
GB3	2.5			0.006										SC	Clayey SAND
GB3	10				21	35	44							SC	Clayey SAND with Gravel
GB3	30									6,327	911,088			BR	Sandy Limestone
GB3	35									8,451	1,216,944			BR	Sandy Limestone
GB4	10	13.2	97.3	0.002	25	27	48	32	9			303	0.0	SC	Clayey SAND with Gravel
GB4	15						18							SC	Clayey SAND with Gravel
GB4	30						48	40	21					GC	Clayey GRAVEL with Sand
GB5	2.5						27							SC	Clayey SAND with Gravel
GB6	10				24	42	34	33	7					SM	Silty SAND with Gravel
GB7	5				25	30	45							SC	Clayey SAND with Gravel
GB8	5				26	33	41							SC	Clayey SAND with Gravel
GB9	10						60	46	27					CL	Sandy LEAN CLAY
GB10	15						47	51	34					GC	Clayey GRAVEL
GB11	2.5				23	44	33	NV	NP					SM	Silty SAND with Gravel
GB12	10				27	39	34	NV	NP					SM	Silty SAND with Gravel
GB13	15				36	47	17							SC	Clayey SAND with Gravel
GB14	5				44	35	21	NV	NP					GM	Silty GRAVEL with Sand
GB15	5				63	8	29	47	26					GC	Clayey GRAVEL
GB15	15	18.2					53							CL	Sandy LEAN CLAY with Gravel
GB16	15						47	61	39					SC	Clayey SAND
GB16	20				7	27	66	48	28					CL	Sandy LEAN CLAY
GB16	33									7,506	1,080,864			BR	Sandy Limestone
GB17	2				25	47	28	27	9					SC	Clayey SAND with Gravel
GB17	5	22.1	91.0		49	17	34	39	20			0	0.0	GC	Clayey GRAVEL with Sand
GB17	15						62	47	27					CL	Sandy LEAN CLAY
GB17	18									8,086	1,164,384			BR	Sandy Limestone
GB18	5						44	55	35					SC	Clayey SAND
GB18	10						31	58	37					SC	Clayey SAND
GB18	15			0.003										CL	Sandy LEAN CLAY
GB18	25									6,497	935,568			BR	Sandy Limestone
GB19	5				33	39	28	NV	NP					SM	Silty SAND with Gravel
GB19	10						41							SC	Clayey SAND with Gravel
GB19	20	19.8	93.0				67	31	8	10	1,480			ML	Sandy SILT
GB19	25							21	12					CL	Sandy LEAN CLAY with Gravel
GB19	33.5									7,685	1,106,640			BR	Sandy Limestone

Sample Identification		In-Place States		Corrosion	Particle Size			Atterberg Limits		Strength		Swell		Soil/Rock Classification[1]	
Boring	Sample Depth (ft)	Moisture Content (%)	Dry Density (pcf)	Water Soluble Sulfates (%)	Gravel (%)	Sand (%)	Fines (%)	Liquid Limit (%)	Plasticity Index (%)	Unconfined Compressive Strength (psi)	Unconfined Compressive Strength (psf)	Swell Pressure (psf)	Swell Percent (%)	USCS Symbol	Description
GB20	5			0.008										GM	Silty GRAVEL with Sand
GB20	10				50	30	20	NV	NP					GM	Silty GRAVEL with Sand
GB20	29.5									7,474	1,076,256			BR	Sandy Limestone
GB21	5				35	36	29							SC	Clayey SAND with Gravel
GB21	15	5.8			45	36	19							GC	Clayey GRAVEL with Sand
GB21	25				43	16	41	30	3					GM	Silty GRAVEL with Sand
GB21	30	14.7						29	7					CL-ML	Sandy SILTY CLAY with Gravel
GB21	37									1,460	210,240			BR	Sandy Limestone
GB22	5				54	22	24	NV	NP					GM	Silty GRAVEL with Sand
GB22	25				68	12	20	NV	NP					GM	Silty GRAVEL
GB22	40				30	31	39							SC	Clayey SAND with Gravel
GB23	5			0.003										SC	Clayey SAND with Gravel
GB23	10				31	23	46	35	13					GC	Clayey GRAVEL with Sand
GB23	25						52	35	12					CL	Sandy LEAN CLAY
GB23	34									9,273	1,335,312			BR	Sandy Limestone
GB24	5				28	11	61	42	21					CL	Gravelly LEAN CLAY
GB24	20				46	39	15	26	6					GC-GM	Silty Clayey GRAVEL with Sand
GB24	32.5									5,326	766,944			BR	Sandy Limestone
GB25	5				2	17	81	37	16					CL	LEAN CLAY with Sand
GB26	2.5			0.003										CL	LEAN CLAY with Sand
GB26	5	10.5	125.6				82	29	14	76	10,978			CL	LEAN CLAY with Sand
GB27	5						59							CL	Sandy LEAN CLAY with Gravel
GB28	2.5				39	35	26	NV	NP					GM	Silty GRAVEL with Sand
GB29	2.5			0.002										CL	LEAN CLAY with Gravel
Tank Farm	5				20	34	46	25	3					SM	Silty SAND with Gravel
Tank Farm	10			0.004										CL	Sandy LEAN CLAY with Gravel
Ballfield	2.5			0.008										CL	Gravelly LEAN CLAY
Ballfield	5						58	39	23					CL	Gravelly LEAN CLAY
Ballfield	12									5,892	848,448			BR	Sandy Limestone
Ballfield	14									3,249	467,856			BR	Sandy Limestone
NR Lodge	2.5			0.006										SP-SC	Poorly Graded SAND with Clay and Gravel
NR Lodge	10						24							SC	Clayey SAND with Gravel

Notes:

1. Where Atterberg Limits and Fines Content testing was not performed, USCS classifications visually/manually determined in the field during the subsurface investigation
2. USCS Symbol column is provided for soil. "BR" indicates bedrock.

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○							50	19	31
□							47	20	27
△									

SIEVE inches size	PERCENT FINER ○	□	△
GRAIN SIZE			
D_{60}			
D_{30}			
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#200	36.8	57.4	19.1

Material Description

○

□

△

REMARKS:

○ Natural Moisture Content: 16.9%
Dry Density: 127.2 pcf

□

△ Natural Moisture Content: 6.6%

○ Location: GB1 C2 1 Depth: 10 Sample Number: 11463
□ Location: GB1 S3 Depth: 20 Sample Number: 11464
△ Location: GB2 C1 Depth: 5 Sample Number: 11465

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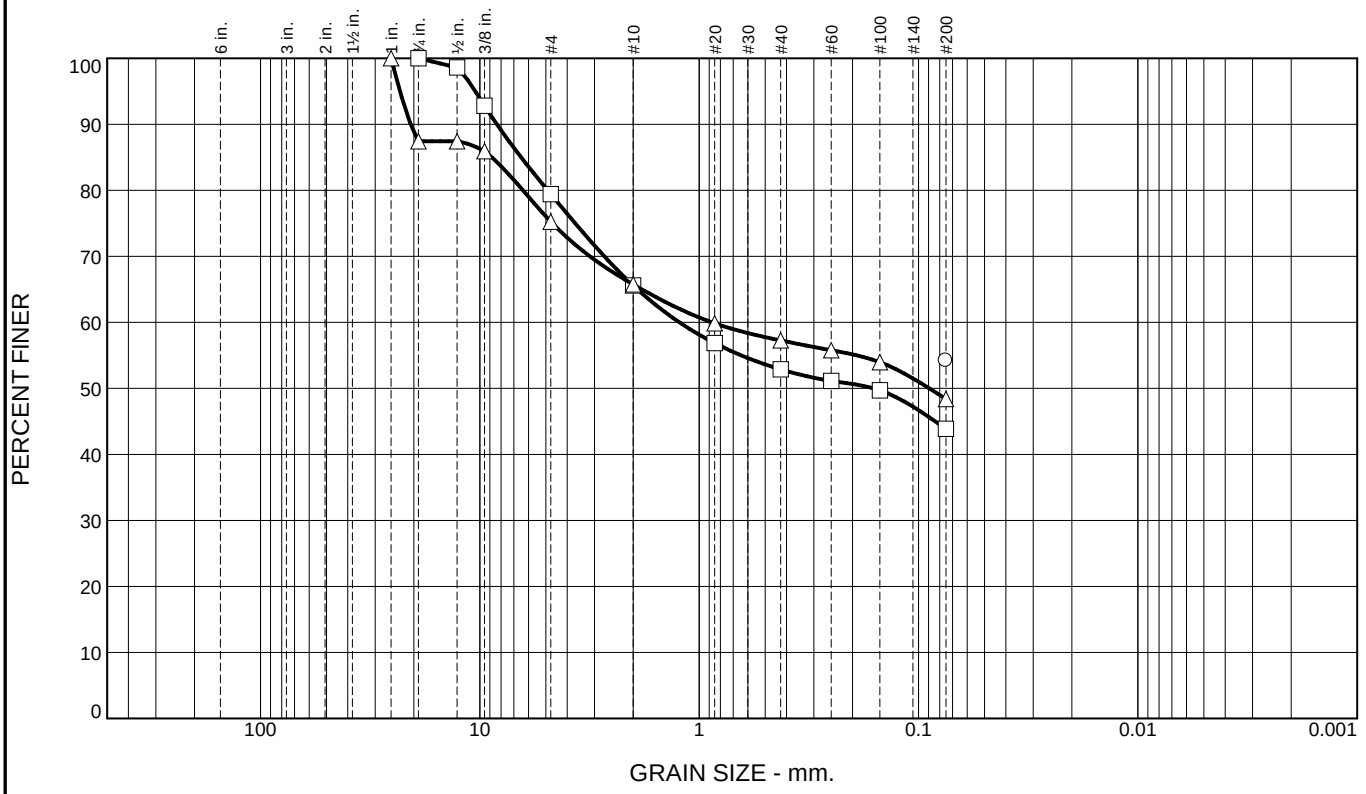
Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186

Project No.: 22-0232

Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○							42	18	24
□	0.0	20.6	35.5	43.9					
△	0.0	24.7	26.9	48.4		SC	32	23	9

SIEVE inches size	PERCENT FINER ○	□	△
1"			100.0
3/4"		100.0	87.4
1/2"		98.6	87.4
3/8"		92.8	85.9
GRAIN SIZE			
D_{60}		1.2294	0.8739
D_{30}			
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4		79.4	75.3
#10		65.6	65.8
#20		56.9	59.9
#40		52.9	57.3
#60		51.1	55.8
#100		49.7	54.0
#200	54.2	43.9	48.4

Material Description

○

□

△clayey sand with gravel

REMARKS:

○ Natural Moisture Content: 18.8%
Dry Density: 101.4 pcf

□

△ Natural Moisture Content: 13.2%
Dry Density: 97.3 pcf

○ Location: GB2 C4 Depth: 35 Sample Number: 11469
□ Location: GB3 S3 Depth: 10 Sample Number: 11470
△ Location: GB4 C1 Depth: 10 Sample Number: 11471

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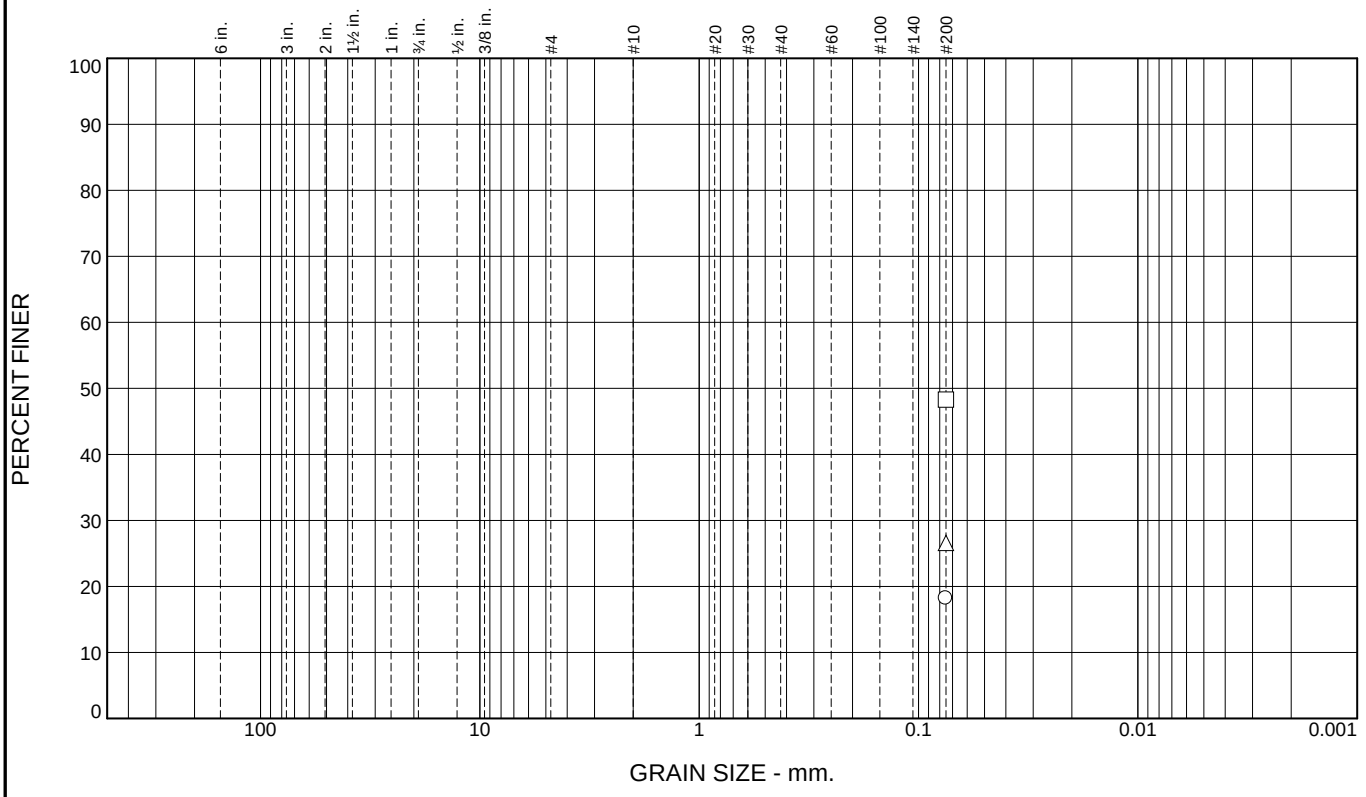
Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186

Project No.: 22-0232

Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○									
□							40	19	21
△									

SIEVE inches size	PERCENT FINER ○	□	△
	GRAIN SIZE		
D_{60}			
D_{30}			
D_{10}			
	COEFFICIENTS		
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#200	18.2	48.3	26.6

Material Description
○
□
△

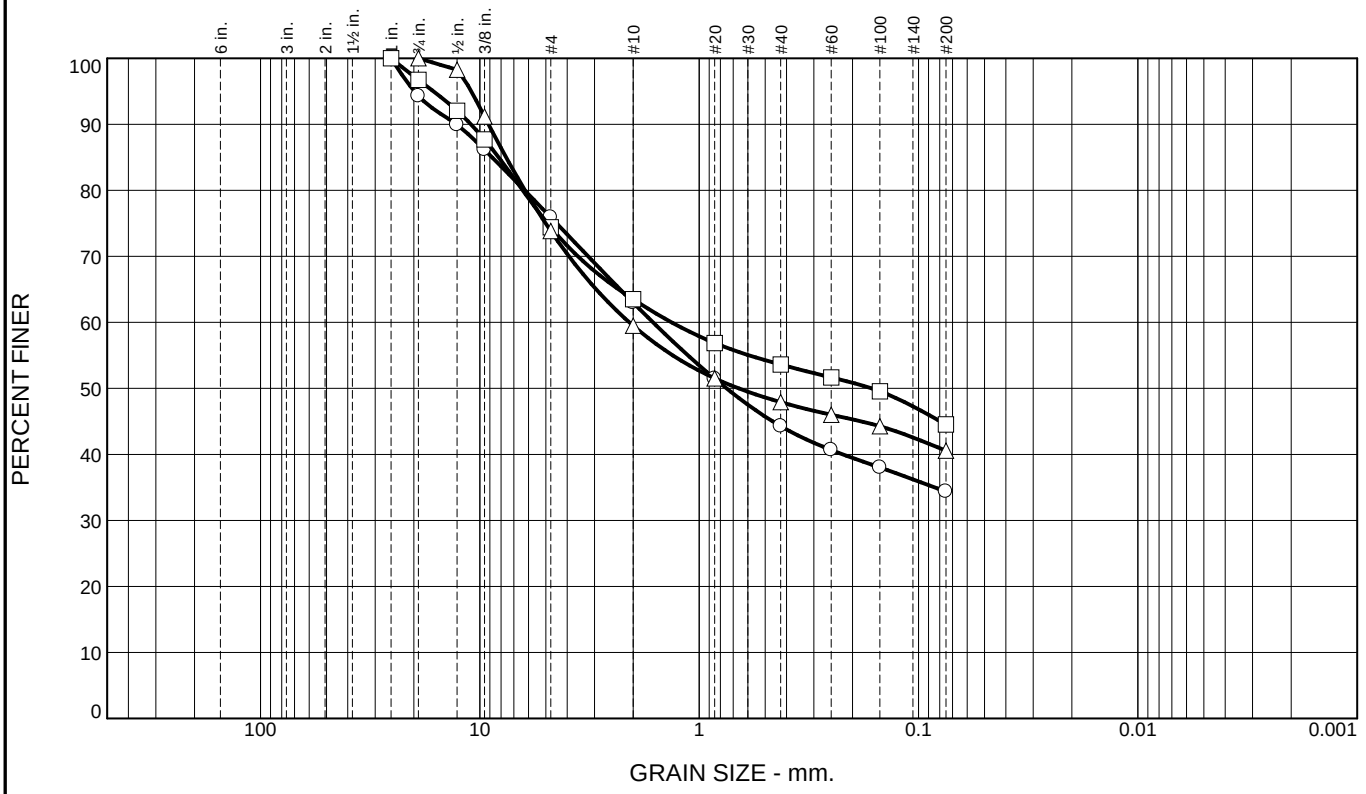
REMARKS:
○
□
△

○ Location: GB4 S3 Depth: 15 Sample Number: 11472
□ Location: GB4 S6 Depth: 30 Sample Number: 11473
△ Location: GB5 S1 Depth: 2.5 Sample Number: 11474

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Client: Lithos Engineering
Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186
Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0.0	24.1	41.5	34.4		SM	33	26	7
□	0.0	25.6	29.9	44.5					
△	0.0	26.2	33.2	40.6					

SIEVE inches size	PERCENT FINER ○	□	△
1"	100.0	100.0	
3/4"	94.3	96.7	100.0
1/2"	89.9	92.0	98.3
3/8"	86.2	87.7	91.2
GRAIN SIZE			
D_{60}	1.6237	1.3346	2.0812
D_{30}			
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	75.9	74.4	73.8
#10	63.0	63.5	59.5
#20	51.5	56.9	51.5
#40	44.3	53.6	47.9
#60	40.7	51.7	46.0
#100	38.0	49.6	44.3
#200	34.4	44.5	40.6

Material Description
○ silty sand with gravel
□
△

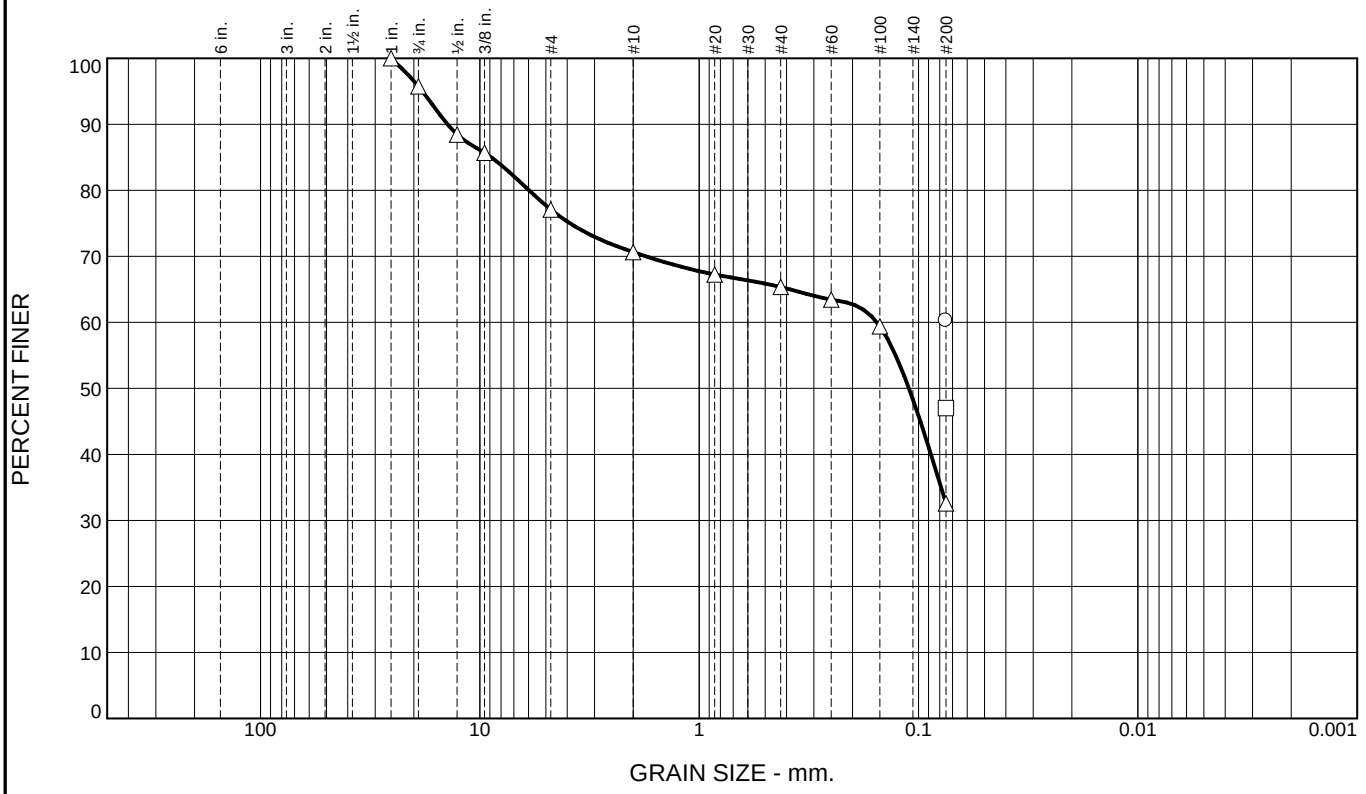
REMARKS:
○
□
△

○ Location: GB6 S3 Depth: 10 Sample Number: 11475
□ Location: GB7 S2 Depth: 5 Sample Number: 11476
△ Location: GB8 S2 Depth: 5 Sample Number: 11477

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Client: Lithos Engineering
Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186
Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○							46	19	27
□							51	17	34
△	0.0	22.9	44.5	32.6		SM	NV	NP	NP

SIEVE inches size	PERCENT FINER ○	□	△
1"			100.0
3/4"			95.7
1/2"			88.4
3/8"			85.7
GRAIN SIZE			
D_{60}			0.1547
D_{30}			
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4			77.1
#10			70.6
#20			67.2
#40			65.4
#60			63.4
#100			59.4
#200	60.3	47.0	32.6

Material Description

○

□

△silty sand with gravel

REMARKS:

○

□

△

○Location: GB9 S3 Depth: 10 Sample Number: 11478

□Location: GB10 C1 Depth: 15 Sample Number: 11479

△Location: GB11 S1 Depth: 2.5 Sample Number: 11480

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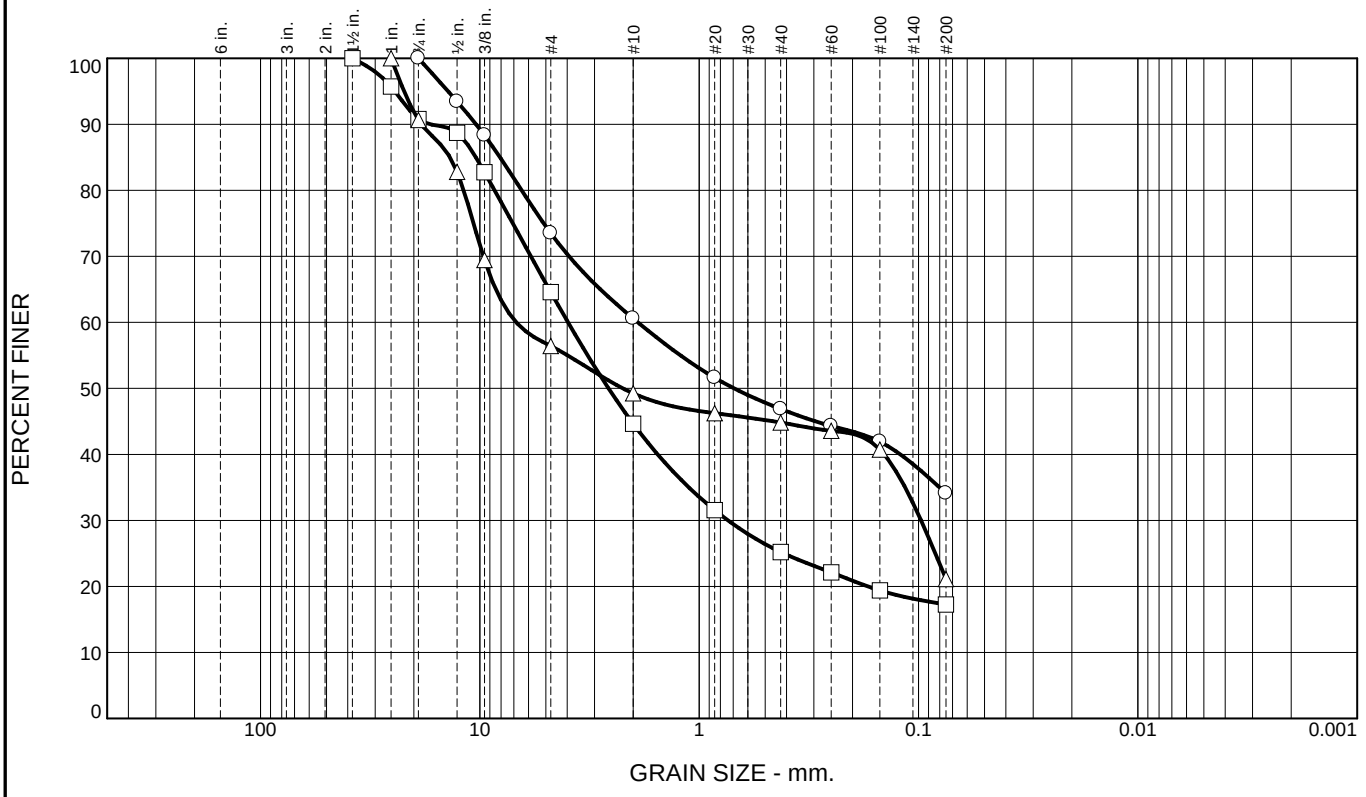
Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186

Project No.: 22-0232

Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0.0	26.5	39.4	34.1		SM	NV	NP	NP
□	0.0	35.4	47.3	17.3					
△	0.0	43.6	35.3	21.1		GM	NV	NP	NP

SIEVE inches size	PERCENT FINER ○	□	△
1 1/2"		100.0	
1"		95.7	100.0
3/4"	100.0	90.8	90.7
1/2"	93.4	88.7	82.8
3/8"	88.3	82.7	69.4
	GRAIN SIZE		
D_{60}	1.9014	3.9630	6.8244
D_{30}		0.7386	0.0974
D_{10}			
	COEFFICIENTS		
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	73.5	64.6	56.4
#10	60.6	44.7	49.2
#20	51.6	31.6	46.2
#40	46.9	25.2	44.8
#60	44.3	22.1	43.6
#100	41.9	19.4	40.7
#200	34.1	17.3	21.1

Material Description
○ silty sand with gravel
□
△ silty gravel with sand

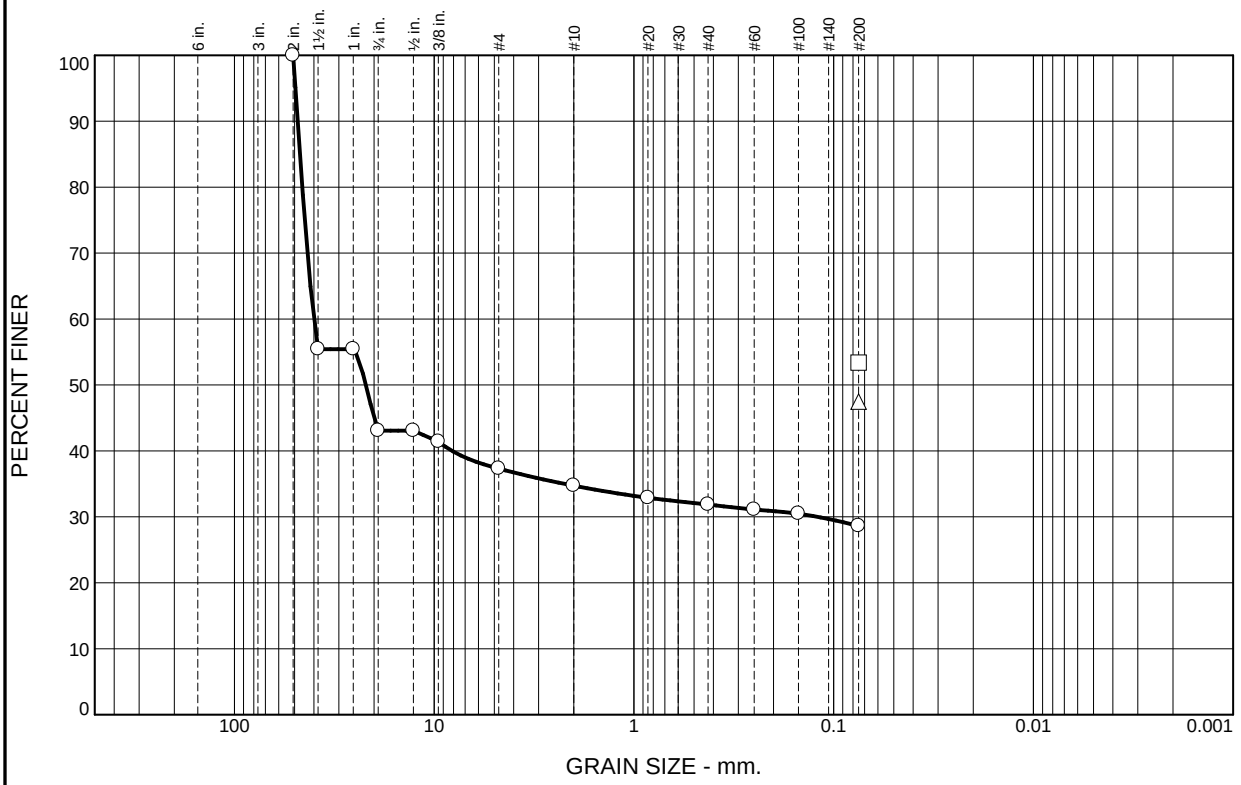
REMARKS:
○
□
△

○ Location: GB12 S3 Depth: 10 Sample Number: 11481
□ Location: GB13 S4 Depth: 15 Sample Number: 11482
△ Location: GB14 C1 Depth: 5 Sample Number: 11483

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Client: Lithos Engineering
Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186
Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	63	8	29		GC	47	21	26
□									
△							61	22	39

SIEVE inches size	PERCENT FINER ○	□	△
2"	100		
1 1/2"	55		
1"	55		
3/4"	43		
1/2"	43		
3/8"	41		
GRAIN SIZE			
D_{60}	39.9705		
D_{30}	0.1201		
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	37		
#10	35		
#20	33		
#40	32		
#60	31		
#100	30		
#200	29	53	47

Material Description

○ USCS: clayey gravel

□

△

REMARKS:

○

□ Dry Density: Not Tested Sample Crumbled
Moisture Content: 18.2%

△

○ Location: GB-15; C1 Depth: 5 Sample Number: 12603
□ Location: GB-15; C3 Depth: 15 Sample Number: 12575
△ Location: GB-16; S3 Depth: 15 Sample Number: 12576

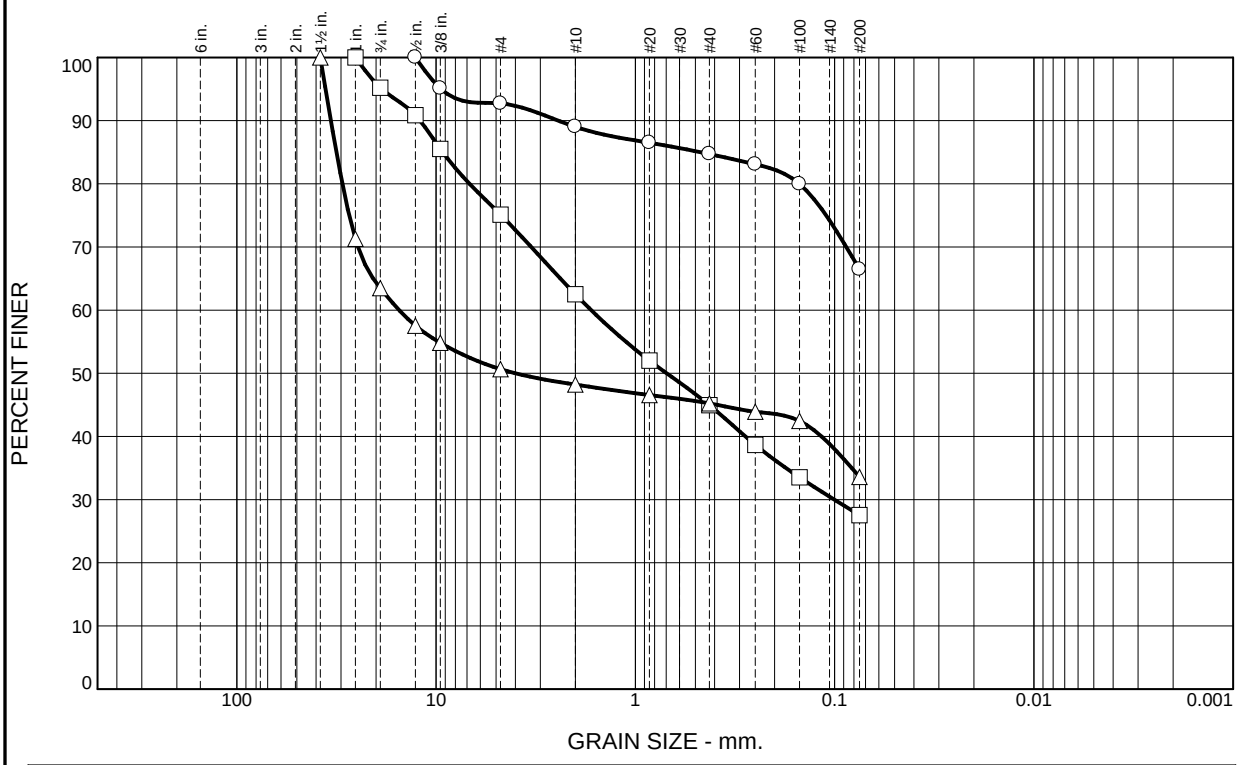
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Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	7	27	66		CL	48	20	28
□	0	25	47	28		SC	27	18	9
△	0	49	17	34		GC	39	19	20

SIEVE inches size	PERCENT FINER ○	□	△
1 1/2"			100
1"		100	71
3/4"		95	63
1/2"	100	91	58
3/8"	95	86	55
GRAIN SIZE			
D_{60}		1.6657	15.3180
D_{30}		0.1003	
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	93	75	51
#10	89	63	48
#20	86	52	47
#40	85	45	45
#60	83	39	44
#100	80	34	42
#200	66	28	34

Material Description

○ USCS: sandy lean clay

□ USCS: clayey sand with gravel

△ USCS: clayey gravel with sand

REMARKS:

○

□

△

○ Location: GB-16; C4 Depth: 20 Sample Number: 12577

□ Location: GB-17; B1 Depth: 2-3 Sample Number: 12578

△ Location: GB-17; C2 Depth: 5 Sample Number: 12579

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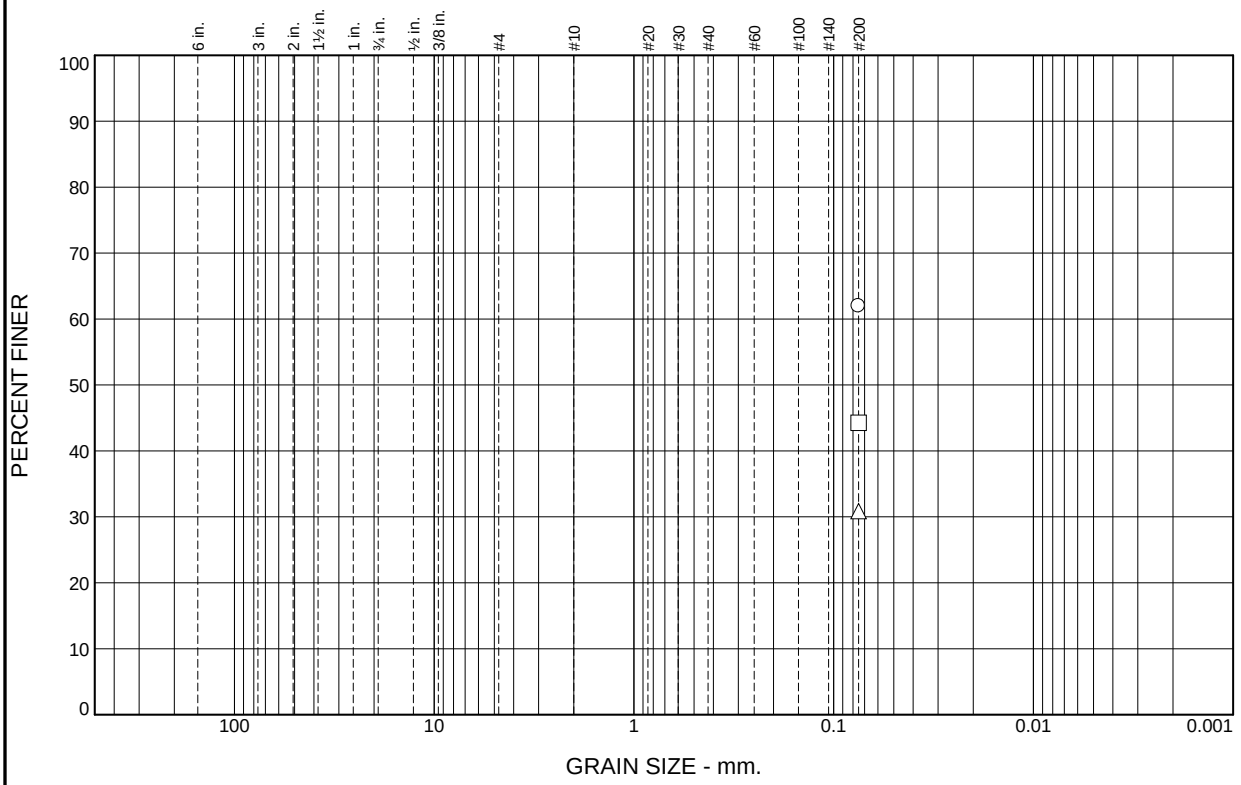
Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Tested By: A. Dudney **Checked By:** K. Runner

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○							47	20	27
□							55	20	35
△							58	21	37

SIEVE inches size	PERCENT FINER ○	□	△
	GRAIN SIZE		
D_{60}			
D_{30}			
D_{10}			
	COEFFICIENTS		
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#200	62	44	31

Material Description

○

□

△

REMARKS:

○

□

△

○Location: GB-17; C4 Depth: 15 Sample Number: 12580

□Location: GB-18; S2 Depth: 5 Sample Number: 12581

△Location: GB-18; C3 Depth: 10 Sample Number: 12582

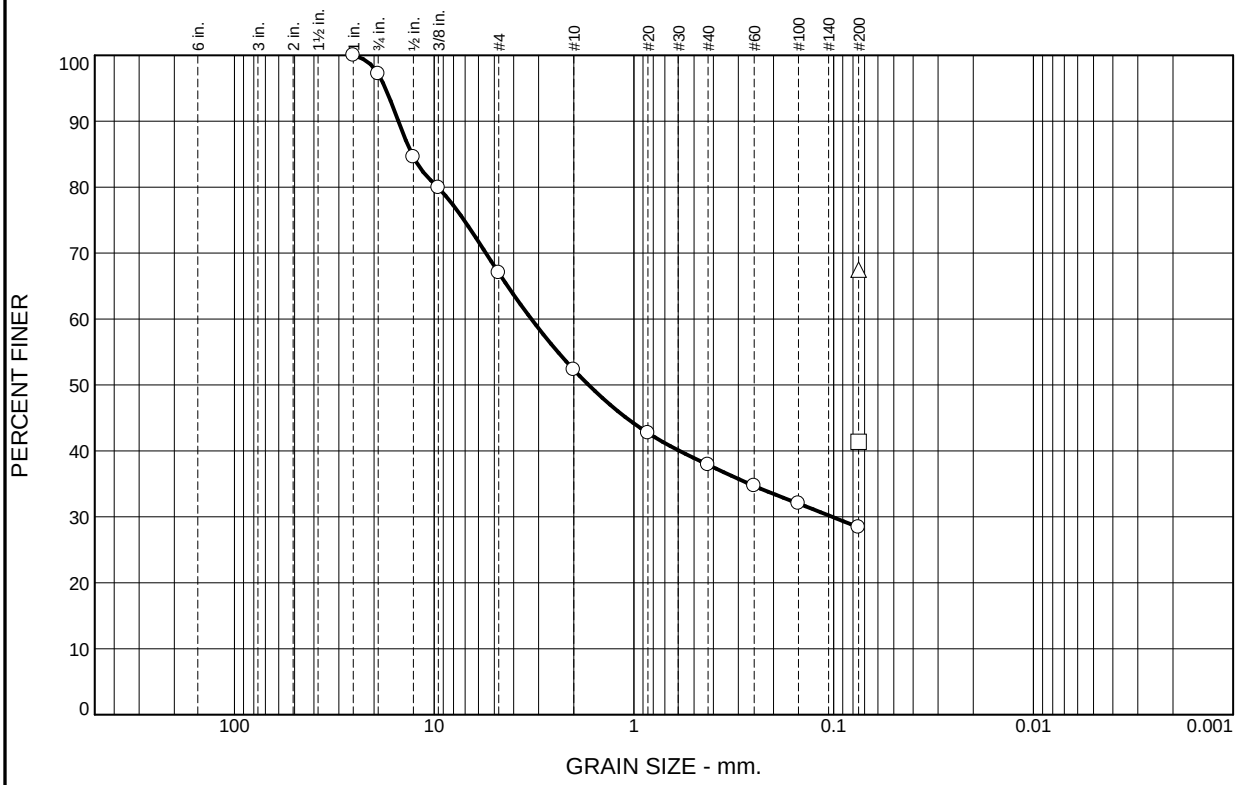
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Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	33	39	28		SM	NV	NP	NP
□									
△							31	23	8

SIEVE inches size	PERCENT FINER ○	□	△
1"	100		
3/4"	97		
1/2"	85		
3/8"	80		
GRAIN SIZE			
D_{60}	3.2593		
D_{30}	0.1015		
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	67		
#10	52		
#20	43		
#40	38		
#60	35		
#100	32		
#200	28	41	67

Material Description

○ USCS: silty sand with gravel

□

△

REMARKS:

○

□

△

○ Location: GB-19; S1 Depth: 5 Sample Number: 12583

□ Location: GB-19; C2 Depth: 10 Sample Number: 12584

△ Location: GB-19; C4 Depth: 20 Sample Number: 12585

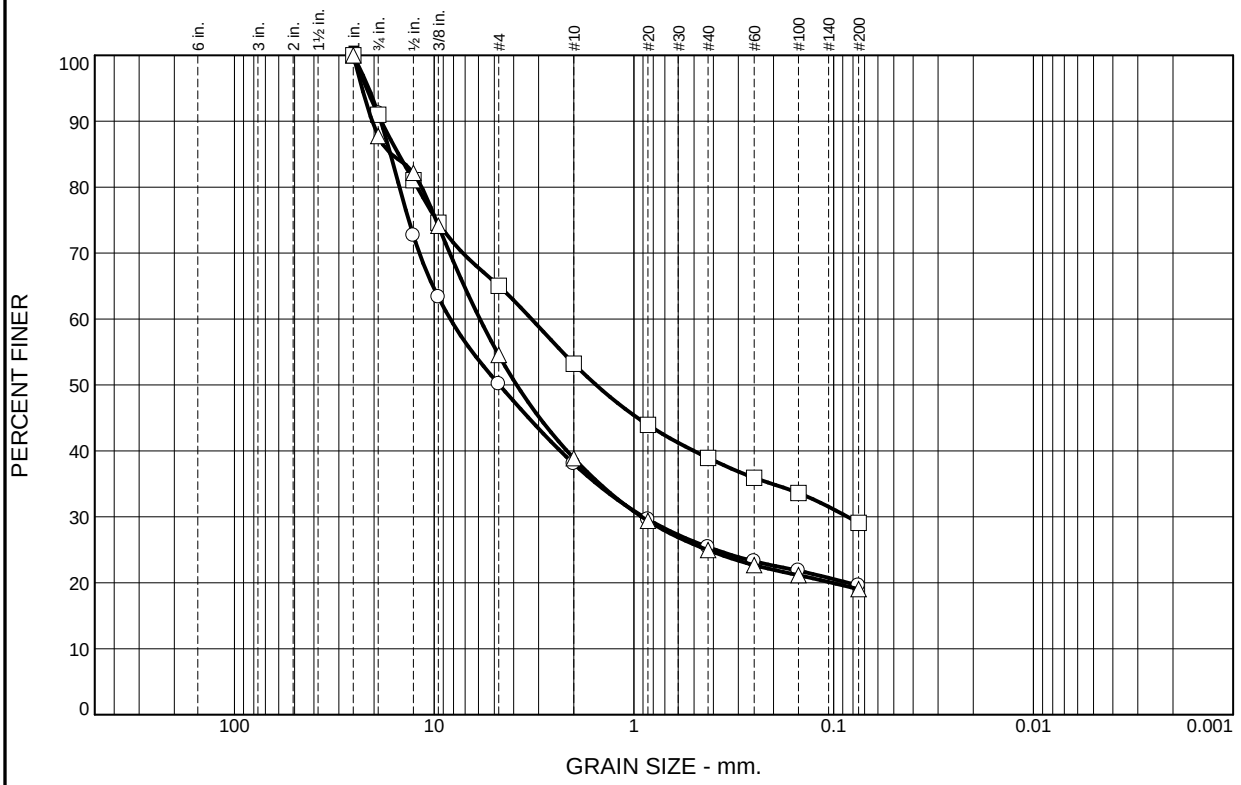
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Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	50	30	20		GM	NV	NP	NP
□	0	35	36	29					
△	0	45	36	19					

SIEVE inches size	PERCENT FINER ○	□	△
1"	100	100	100
3/4"	91	91	88
1/2"	73	81	82
3/8"	63	75	74
GRAIN SIZE			
D_{60}	8.2983	3.2508	5.9002
D_{30}	0.8940	0.0851	0.9132
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	50	65	55
#10	38	53	39
#20	30	44	29
#40	25	39	25
#60	23	36	23
#100	22	34	21
#200	20	29	19

Material Description

○ USCS: silty gravel with sand

□

△

REMARKS:

○

□

△ Dry Density: Not tested sample crumbled
Moiture Content: 5.8%

○ Location: GB-20; S2 Depth: 10 Sample Number: 12587
□ Location: GB-21; S1 Depth: 5 Sample Number: 12588
△ Location: GB-21; C3 Depth: 15 Sample Number: 12589

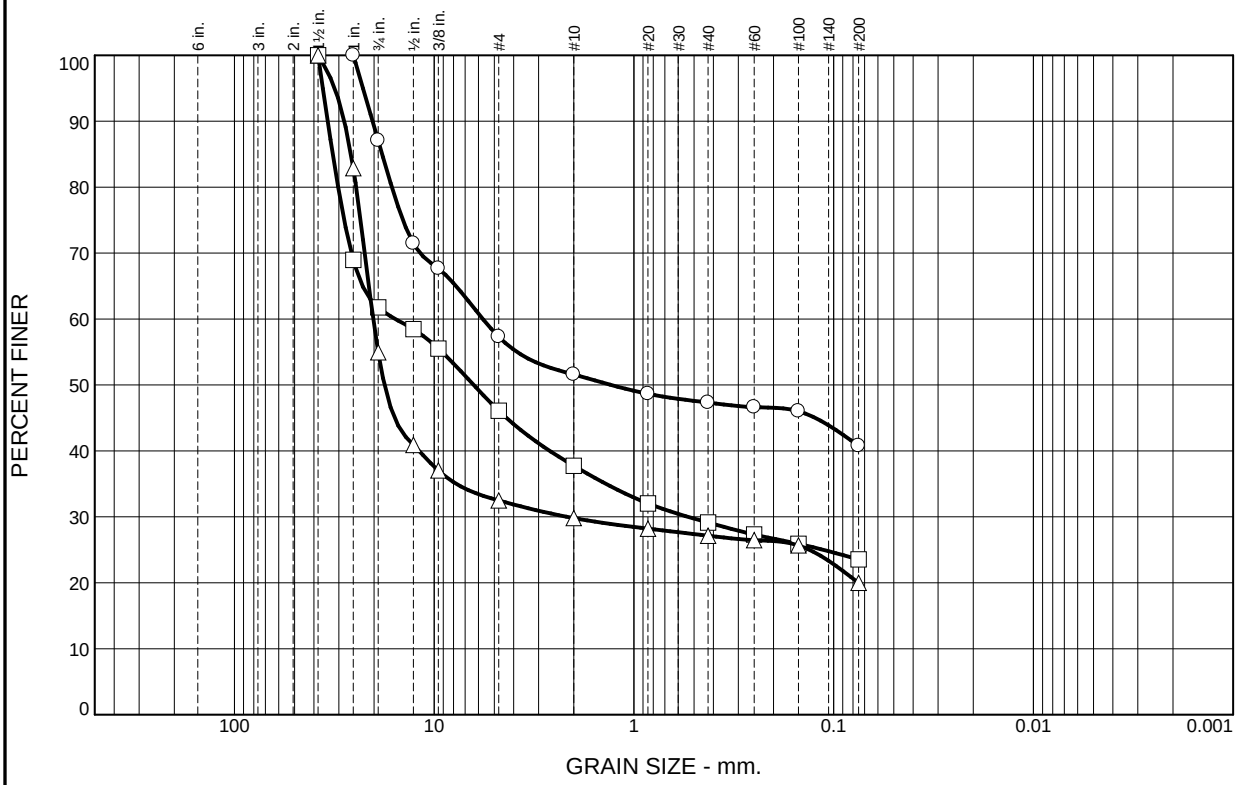
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Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	43	16	41		GM	30	27	3
□	0	54	22	24		GM	NV	NP	NP
△	0	68	12	20		GM	NV	NP	NP

SIEVE inches size	PERCENT FINER ○	□	△
1 1/2"		100	100
1"	100	69	83
3/4"	87	62	55
1/2"	71	58	41
3/8"	68	56	37
GRAIN SIZE			
D_{60}	5.6988	15.6413	20.1766
D_{30}		0.5359	2.1567
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	57	46	32
#10	52	38	30
#20	49	32	28
#40	47	29	27
#60	47	27	26
#100	46	26	26
#200	41	24	20

Material Description

○ USCS: silty gravel with sand

□ USCS: silty gravel with sand

△ USCS: silty gravel

REMARKS:

○ Dry Density: Not tested sample crumbled
Moisture Content: Not tested

□

△

○ Location: GB-21; C5 Depth: 25 Sample Number: 12590

□ Location: GB-22; S-1 Depth: 5 Sample Number: 12592

△ Location: GB-22; S5 Depth: 25 Sample Number: 12593

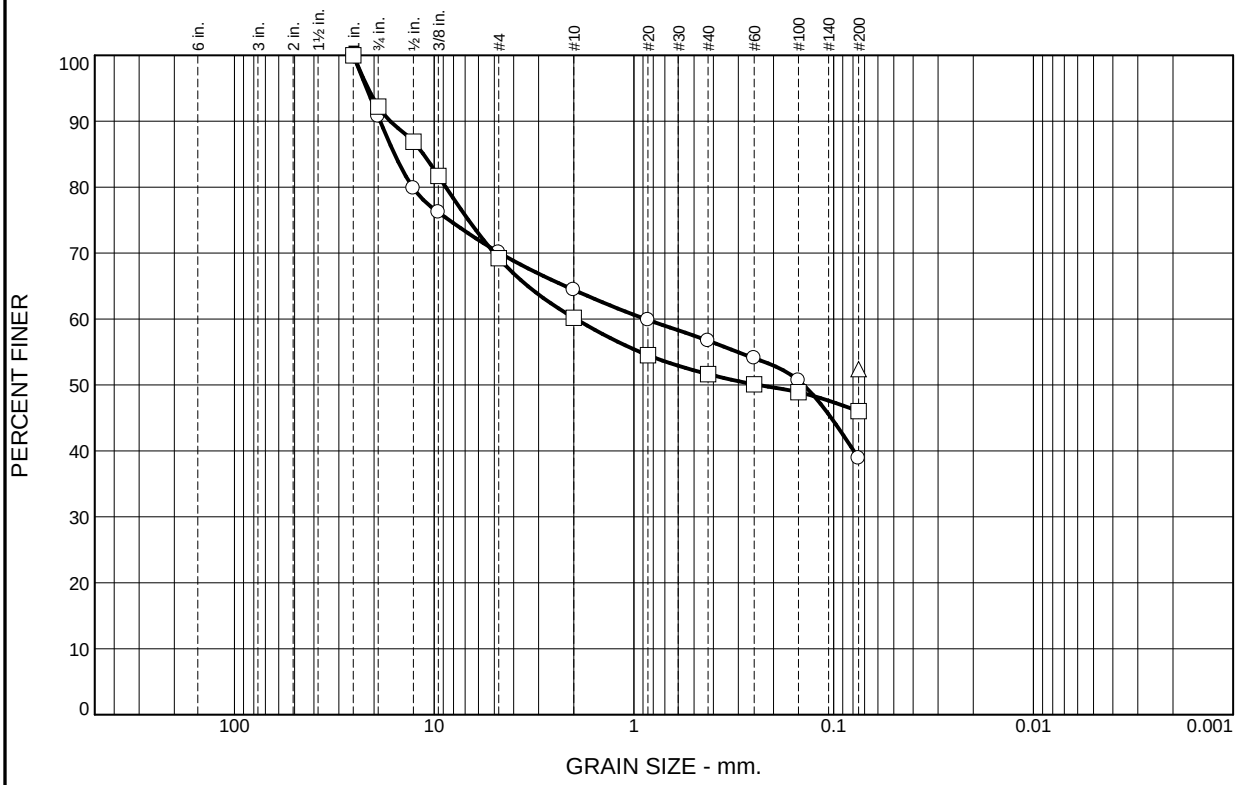
Martinez Associates

Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	30	31	39					
□	0	31	23	46		GC	35	22	13
△							35	23	12

SIEVE inches size	PERCENT FINER ○	□	△
1"	100	100	
3/4"	91	92	
1/2"	80	87	
3/8"	76	82	
	GRAIN SIZE		
D_{60}	0.8762	1.9505	
D_{30}			
D_{10}			
	COEFFICIENTS		
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	70	69	
#10	64	60	
#20	60	55	
#40	57	52	
#60	54	50	
#100	51	49	
#200	39	46	52

Material Description

○

□USCS: clayey gravel with sand

△

REMARKS:

○

□

△

○Location: GB-22; S8 Depth: 40 Sample Number: 12594

□Location: GB-23; C2 Depth: 10 Sample Number: 12595

△Location: GB-23; S5 Depth: 25 Sample Number: 12596

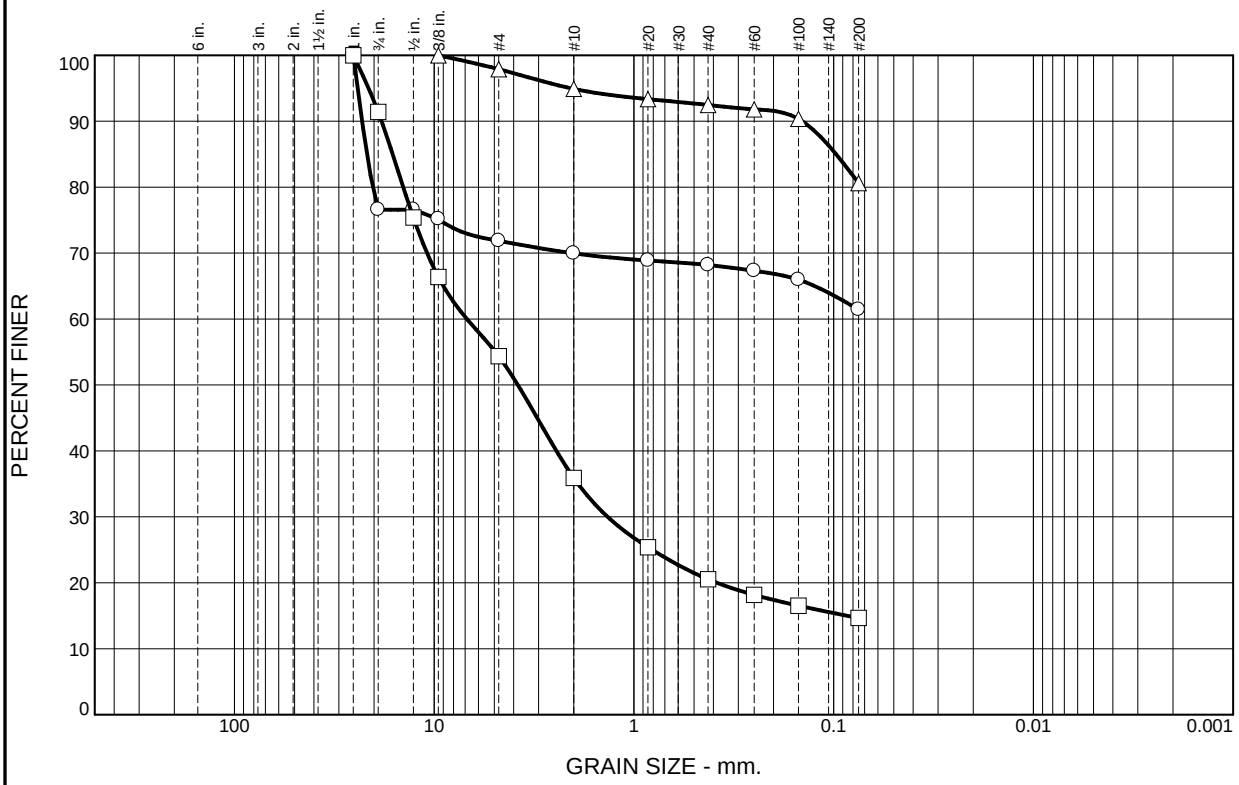
Martinez Associates

Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0	28	11	61		CL	42	21	21
□	0	46	39	15		GC-GM	26	20	6
△	0	2	17	81		CL	37	21	16

SIEVE inches size	PERCENT FINER ○	□	△
1"	100	100	
3/4"	77	91	
1/2"	77	75	
3/8"	75	66	100
GRAIN SIZE			
D_{60}		6.8488	
D_{30}		1.3550	
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	72	54	98
#10	70	36	95
#20	69	25	93
#40	68	21	92
#60	67	18	92
#100	66	17	90
#200	61	15	81

Material Description

○USCS: gravelly lean clay

□USCS: silty clayey gravel with sand

△USCS: lean clay with sand

REMARKS:

○

□

△

○Location: GB-24; C1 Depth: 5 Sample Number: 12597

□Location: GB-24; S4 Depth: 20 Sample Number: 12598

△Location: GB-25; S2 Depth: 5 Sample Number: 12599

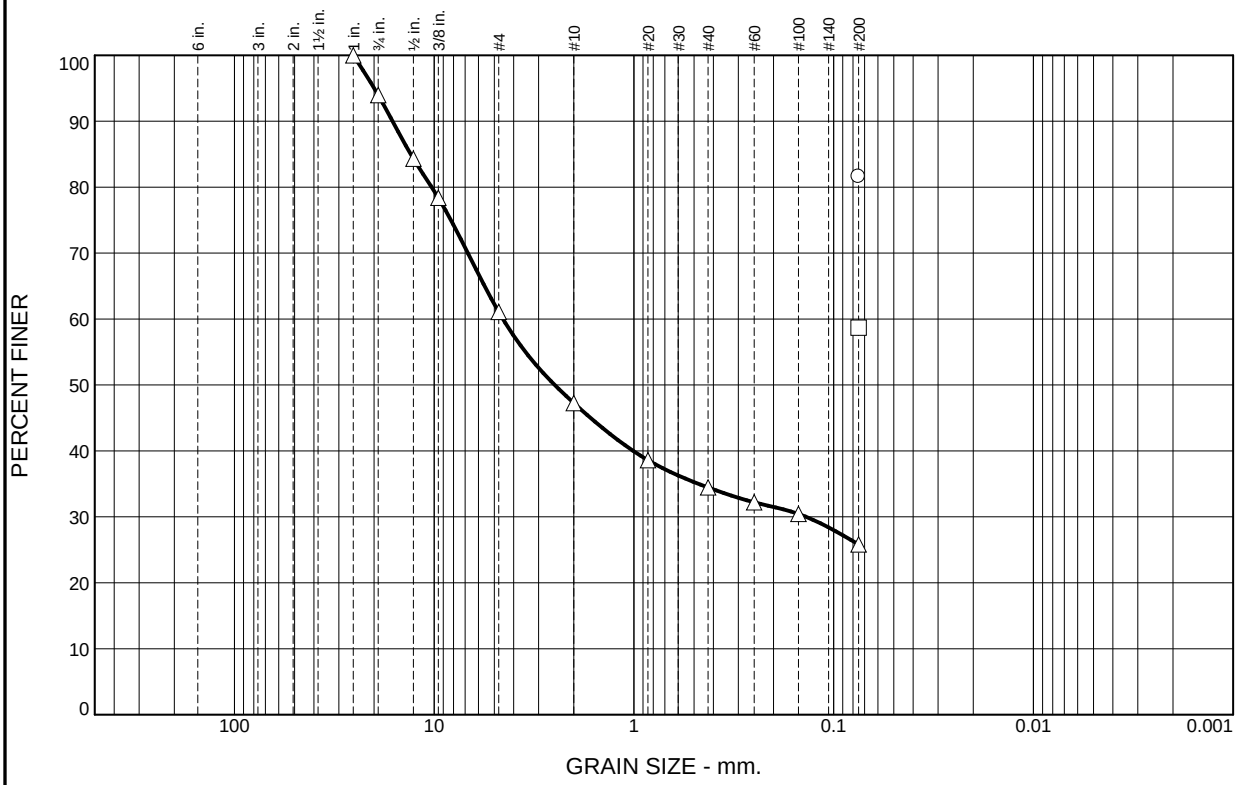
Martinez Associates

Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○							29	15	14
□									
△	0	39	35	26		GM	NV	NP	NP

SIEVE inches size	PERCENT FINER ○	□	△
1"			100
3/4"			94
1/2"			84
3/8"			78
GRAIN SIZE			
D_{60}			4.5242
D_{30}			0.1373
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4			61
#10			47
#20			39
#40			34
#60			32
#100			30
#200	82	59	26

Material Description

○

□

△USCS: silty gravel with sand

REMARKS:

○

□

△

○Location: GB-26; C1 Depth: 5 Sample Number: 12600

□Location: GB-27; S2 Depth: 5 Sample Number: 12601

△Location: GB-28; S1 Depth: 2.5 Sample Number: 12602

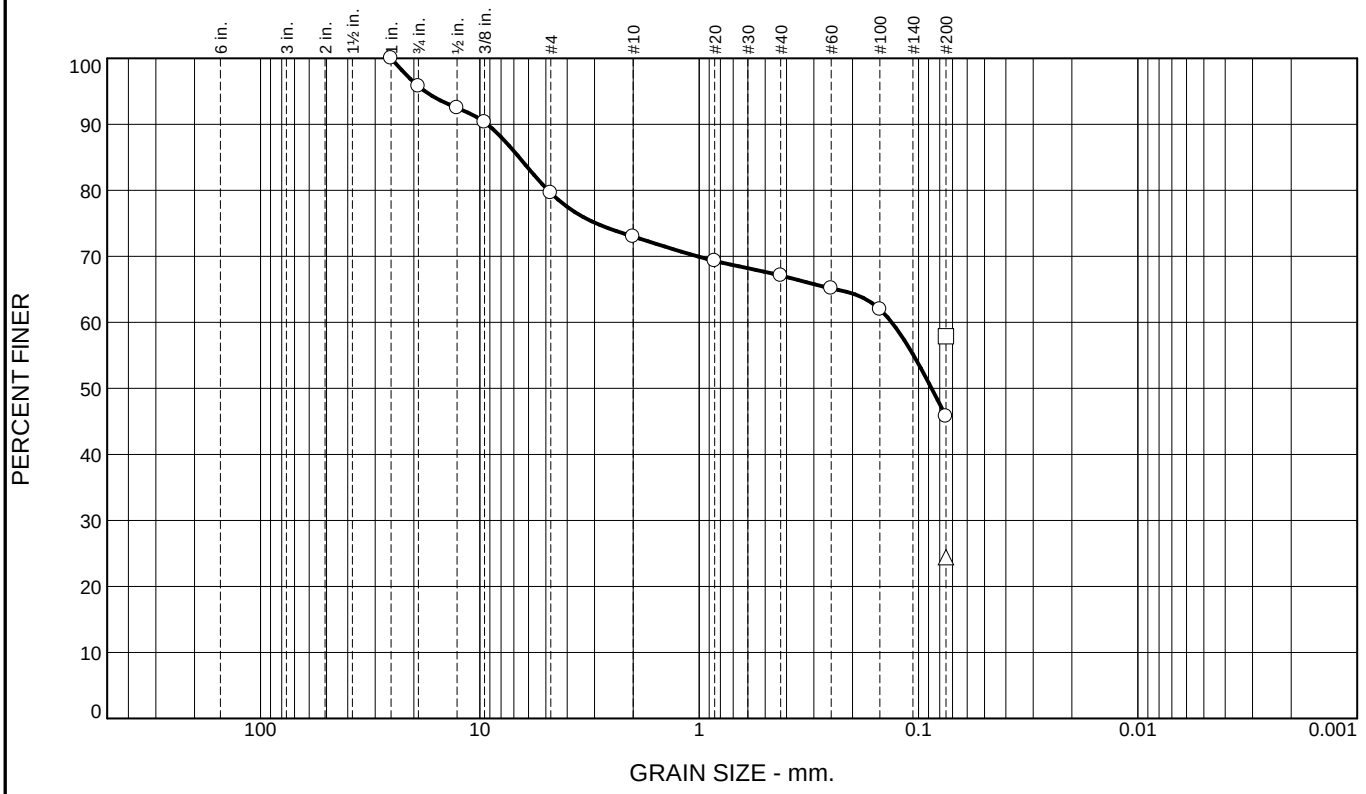
Martinez Associates

Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 Figure

Particle Size Distribution Report



	+3"	% GRAVEL	% SAND	% SILT	% CLAY	USCS	LL	PL	PI
○	0.0	20.4	33.8	45.8		SM	25	22	3
□							39	16	23
△									

SIEVE inches size	PERCENT FINER ○	□	△
1"	100.0		
3/4"	95.8		
1/2"	92.5		
3/8"	90.3		
GRAIN SIZE			
D_{60}	0.1324		
D_{30}			
D_{10}			
COEFFICIENTS			
C_c			
C_u			

SIEVE number size	PERCENT FINER ○	□	△
#4	79.6		
#10	73.0		
#20	69.3		
#40	67.1		
#60	65.2		
#100	62.0		
#200	45.8	57.9	24.4

Material Description

○ silty sand with gravel

□

△

REMARKS:

○

□

△

○ Location: Tank Farm BP S1 Depth: 5 Sample Number: 11484

□ Location: Ballfield BP S2 Depth: 5 Sample Number: 11485

△ Location: NR Lodge BP S3 Depth: 10 Sample Number: 11486

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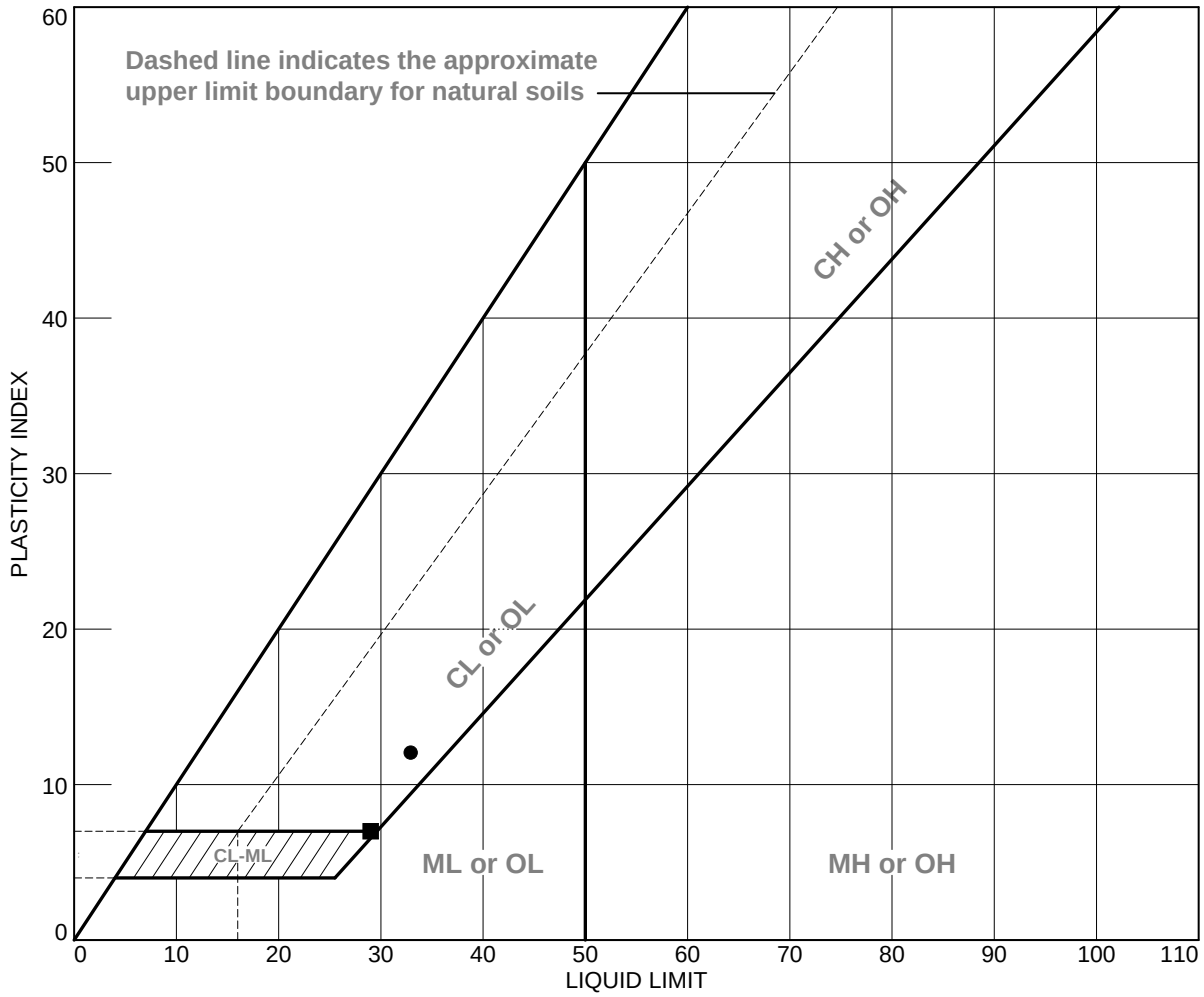
Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement
Lithos Project # 21186

Project No.: 22-0232

Figure

LIQUID AND PLASTIC LIMITS TEST REPORT



SOIL DATA								
SYMBOL	SOURCE	SAMPLE NO.	DEPTH	NATURAL WATER CONTENT (%)	PLASTIC LIMIT (%)	LIQUID LIMIT (%)	PLASTICITY INDEX (%)	USCS
●		12586-GB-19;S5	25		21	33	12	
■		12591-GB 21:S6	30	14.7	22	29	7	

Martinez Associates

Client: Lithos Engineering

Project: GRCA North Rim Water Supply Line Replacement (LE 21186)

Project No.: 22-0232 **Figure**

MOISTURE CONTENT & DENSITY DETERMINATION

PROJECT: GRCA North Rim Water Supply Line Replacement JOB NO. 22-0232

TESTED BY RSK DATE 12/6/2022 CLIENT JOB NO.: 21186

LAB NO.	11463	11469	11471			
Boring No.	GB1	GB2	GB4			
Depth, ft.	10.0	35.0	10.0			
Diameter, in.	1.928	1.880	1.930			
Length, in.	0.750	3.720	4.050			
Wet Soil + Tare, gm.	310.6	334.5	350.7			
Tare, gm.	237.4	7.5	7.8			
Wet Soil, gm.	73.2	327.0	342.9			
Volumn, cu. ft.	0.001267	0.005976	0.006857			
Wet Density, pcf	***127.2***	***120.5***	***110.2***			
Cup No.	34	8	52			
Cup Weight, gm	7.9	7.5	7.8			
Wet Soil + Cup, gm.	256.8	343.5	285.4			
Dry Soil + Cup, gm.	220.8	290.3	253.1			
Water, gm.	36.0	53.2	32.3			
Dry Soil, gm.	212.9	282.8	245.3			
Moisture Content, %	***16.9***	***18.8***	***13.2***			
Dry Density, pcf	***108.8***	***101.4***	***97.3***			

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One Dimensional Swell/Consolidation (ASTM D 4546)

(Denver Area Swell/Consolidation Test)

Client Project No.:	21186	Proj. Name:	GRCA North Rim Water Supply	Sampled By:	NP
Martinez Job No.:	22-0232		Line Replacement	Sample Date:	NP
Sample ID:	11463			Reviewed By:	K. Runner
Sample Location:	GB1 C21 @ 10'				
Soil Description:					
USCS:					

Sample Data:

Ring No:	H	Dish No:	34
Ring Mass (g):	237.4	Dish Mass (g):	7.9
Sample Height (in):	0.75	Swell Machine #:	8

Pre-test Sample		Post-test Sample	
Ring + Sample (g):	310.6	Ring + Sample (g):	311.3
Dish wt:	7.9	Dish wt:	8.0
Wet wt (g):	256.8	Wet wt (g):	81.9
Dry wt (g):	220.8	Dry wt (g):	68.8

Results:

Pre-test Sample		Post-test Sample	
Moisture Content:	16.9%	Moisture Content:	21.6%
Wet Density (pcf):	126.3	Wet Density (pcf):	127.7
Dry Density (pcf):	108.0	Dry Density (pcf):	105.1

Swell/Consolidation

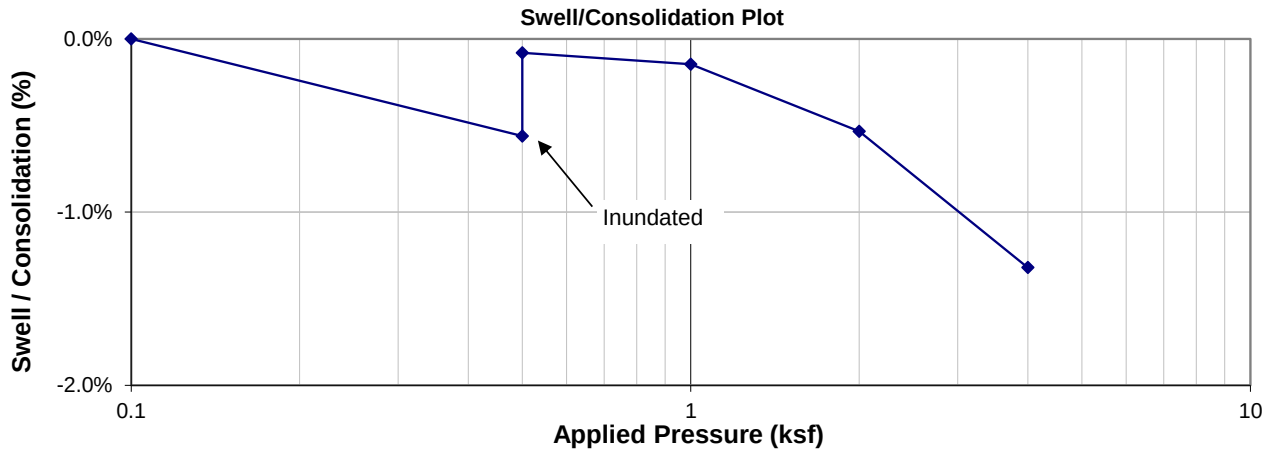
Load (ksf):	0.1	0.5	Add Water	0.5	1	2	4		
Correction (x 10-4):	0	12		12	27	46	68		
Dial Reading (x 10-4):	2757	2703		2739	2719	2671	2590		
Swell/Consolidation %:	**0.0%**	**-0.6%**		**-0.1%**	**-0.1%**	**-0.5%**	**-1.3%**		

Results:

Swell Upon Wetting @ 500 psf: **0.5%**

Swell Pressure (psf): **1,250**

Tested By: R. S. Kay
Checked By K. Runner



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One Dimensional Swell/Consolidation (ASTM D 4546)

(Denver Area Swell/Consolidation Test)

Client Project No.:	21186	Proj. Name:	GRCA North Rim Water Supply	Sampled By:	NP
Martinez Job No.:	22-0232		Line Replacement	Sample Date:	NP
Sample ID:	11466			Reviewed By:	K. Runner
Sample Location:	GB2 C2 @ 15'				
Soil Description:					
USCS:					

Sample Data:

Ring No:	F	Dish No:	NOT
Ring Mass (g):	237.3	Dish Mass (g):	8.1
Sample Height (in):	0.75	Swell Machine #:	6

Pre-test Sample		Post-test Sample	
Ring + Sample (g):	308.1	Ring + Sample (g):	309.0
Dish wt:	8.1	Dish wt:	8.1
Wet wt (g):	286.6	Wet wt (g):	70.7
Dry wt (g):	235.9	Dry wt (g):	56.8

Results:

Pre-test Sample		Post-test Sample	
Moisture Content:	22.3%	Moisture Content:	28.6%
Wet Density (pcf):	122.2	Wet Density (pcf):	125.5
Dry Density (pcf):	99.9	Dry Density (pcf):	97.6

Swell/Consolidation

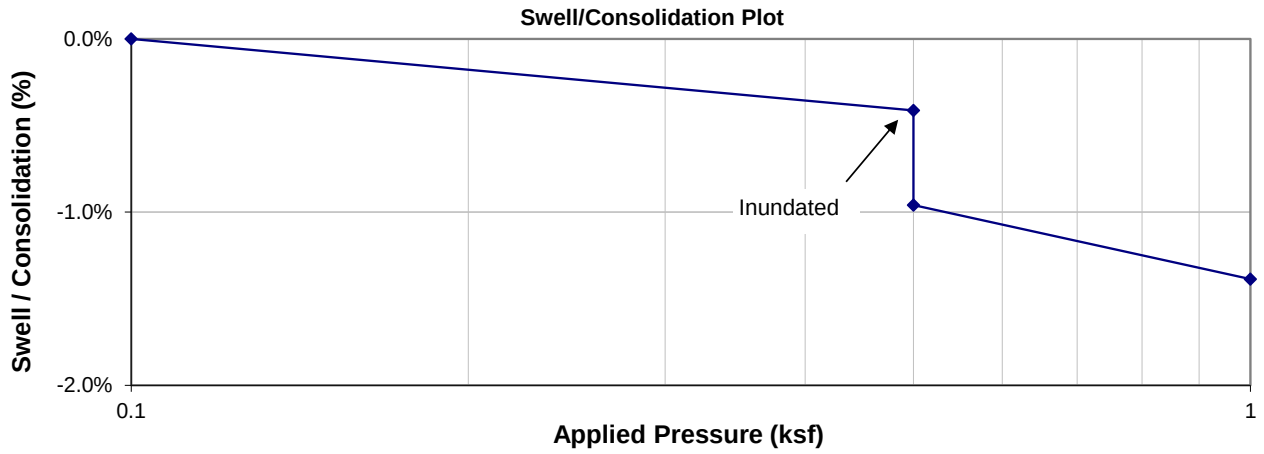
Load (ksf):	0.1	0.5	Add Water	0.5	1				
Correction (x 10-4):	0	18		18	40				
Dial Reading (x 10-4):	2339	2290		2249	2195				
Swell/Consolidation %:	**0.0%**	**-0.4%**		**-1.0%**	**-1.4%**				

Results:

Settlement Upon Wetting @ 500 psf: **-0.5%**

Swell Pressure (psf): **105**

Tested By: R. S. Kay
Checked By K. Runner



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One Dimensional Swell/Consolidation (ASTM D 4546)

(Denver Area Swell/Consolidation Test)

Client Project No.:	21186	Proj. Name:	GRCA North Rim Water Supply	Sampled By:	NP
Martinez Job No.:	22-0232		Line Replacement	Sample Date:	NP
Sample ID:	11467			Reviewed By:	K. Runner
Sample Location:	GB2 C3 #3 @ 20'				
Soil Description:					
USCS:					

Sample Data:

Ring No:	E	Dish No:	2
Ring Mass (g):	236.8	Dish Mass (g):	7.9
Sample Height (in):	0.75	Swell Machine #:	5

Pre-test Sample		Post-test Sample	
Ring + Sample (g):	307.0	Ring + Sample (g):	307.9
Dish wt:	7.9	Dish wt:	8.0
Wet wt (g):	289.6	Wet wt (g):	71.1
Dry wt (g):	244.8	Dry wt (g):	57.3

Results:

Pre-test Sample		Post-test Sample	
Moisture Content:	18.9%	Moisture Content:	28.1%
Wet Density (pcf):	121.1	Wet Density (pcf):	124.2
Dry Density (pcf):	101.9	Dry Density (pcf):	97.0

Swell/Consolidation

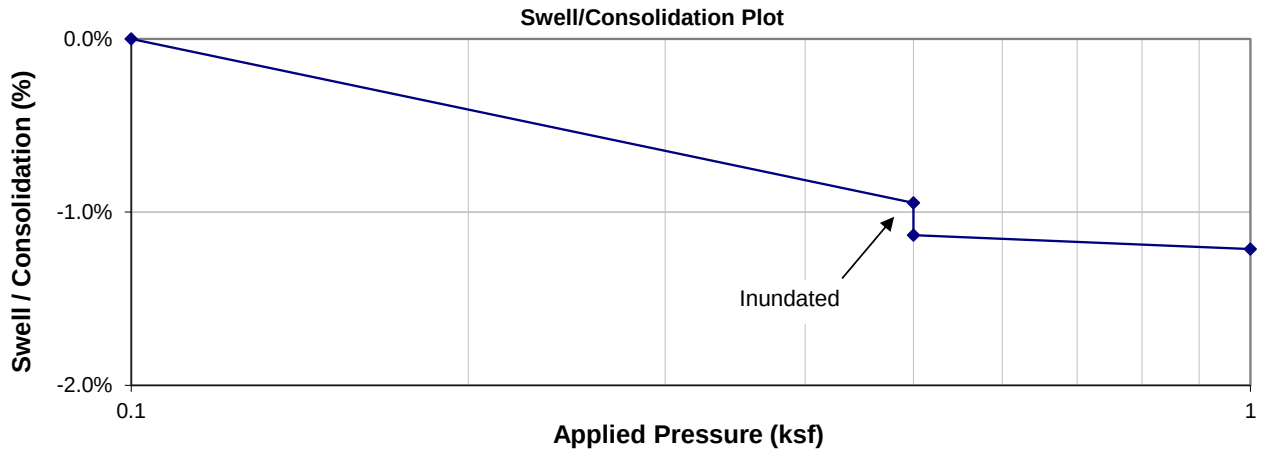
Load (ksf):	0.1	0.5	Add Water	0.5	1				
Correction (x 10-4):	0	14		14	32				
Dial Reading (x 10-4):	2860	2775		2761	2737				
Swell/Consolidation %:	**0.0%**	**-0.9%**		**-1.1%**	**-1.2%**				

Results:

Settlement Upon Wetting @ 500 psf: **-0.2%**

Swell Pressure (psf): **0**

Tested By: R. S. Kay
Checked By K. Runner



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Martinez Associates

One Dimensional Swell/Consolidation (ASTM D 4546)

(Denver Area Swell/Consolidation Test)

Client Project No.:	21186	Proj. Name:	GRCA North Rim Water Supply	Sampled By:	NP
Martinez Job No.:	22-0232		Line Replacement	Sample Date:	NP
Sample ID:	11471			Reviewed By:	K. Runner
Sample Location:	GB4 C1 @ 10'				
Soil Description:					
USCS:					

Sample Data:

Ring No:	G	Dish No:	52
Ring Mass (g):	237.8	Dish Mass (g):	7.8
Sample Height (in):	0.75	Swell Machine #:	7

Pre-test Sample		Post-test Sample	
Ring + Sample (g):	301.7	Ring + Sample (g):	307.5
Dish wt:	7.8	Dish wt:	8.2
Wet wt (g):	277.6	Wet wt (g):	69.3
Dry wt (g):	245.3	Dry wt (g):	56.2

Results:

Pre-test Sample		Post-test Sample	
Moisture Content:	13.6%	Moisture Content:	27.2%
Wet Density (pcf):	110.3	Wet Density (pcf):	120.8
Dry Density (pcf):	97.1	Dry Density (pcf):	94.9

Swell/Consolidation

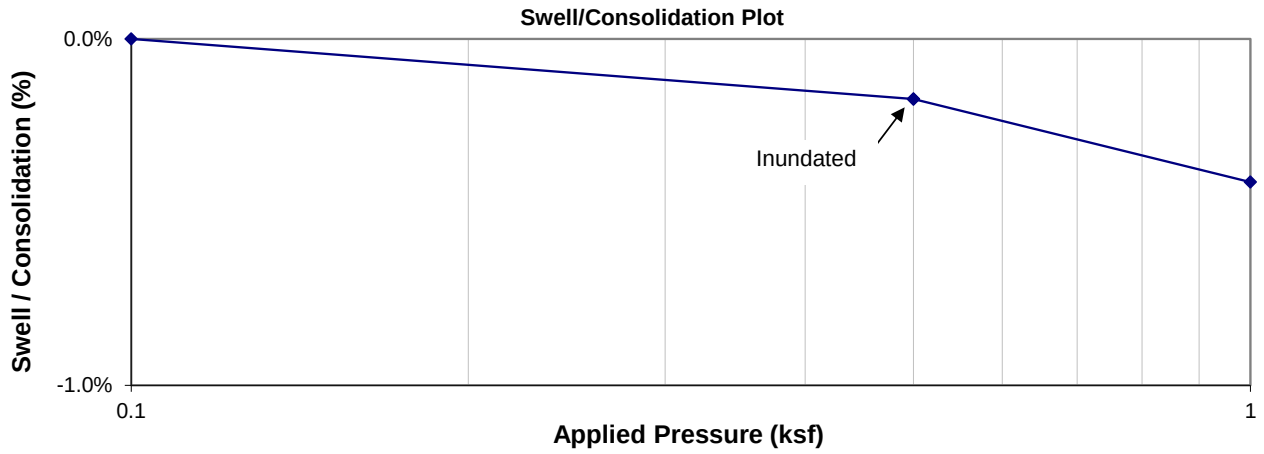
Load (ksf):	0.1	0.5	Add Water	0.5	1				
Correction (x 10-4):	0	20		20	36				
Dial Reading (x 10-4):	2262	2229		2229	2195				
Swell/Consolidation %:	**0.0%**	**-0.2%**		**-0.2%**	**-0.4%**				

Results:

Settlement Upon Wetting @ 500 psf: **0.0%**

Swell Pressure (psf): **303**

Tested By: R. S. Kay
Checked By K. Runner



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One Dimensional Swell/Consolidation (ASTM D 4546)

(Denver Area Swell/Consolidation Test)

Project Name: GRCA North Rim Water Supply Line Replacement File No.: 22-0232 Client Project 21186

Lab ID No.: 12579 Sample Date: 11/22/2023 Sampled By: Lithos Reviewed By: K. Runner

Sample Location: GB-17; C2 @ 5' USCS:

Sample Data:

Ring No:	A	Dish No:	Catishan
Ring Mass (g):	237.5	Dish Mass (g):	247.7
Sample Height (in):	0.75	Swell Machine #:	1

Pre-test Sample		Post-test Sample	
Ring + Sample (g):	301.9	Ring + Sample (g):	302.6
Dish wt:	247.7	Dish wt:	6.8
Wet wt (g):	558.6	Wet wt (g):	71.7
Dry wt (g):	502.4	Dry wt (g):	56.1

Results:

Pre-test Sample		Post-test Sample	
Moisture Content:	22.1%	Moisture Content:	31.6%
Wet Density (pcf):	111.1	Wet Density (pcf):	116.9
Dry Density (pcf):	91.0	Dry Density (pcf):	88.8

Swell/Consolidation

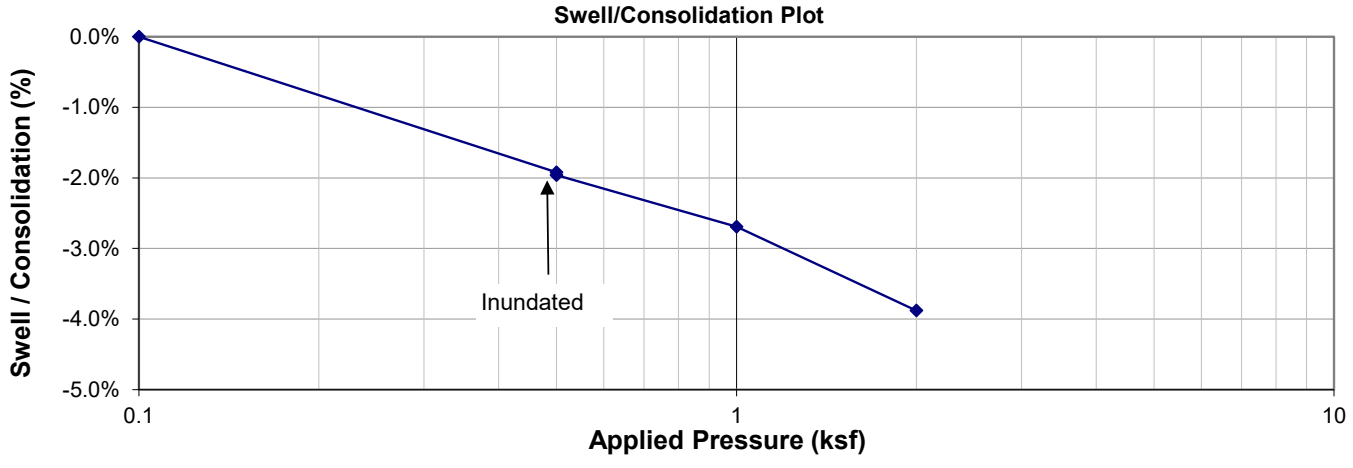
Load (ksf):	0.1	0.5	Add Water	0.5	1	2			
Correction (x 10-4):	0	16		16	38	89			
Dial Reading (x 10-4):	5950	5790		5787	5710	5570			
Swell/Consolidation %:	**0.0%**	**-1.9%**		**-2.0%**	**-2.7%**	**-3.9%**			

Results:

Swell Upon Wetting @ 500 psf: **0.0%**

Swell Pressure (psf): **NA**

Tested By: A. Dudney
Checked By: K. Runner



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Martinez Associates

Unconfined Compressive Strength of Cohesive Soil (ASTM 2166)

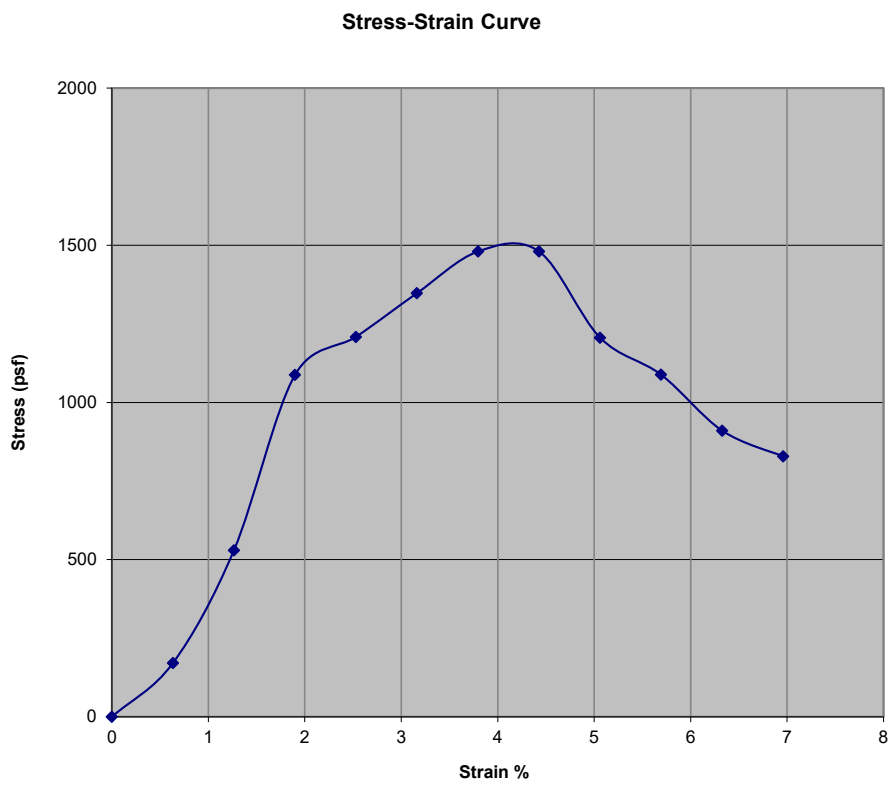
Client Project No.: 21186
Martinez Job No.: 22-0232
Lab ID Number: 12585
Sample Location: GB-19; C4 @ 20'
Soil Description:
USCS:

Proj. Name: GRCA North Rim Water Supply Line Replacement
Lab Tech: A. Dudney Test Date: 12/8/23

Sampled By: Lithos
Sample Date: 11/22/23
Reviewed By: K. Runner

Unconfined Compressive Strength (q_u): 1,480 psf

Axial Strain (%)	Axial Load (lb)
0.0	0
0.6	171
1.3	530
1.9	1087
2.5	1209
3.2	1348
3.8	1480
4.4	1480
5.1	1206
5.7	1088
6.3	910
7.0	829



Martinez Associates
14025 W. 66th Avenue
Arvada, Colorado 80004
Phone: (303) 459-2216
Fax: (303) 482-2230

Martinez Associates

Unconfined Compressive Strength of Cohesive Soil (ASTM 2166)

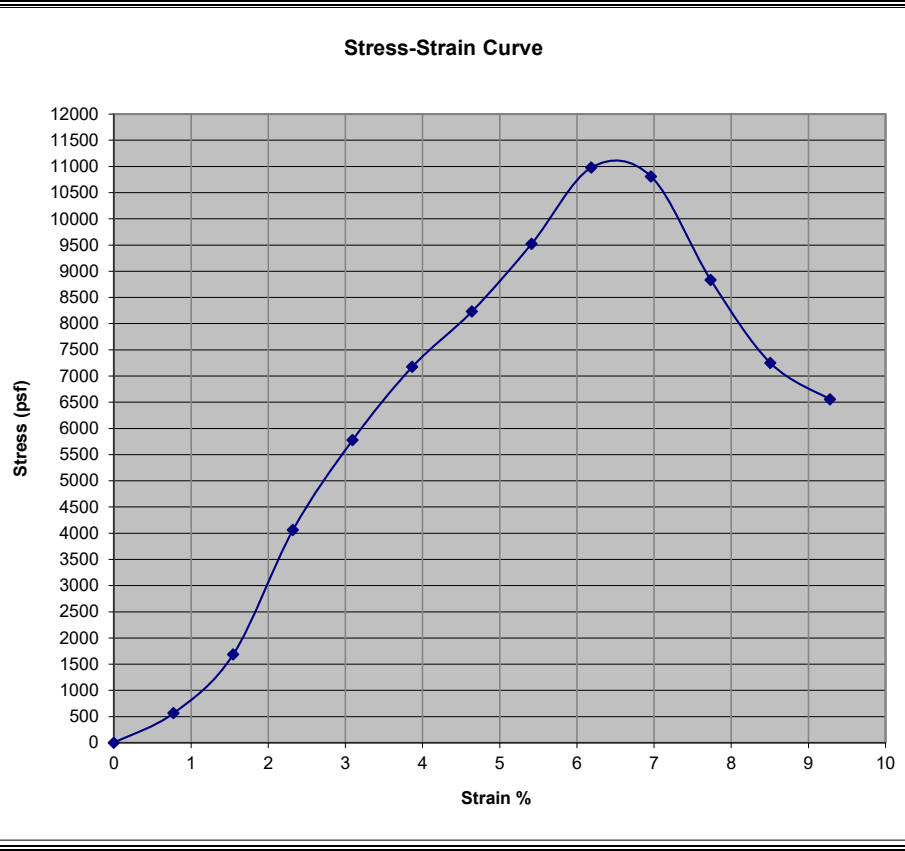
Client Project No.: 21186
Martinez Job No.: 22-0232
Lab ID Number: 12600
Sample Location: GB-26; C2 @ 5'
Soil Description:
USCS:

Proj. Name: GRCA North Rim Water Supply Line Replacement
Lab Tech: A. Dudney Test Date: 12/8/23

Sampled By: Lithos
Sample Date: 11/22/23
Reviewed By: K. Runner

Unconfined Compressive Strength (q_u): 10,978 psf

Axial Strain (%)	Axial Load (lb)
0.0	0
0.8	565
1.5	1687
2.3	4061
3.1	5778
3.9	7175
4.6	8233
5.4	9524
6.2	10978
7.0	10808
7.7	8835
8.5	7249
9.3	6556



ADVANCED TERRA TESTING

Unconfined Compressive Strength
ASTM D7012 Method C

CLIENT	Lithos Engineering			JOB NO.	2962-034
PROJECT	GRCA North Rim			LOCATION	North Rim, Arizona
PROJECT NO.	21186				
BORING NO.		GB3	GB3	BALLFIELD	BALLFIELD
DEPTH		30'	35'	12'	14'
SAMPLE NO.		Run 4	Run 5	Run 2A	Run 2B
DATE SAMPLED					
DATE TESTED		12/12/22	12/12/22	12/12/22	12/12/22
TECHNICIAN		DL	DL	DL	DL
ROCK TYPE					
Diameter (in):		2.391	2.389	2.375	2.387
Height (in):		4.909	4.987	4.998	5.101
Mass of Wet Rock (g):		839.30	859.20	795.30	846.00
Wet Density (lbs/ft³):		145.1	146.4	136.8	141.2
Wet Density (g/cm³):		2.324	2.345	2.192	2.262
Peak Load (lbs):		28408	37883	26104	14537
Compressive Strength (psi)		6327	8451	5892	3249
Compressive Strength (MPa)		44	58	41	22
Failure Type:		Fracture / Bedding	Fracture / Bedding	Shear / Fracture	Fracture / Bedding
BORING NO.					
DEPTH					
SAMPLE NO.					
DATE SAMPLED					
DATE TESTED					
TECHNICIAN					
ROCK TYPE					
Diameter (in):					
Height (in):					
Mass of Wet Rock (g):					
Wet Density (lbs/ft³):					
Wet Density (g/cm³):					
Peak Load (lbs):					
Compressive Strength (psi)					
Compressive Strength (MPa)					
Failure Type:					
NOTES					

Data entry by:	DL	Date: 12/12/22
Checked by:	HN	Date: 12/12/22
File name:	2962034__Rock UCS-TCS ASTM D7012 Method A and C_0.xlsm	

ADVANCED TERRA TESTING

Unconfined Compressive Strength
ASTM D7012 Method C

CLIENT	Lithos Engineering			JOB NO.	2962-052	
PROJECT	GRCA North Rim			LOCATION	North Rim, AZ	
PROJECT NO.	21186					
BORING NO.		GB-16	GB-17	GB-18	GB-19	GB-20
DEPTH		33	18	25	33.5	29.5
SAMPLE NO.		Run 2	Run 1	Run 2	Run 2	Run 2
DATE SAMPLED						
DATE TESTED		01/08/24	01/08/24	01/08/24	01/08/24	01/08/24
TECHNICIAN		DL	DL	DL	DL	DL
ROCK TYPE		Limestone	Limestone	Limestone	Limestone	Limestone
Diameter (in):		2.394	2.391	2.384	2.371	2.393
Height (in):		5.132	5.039	5.142	4.885	5.069
Mass of Wet Rock (g):		886.10	906.00	918.60	850.40	884.80
Wet Density (lbs/ft³):		146.1	152.6	152.5	150.2	147.9
Wet Density (g/cm³):		2.341	2.444	2.442	2.406	2.368
Peak Load (lbs):		33788	36306	29001	33930	33613
Compressive Strength (psi)		7506	8086	6497	7685	7474
Compressive Strength (MPa)		52	56	45	53	52
Failure Type:		Fracture / Bedding	Fracture / Bedding	Fracture / Shear	Shear / Fracture	Fracture / Shear

BORING NO.	GB-21	GB-23	GB-24	
DEPTH	37	34	32.5	
SAMPLE NO.	Run 2	Run 1	Run 2	
DATE SAMPLED				
DATE TESTED	01/08/24	01/08/24	01/08/24	
TECHNICIAN	DL	DL	DL	
ROCK TYPE	Limestone	Limestone	Limestone	
Diameter (in):	2.403	2.396	2.392	
Height (in):	5.086	5.054	4.744	
Mass of Wet Rock (g):	831.80	899.30	791.70	
Wet Density (lbs/ft³):	137.4	150.3	141.5	
Wet Density (g/cm³):	2.201	2.408	2.266	
Peak Load (lbs):	6619	41812	23957	
Compressive Strength (psi)	1460	9273	*5326	
Compressive Strength (MPa)	10	64	*37	
Failure Type:	Fracture / Bedding	Fracture / Bedding	Fracture / Bedding	

NOTES	* Despite the ASTM recommendation, due to sample dimensions a correction is applied with the equation CF= 1/(0.88+0.24(D/H))

Data entry by:	DL	Date: 01/09/24
Checked by:	HN	Date: 01/09/24
File name:	2962052__Rock UCS-TCS ASTM D7012 Method A and C_0.xlsm	

Appendix E Laboratory Corrosion Results

Soil Analysis Lab Results

Client: LITHOS Engineering
Job Name: GRCA North Rim Water Supply Line Replacement
Client Job Number: 21186
Project X Job Number: S221128A
November 29, 2022

	Method	ASTM D4327		ASTM D4327		ASTM G187		ASTM G51	ASTM G200	SM 4500-D	ASTM D4327	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D4327	ASTM D4327
Bore# / Description	**Depth**	**Sulfates** SO_4^{2-}		**Chlorides** Cl^-		**Resistivity** As Rec'd | Minimum		**pH**	**Redox**	**Sulfide** S^{2-}	**Nitrate** NO_3^-	**Ammonium** NH_4^+	**Lithium** Li^+	**Sodium** Na^+	**Potassium** K^+	**Magnesium** Mg^{2+}	**Calcium** Ca^{2+}	**Fluoride** F_2^{--}	**Phosphate** PO_4^{3-}
	(ft)	(mg/kg)	(wt%)	(mg/kg)	(wt%)	(Ohm-cm)	(Ohm-cm)		(mV)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)
NRL BP - S1 Poorly graded sand with little gravel, few clay	2.5-4	59.5	0.0059	28.1	0.0028	515,900	13,400	6.5	183	1.32	0.6	0.3	0.04	31.8	4.4	31.3	112.0	0.5	9.9
Ballfield BP - S1 Lean clay with some gravel	2.5-4	80.8	0.0081	41.8	0.0042	4,221	1,474	6.6	252	6.39	5.5	0.1	0.02	53.7	5.7	78.9	278.8	3.2	3.8
TF BP - S2 Lean clay with some gravel	10-11.5	39.9	0.0040	178.9	0.0179	8,040	3,551	5.5	266	2.60	0.9	0.2	0.01	133.4	2.3	32.3	193.0	7.2	42.5
GB3 - S1 Poorly graded sand with some clay, few gravel	2.5-4	62.0	0.0062	28.8	0.0029	201,000	27,470	7.0	231	1.14	0.9	0.0	0.02	31.5	6.7	20.6	102.0	0.2	n.a.
GB1 - S3 Lean clay, reddish brown	20-21.5	40.5	0.0041	35.3	0.0035	2,211	2,144	5.7	249	5.79	0.2	0.1	ND	25.8	3.3	125.9	72.3	12.4	25.9
GB2 - C1 #1 Poorly graded sand with little gravel, few clay	5-6	54.0	0.0054	18.3	0.0018	66,000	22,500	6.3	184	6.49	1.4	0.3	5.38	2.5	n.a.	18.4	85.9	0.0	0.3
GB4 - C1 #2 Clayey sand with little gravel	10-11	14.8	0.0015	819.8	0.0820	100,500	7,370	5.6	252	0.54	0.4	0.1	ND	29.3	934.8	12.5	72.3	0.4	0.4

Cations and Anions, except Sulfide and Bicarbonate, tested with Ion Chromatography
mg/kg = milligrams per kilogram (parts per million) of dry soil weight
ND = 0 = Not Detected | NT = Not Tested | Unk = Unknown
Chemical Analysis performed on 1:3 Soil-To-Water extract
PPM = mg/kg (soil) = mg/L (Liquid)

Soil Analysis Lab Results

Client: LITHOS Engineering
Job Name: GRCA North Rim Water Supply Line Replacement
Client Job Number: 21186
Project X Job Number: S231204K
December 6, 2023

	Method	ASTM D4327		ASTM D4327		ASTM G187		ASTM G51	ASTM G200	SM 4500-D	ASTM D4327	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D6919	ASTM D4327	ASTM D4327
Bore# / Description	**Depth**	**Sulfates** SO_4^{2-}		**Chlorides** Cl^-		**Resistivity** As Rec'd | Minimum		**pH**	**Redox**	**Sulfide** S^{2-}	**Nitrate** NO_3^-	**Ammonium** NH_4^+	**Lithium** Li^+	**Sodium** Na^+	**Potassium** K^+	**Magnesium** Mg^{2+}	**Calcium** Ca^{2+}	**Fluoride** F_2^{--}	**Phosphate** PO_4^{3-}
	(ft)	(mg/kg)	(wt%)	(mg/kg)	(wt%)	(Ohm-cm)	(Ohm-cm)		(mV)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)	(mg/kg)
GB-18, S4 Sandy Lean Clay	15.0	33.5	0.0034	53.8	0.0054	1,608	1,474	7.2	122	7.4	2.1	2.2	ND	57.8	10.5	69.0	302.1	14.7	100.9
GB-20, C1 Well Graded Sand with Silt and Gravel	5.0	75.8	0.0076	109.6	0.0110	127,300	10,050	7.4	140	0.6	0.7	2.9	ND	82.7	6.0	26.2	112.5	4.9	1.8
GB-23, S1 Clayey Sand with Gravel	5.0	33.6	0.0034	69.5	0.0069	201,000	14,740	6.3	157	5.5	0.6	2.8	ND	63.0	6.0	39.9	193.4	10.7	8.6
GB-26, S1 Silty Sand	2.5	26.9	0.0027	40.7	0.0041	227,800	14,070	7.1	151	4.3	34.1	0.1	ND	34.1	4.0	27.4	127.6	6.1	2.1
GB-29, C1 lean Clay with Gravel	2.5	23.1	0.0023	44.5	0.0045	3,685	2,613	6.1	150	2.4	1.7	1.3	ND	39.2	1.9	27.4	176.6	6.7	15.0

Cations and Anions, except Sulfide and Bicarbonate, tested with Ion Chromatography
mg/kg = milligrams per kilogram (parts per million) of dry soil weight
ND = 0 = Not Detected | NT = Not Tested | Unk = Unknown
Chemical Analysis performed on 1:3 Soil-To-Water extract
PPM = mg/kg (soil) = mg/L (Liquid)

Note: Sometimes a bad sulfate hit is a contaminated spot. Typical fertilizers are Potassium chloride, ammonium sulfate or ammonium sulfate nitrate (ASN). So this is another reason why testing full corrosion series is good because we then have the data to see if those other ingredients are present meaning the soil sample is just fertilizer-contaminated soil. This can happen often when the soil samples collected are simply surface scoops which is why it's best to dig in a foot, throw away the top and test the deeper stuff. Dairy farms are also notorious for these items.

Appendix F Percolation Test Results Memo

November 2, 2022
Project No. 21186

2750 S. Wadsworth Blvd., Suite D-200
Denver, Colorado 80227
303.625.9502
www.LithosEng.com

HDR
419 Canyon Ave., Suite 316
Fort Collins, Colorado 80521

Attention: Ryan Rhoades, PE

Regarding: Grand Canyon 318719 North Rim Utilities Improvements
Percolation Test Results

Mr. Rhoades,

Between September 19th and 20th Lithos Engineering conducted five percolation tests at locations directed by HDR, with the test being set up on the 19th and conducted on the 20th. These tests were conducted in the vicinity of Grand Canyon National Park's North Rim wastewater treatment plant with the intent on providing data that may guide the design of a future leach field for wastewater disposal. The location of each of these percolation tests is shown in Figure 1.



Figure 1: Location of each percolation test conducted on the North Rim of Grand Canyon National Park. Blue dots represent locations of percolation tests with identifying labels next to them.

Percolation tests were conducted using the EPA's falling head percolation test guidelines. The testing procedure consisted of the following:

- Dig a six-inch diameter hole three feet deep.
- Soak hole with water overnight so as to conduct the test with saturated soil conditions.
- Measure depths to the bottom of the hole and the top of the approximately two-inch gravel layer placed at the bottom of the hole. Water added to the hole for testing should be six inches above the top of the gravel layer.
- Add water to the hole until the water surface is six inches above the top of the gravel or eight inches above the bottom of the hole.
- Wait specified amount of time (10 min for sandy soils, 30 min for soils with a higher clay content) and measure how far the water level had dropped.
- Repeat measurements every 10 minutes for an hour if sandy soil, or every 30 minutes until two consecutive water level drops differ by less than 1/16 of an inch for more clay rich soils.

Upon soaking the holes it was found that percolation test 1 (PT1) was sandy, with water percolating quickly out of the hole, and as such the sandy soil testing procedure was used. The remaining test locations possessed enough clay to warrant the use of the clay testing procedure.

Results from the percolation tests are shown in the following tables. Photos of the testing setup for each location are also provided. A drawing of the test setup used is found in the attached document.

Percolation Test 1 (PT1)

Observed Soil Type = sand and cobbles for first two feet followed by one foot of stiff clay
Depth to Bottom = 3’ 1.5”
Depth to Gravel = 2’ 9.5”
Depth to Top Water = 2’ 3.5”

Table 1: Percolation test data for Percolation test 1.

Time	Interval	Depth	Drop	Rate
8:50	-	2’ 3.5”	-	
9:00	10	2’ 9.5”	6” (in 7 min)	
9:10	10	2’ 8.5”	5”	
9:20	10	2’ 7.5”	4”	
9:30	10	2’ 7”	3.5”	
9:40	10	2’ 8”	4.5”	
9:50	10	2’ 7.5”	4”	**2.5 min/inch**



Figure 2: Test setup for percolation test 1.

Percolation Test 2 (PT 2)

Observed Soil Type = predominately sand with few cobbles and few fines
Depth to Bottom = 3’ 5”
Depth to Gravel = 3’ 2”
Depth to Top Water = 2’ 8”

Table 2: Percolation test data for Percolation test 2.

Time	Interval	Depth	Drop	Rate
9:05	-	2’ 8”	-	
9:35	30	2’ 8.5”	0.5”	
10:05	30	2’ 8.625”	0.625”	
10:35	30	2’ 9”	1”	
11:05	30	2’ 8.5”	0.5”	
11:35	30	2’ 8.5”	0.5”	**60 min/inch**



Figure 3: Test setup for percolation test 2.

Percolation Test 3 (PT 3)

Observed Soil Type = sand and clay with few cobbles
Depth to Bottom = 3’ 6”
Depth to Gravel = 3’ 3”
Depth to Top Water = 2’ 9”

Table 3: Percolation test data for Percolation test 3.

Time	Interval	Depth	Drop	Rate
9:25	-	2’ 8”	-	
9:55	30	2’ 8”	0”	
10:25	30	2’ 8.25”	0.25”	
11:00	35	2’ 8.25”	0.25”	
11:30	30	2’ 8.1875”	0.1875”	**160 min/inch**



Figure 4: Test setup for percolation test 3.

Percolation Test 4 (PT 4)

Observed Soil Type = sand and abundant cobbles

Depth to Bottom = 3’

Depth to Gravel = 2’ 10”

Depth to Top Water = 2’ 4”

Table 4: Percolation test data for Percolation test 4.

Time	Interval	Depth	Drop	Rate
12:50	-	2’ 4”	-	
13:20	30	2’ 5.5”	1.5”	
13:50	30	2’ 5”	1”	
14:20	30	2’ 5”	1”	**30 min/inch**



Figure 5: Test setup for percolation test 4.

Percolation Test 5 (PT 5)

Observed Soil Type = sand and abundant cobbles
Depth to Bottom = 3’ 3”
Depth to Gravel = 3’ 1”
Depth to Top Water = 2’ 7”

Table 5: Percolation test data for Percolation test 5.

Time	Interval	Depth	Drop	Rate
13:00	-	2’ 6.5”	-	
13:30	30	2’ 7.5”	1”	
14:00	30	2’ 7.25”	0.75”	
14:30	30	2’ 7.25”	0.75”	**40 min/inch**



Figure 6: Test setup for percolation test 5.

If you have any questions about the contents of this memorandum, please do not hesitate to contact the undersigned.

Sincerely,
Lithos Engineering

Adam Malsam, EIT
Staff Engineer

Matt Olson, PE
Project Manager

LITHOS ENGINEERING